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DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

Division of Electronics and Computers
Applied Electronics Laboratory

Development/Implementation of a Microwave Transceiver for a Telecommunication System.

D I P L O M A T H E S I S

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ABSTRACT

Development/Implementation of a Microwave Transceiver for a Telecommunication System

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This diploma thesis presents the PCB design of a dual-polarization mmWave transceiver platform for the 28 GHz to 31 GHz band. This range partially overlaps the 5G NR FR2 spectrum and is also relevant to fixed wireless access and point-to-point links in the microwave/mmWave region.

The platform operates in Time Division Duplex (TDD) mode. The hardware consists of two AKRONIC ALTUS28TR transceivers, one per polarization branch, and an AKRONIC NEXUS28Z PLL that provides a shared local oscillator for both branches. Two Red Pitaya STEMlab 125-14 boards generate and acquire the analog I/Q baseband signals, with fully differential amplifier stages adapting the single-ended Red Pitaya ports to the differential transceiver interface.

The thesis covers the system architecture, component selection, interface circuits, power distribution, and PCB layout in KiCad on a six-layer hybrid RO4350B/FR-4 stackup. The power network and the analog baseband interfaces are verified in LTspice, the critical high-frequency interconnects are extracted with full-wave EM simulation in RFPro, and a Python framework implements host-side TDD control of the Red Pitaya boards, including timing parameters, guard intervals, and GPIO switch logic.

The simulations confirm the design targets. All supply rails settle at their nominal values, with converter-side switching ripple filtered to sub-millivolt levels at the point-of-load nodes. The transmit interface presents an approximately 50 Ω load to the Red Pitaya DAC and delivers a 1.6 Vpp differential signal at a 0.5 V common-mode level, while the receive interface drives the ADC with a ground-centered 2 Vpp signal. The EM extractions show a well-matched LO path, with return loss near 30 dB and insertion loss below 0.1 dB across the LO band, and an RF/TDD front-end whose through loss and branch coupling remain within the levels expected for a dense mmWave switching region. The result is a layout-complete, simulation-verified PCB design package, ready for fabrication, bring-up, and laboratory evaluation.

ΕΚΤΕΤΑΜΕΝΗ ΕΛΛΗΝΙΚΗ ΠΕΡΙΛΗΨΗ

ΣΧΕΔΙΑΣΗ ΚΑΙ ΑΝΑΠΤΥΞΗ ΜΙΚΡΟΚΥΜΑΤΙΚΟΥ ΠΟΜΠΟΔΕΚΤΗ ΓΙΑ ΤΗΛΕΠΙΚΟΙΝΩΝΙΑΚΟ ΣΥΣΤΗΜΑ

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ΟΝΟΜΑΤΕΠΩΝΥΜΟ ΕΠΙΒΛΕΠΟΝΤΟΣ:
ΓΡΗΓΟΡΙΟΣ ΚΑΛΥΒΑΣ

Η παρούσα διπλωματική εργασία ασχολείται με τον σχεδιασμό και την προσομοιωτική αξιολόγηση μιας πλακέτας τυπωμένου κυκλώματος (PCB) για πομποδέκτη mmWave διπλής πόλωσης στην περιοχή από 28 GHz έως 31 GHz. Η περιοχή αυτή είναι χρήσιμη για ζεύξεις υψηλής χωρητικότητας, fixed wireless access και point-to-point μικροκυματικές συνδέσεις. Επικαλύπτεται επίσης με τμήματα των ζωνών 5G NR FR2, χωρίς όμως να αποτελεί στο σύνολό της μία ενιαία ζώνη 5G NR. Για αυτό, στο κείμενο αντιμετωπίζεται ως mmWave περιοχή ενδιαφέροντος για τηλεπικοινωνιακές εφαρμογές και όχι ως αποκλειστικά 5G band.

Η πλατφόρμα βασίζεται σε δύο ολοκληρωμένους πομποδέκτες ALTUS28TR της εταιρείας AKRONIC, έναν για κάθε κλάδο πόλωσης. Για το LO χρησιμοποιείται ένα κοινό PLL NEXUS28Z. Με αυτή τη διάταξη μπορούν να υλοποιηθούν δύο κλάδοι πόλωσης, κατακόρυφος και οριζόντιος, με κοινή συχνοτική αναφορά. Η τελική σχετική φάση των δύο κλάδων δεν καθορίζεται μόνο από το PLL. Επηρεάζεται επίσης από τη διανομή του LO, τα μήκη των διαδρομών, τα εξαρτήματα και τις ανοχές της πλακέτας. Για την παραγωγή και την ψηφιοποίηση των σημάτων βασικής ζώνης χρησιμοποιούνται δύο Red Pitaya STEMlab 125-14 boards, τα οποία λειτουργούν ως εξωτερικό baseband subsystem.

Το σύστημα λειτουργεί σε Time Division Duplex (TDD) mode. Η εκπομπή και η λήψη χρησιμοποιούν το ίδιο φασματικό κανάλι, αλλά σε διαφορετικές χρονικές θυρίδες. Χρειάζονται επομένως guard intervals και έλεγχος των RF switches, ώστε να μη συμπίπτουν οι καταστάσεις εκπομπής και λήψης. Για τον έλεγχο αυτής της ακολουθίας γράφτηκε Python host-side κώδικας. Ο κώδικας διαμορφώνει τα Red Pitaya boards, ορίζει τα βασικά time slots και οδηγεί μέσω GPIO τα σήματα που αλλάζουν την κατάσταση των RF switches.

Η εργασία σταματά πριν από την κατασκευή της πλακέτας. Το βάρος δίνεται στο PCB layout, στις κυκλωματικές προσομοιώσεις και στην ηλεκτρομαγνητική αξιολόγηση των σημείων που κρίθηκαν πιο ευαίσθητα. Το αποτέλεσμα είναι ένα PCB design package με ολοκληρωμένο layout και τεκμηριωμένους προσομοιωτικούς ελέγχους. Μπορεί να χρησιμοποιηθεί ως βάση για μελλοντική κατασκευή, bring-up και εργαστηριακή αξιολόγηση.

1. Εισαγωγή και Στόχος της Εργασίας

Σε συχνότητες κοντά στα 30 GHz, η πλακέτα δεν είναι απλώς το μέσο που συνδέει τα ολοκληρωμένα. Μικρά τμήματα δρομολόγησης, pads εξαρτημάτων, connector launches, vias και αλλαγές γεωμετρίας μπορούν να επηρεάσουν την προσαρμογή, τις απώλειες και τη σύζευξη μεταξύ κοντινών διαδρομών. Σε χαμηλότερες συχνότητες αρκετά από αυτά θα είχαν μικρότερη σημασία. Στην περιοχή mmWave, όμως, μπαίνουν μέσα στην ίδια τη RF συμπεριφορά του συστήματος.

Ο στόχος της εργασίας είναι ο σχεδιασμός μιας ρεαλιστικής PCB πλατφόρμας πομποδέκτη διπλής πόλωσης στην περιοχή από 28 GHz έως 31 GHz. Η εργασία καλύπτει την αρχιτεκτονική του συστήματος, την επιλογή των βασικών ολοκληρωμένων, τα αναλογικά και ψηφιακά κυκλώματα υποστήριξης, το δίκτυο τροφοδοσίας, το layout της πλακέτας και την προσομοιωτική αξιολόγηση των πιο κρίσιμων τμημάτων. Η έμφαση βρίσκεται στην πρακτική ενσωμάτωση όλων αυτών σε μία κοινή πλακέτα, με τους περιορισμούς που επιβάλλει η λειτουργία σε mmWave συχνότητες.

2. Αρχιτεκτονική της Πλατφόρμας

Η πλατφόρμα χωρίζεται πρακτικά σε RF, LO, baseband, τροφοδοσία και ψηφιακό έλεγχο. Κάθε πόλωση έχει τον δικό της ALTUS28TR. Ο συγκεκριμένος πομποδέκτης συγκεντρώνει τα TX και RX paths, τη διαφορική I/Q baseband διεπαφή και τον SPI έλεγχο, οπότε γύρω από αυτόν οργανώνεται μεγάλο μέρος του layout.

Το κοινό LO παράγεται από το NEXUS28Z και οδηγείται στους δύο πομποδέκτες μέσω επιλογής πηγής, ενίσχυσης και διανομής. Η πλακέτα περιλαμβάνει και είσοδο εξωτερικού LO μέσω SMA. Αυτή η επιλογή κρατήθηκε γιατί το onboard PLL δεν καλύπτει μόνο του όλο το άνω τμήμα του στόχου συχνοτήτων.

Η βασική ζώνη συνδέεται με τα Red Pitaya STEMlab 125-14 boards. Τα Red Pitaya δίνουν και δέχονται single-ended αναλογικά σήματα, ενώ ο ALTUS28TR χρησιμοποιεί διαφορικές baseband θύρες με συγκεκριμένες απαιτήσεις πλάτους και common-mode. Για αυτό χρειάζονται ξεχωριστά στάδια signal conditioning στην εκπομπή και στη λήψη. Στην εκπομπή, το LTC6409 προσαρμόζει το σήμα του Red Pitaya, το μετατρέπει από single-ended σε differential και το οδηγεί προς την είσοδο του ALTUS28TR με το απαιτούμενο common-mode επίπεδο. Στη λήψη, το ADA4927 παίρνει το διαφορικό σήμα από τον ALTUS28TR και το μετατρέπει σε single-ended σήμα κατάλληλο για το ADC του Red Pitaya.

3. Σχεδίαση PCB και Stackup

Η πλακέτα σχεδιάστηκε ως εξαεπίπεδη hybrid κατασκευή. Στα εξωτερικά διηλεκτρικά στρώματα χρησιμοποιείται RO4350B, ώστε οι κρίσιμες RF και LO διαδρομές να βρίσκονται σε υλικό χαμηλότερων απωλειών. Τα εσωτερικά στρώματα βα-

σίζονται σε S1000-2M και S1000-2MB, που είναι πιο πρακτικά για power planes, control routing και μηχανική υποστήριξη. Έτσι το RF υλικό χρησιμοποιείται εκεί όπου έχει ουσιαστική επίδραση στην απόδοση, χωρίς να γίνει ολόκληρη η πλακέτα από ακριβότερο low-loss laminate.

Οι RF και LO διαδρομές δρομολογούνται στο επάνω στρώμα ως grounded coplanar waveguides (GCPW). Το αμέσως επόμενο εσωτερικό στρώμα λειτουργεί ως ground reference. Η μορφή GCPW βοηθά στον έλεγχο της τοπικής επιστροφής ρεύματος και επιτρέπει πυκνό via stitching γύρω από τις κρίσιμες γραμμές. Για τις RF/LO διαδρομές χρησιμοποιήθηκε γεωμετρία περίπου 0.30 mm πλάτους και περίπου 0.10 mm πλευρικού κενού, ώστε η χαρακτηριστική αντίσταση να βρίσκεται κοντά στα 50 Ω.

Η τοποθέτηση των εξαρτημάτων έγινε με βάση τη λειτουργία κάθε περιοχής. Τα RF front-end τμήματα μπήκαν κοντά στους αντίστοιχους connectors και στους ALTUS28TR. Η LO αλυσίδα οργανώθηκε ώστε να τροφοδοτεί τους δύο κλάδους με όσο το δυνατόν πιο συμμετρική διανομή. Τα baseband stages τοποθετήθηκαν κοντά στις αντίστοιχες θύρες, ενώ το ψηφιακό υποσύστημα κρατήθηκε μακριά από τις πιο ευαίσθητες RF και αναλογικές περιοχές.

Στο layout η γείωση αντιμετωπίστηκε ως μέρος της RF σχεδίασης και όχι ως απλή copper fill περιοχή. Γύρω από τις RF και LO διαδρομές χρησιμοποιήθηκε via stitching, ενώ οι αλλαγές στρώματος στις high-frequency διαδρομές κρατήθηκαν όσο το δυνατόν λιγότερες. Έτσι οι return paths μένουν πιο ελεγχόμενες και αποφεύγονται περιττές ασυνέχειες.

Η τελική πλακέτα έχει διαστάσεις 135 mm $\times$ 150 mm και περιλαμβάνει περίπου 130 εξαρτήματα. Το μέγεθος δεν προκύπτει μόνο από τις RF διαδρομές. Χρειάστηκε χώρος για τους RF connectors, τη διανομή του LO, τα αναλογικά στάδια, τις τροφοδοσίες, τα jumpers και τα σημεία πρόσβασης για το bring-up. Η πλακέτα επομένως δεν σχεδιάστηκε ως μικρό RF test board, αλλά ως board-level πλατφόρμα για έλεγχο πολλών υποσυστημάτων.

4. Δίκτυο Τροφοδοσίας

Το δίκτυο τροφοδοσίας παράγει τις απαραίτητες τάσεις από εξωτερική είσοδο 5 V. Οι βασικές τροφοδοσίες είναι η 2.6 V για τους ALTUS28TR, οι 1.8 V γραμμές για το NEXUS28Z, η 3.3 V ψηφιακή τροφοδοσία και οι ± 2.5 V αναλογικές τροφοδοσίες των FDA stages.

Για τις κύριες θετικές τάσεις χρησιμοποιούνται switching regulators. Οι πιο ευαίσθητες ή βοηθητικές τροφοδοσίες υλοποιούνται με LDOs, ενώ η αρνητική αναλογική τάση παράγεται με inverting converter. Έτσι, τα switching stages χρησιμοποιούνται στα rails όπου η απόδοση έχει μεγαλύτερη σημασία, ενώ τα LDOs τοποθετούνται στα πιο ευαίσθητα σημεία.

Στο PDN δεν αρκεί να παραχθεί η σωστή DC τάση. Κάθε rail έχει διαφορετικό ρεύμα, διαφορετική ευαισθησία στον θόρυβο και διαφορετικές απαιτήσεις τοπικής αποσύζευξης. Για αυτό λήφθηκαν υπόψη η πτώση τάσης στους αγωγούς χαλκού,

η κυμάτωση των converters και η απομόνωση μεταξύ ευαίσθητων και πιο θορυβωδών domains.

Στα rails μεγαλύτερου ρεύματος χρησιμοποιήθηκαν πλατύτερες διαδρομές χαλκού, copper pours και πολλαπλά vias στις αλλαγές στρώματος. Με αυτόν τον τρόπο μειώνεται η αντίσταση της διαδρομής και αποφεύγεται η συγκέντρωση ρεύματος σε ένα μόνο via ή σε ένα πολύ στενό τμήμα χαλκού.

Η πλακέτα περιλαμβάνει και πρακτικές δυνατότητες bring-up. Κάθε κύριο rail μπορεί να πάρει τροφοδοσία είτε από το onboard regulator είτε από εξωτερική πηγή μέσω power-injection pad. Η επιλογή γίνεται με jumper. Έτσι μπορεί να σηκωθεί ένα rail ή ένα συγκεκριμένο domain χωρίς να ενεργοποιηθεί όλο το σύστημα.

Τα status LEDs μπήκαν για τον πρώτο έλεγχο της πλακέτας. Με μια ματιά φαίνεται αν οι βασικές τροφοδοσίες ανέβηκαν κατά την αρχική ενεργοποίηση. Η ακριβής τιμή κάθε rail πρέπει φυσικά να μετρηθεί με όργανο, αλλά αυτή η ένδειξη βοηθά στο bring-up και γλιτώνει χρόνο όταν υπάρχει κάποιο βασικό πρόβλημα τροφοδοσίας.

Η προσομοιωτική αξιολόγηση του PDN έγινε στο LTspice. Εξετάστηκαν η εκκίνηση, η σταθεροποίηση των τάσεων, η κυμάτωση και η διαφορά μεταξύ converter-side και point-of-load κόμβων. Οι προσομοιώσεις έδειξαν ότι οι τάσεις συγκλίνουν κοντά στις ονομαστικές τιμές τους.

Ο συνδυασμός ferrite bead και τοπικής αποσύζευξης μείωσε την κυμάτωση από περίπου 2 mV έως 10 mV peak-to-peak στους converter-side κόμβους σε sub-millivolt επίπεδα στα αντίστοιχα point-of-load nodes. Οι βασικές περιπτώσεις που εξετάστηκαν ήταν η 2.6 V RF τροφοδοσία, η 1.8 V core τροφοδοσία και οι ± 2.5 V αναλογικές τροφοδοσίες.

5. Baseband Interface και Ψηφιακός Έλεγχος

Το Red Pitaya και ο ALTUS28TR δεν μπορούν να συνδεθούν απευθείας στη βασική ζώνη. Το Red Pitaya χρησιμοποιεί single-ended αναλογικά I/O, ενώ ο ALTUS28TR απαιτεί διαφορική διασύνδεση με συγκεκριμένο common-mode και συγκεκριμένο επιτρεπτό πλάτος σήματος. Στην εκπομπή, το LTC6409 stage παρουσιάζει φορτίο περίπου 50 Ω προς το DAC του Red Pitaya, μετατρέπει το σήμα σε διαφορικό και ρυθμίζει το common-mode γύρω από 0.5 V. Στη λήψη, το ADA4927 stage δέχεται το διαφορικό σήμα από τον ALTUS28TR και παρέχει single-ended, ground-centered σήμα περίπου 2 V_{pp} προς το ADC του Red Pitaya, χωρίς να φορτώνει έντονα το προηγούμενο buffer.

Το ψηφιακό υποσύστημα ελέγχου έχει δύο τρόπους πρόσβασης. Ο πρώτος είναι το onboard FT232H μέσω micro-USB, που λειτουργεί ως USB-to-SPI/GPIO interface για host-side έλεγχο. Ο δεύτερος είναι το προαιρετικό εξωτερικό Cheetah header. Με αυτό μπορεί να γίνει άμεσο external SPI control και debugging χωρίς να χρησιμοποιείται το FT232H path.

Το host-side domain λειτουργεί στα 3.3 V, ενώ τα RF ICs απαιτούν χαμηλότερα

logic levels. Για αυτό χρησιμοποιούνται level shifters από 3.3 V προς 2.6 V για τους ALTUS28TR και από 3.3 V προς 1.8 V για το NEXUS28Z. Η SPI αρχιτεκτονική χρησιμοποιεί dedicated chip-selects και απομόνωση των shared readback paths. Όπου υπάρχει πιθανότητα δύο ολοκληρωμένα να οδηγήσουν την ίδια γραμμή επιστροφής δεδομένων, χρησιμοποιούνται tri-state buffers.

Το software framework σε Python χρησιμοποιείται για τη διαμόρφωση των Red Pitaya boards, τη δημιουργία της TDD ακολουθίας, τον έλεγχο των time slots και τη μεταγωγή των RF switches μέσω GPIO. Η λύση αυτή είναι χρήσιμη για αρχικό bring-up και λειτουργική διερεύνηση. Δεν δίνει όμως deterministic hardware timing. Για αυστηρό TDD συγχρονισμό, ο κρίσιμος χρονισμός θα πρέπει σε επόμενη φάση να μεταφερθεί σε FPGA ή σε άλλο deterministic hardware.

6. Προσομοιώσεις και Ηλεκτρομαγνητική Αξιολόγηση

Η προσομοιωτική αξιολόγηση δεν έγινε με το ίδιο εργαλείο για όλα τα τμήματα της πλακέτας. Τα κυκλώματα χαμηλότερης συχνότητας, όπως το PDN και τα baseband conditioning stages, εξετάστηκαν στο LTspice. Οι δομές υψηλής συχνότητας που επηρεάζονται έντονα από τη γεωμετρία του layout εξετάστηκαν με full-wave EM extraction στο RFPro.

Στο TX baseband interface, οι προσομοιώσεις έλεγξαν τη μετατροπή του single-ended σήματος σε διαφορικό, τη μετατόπιση του common-mode και την αντίσταση που βλέπει η έξοδος του Red Pitaya. Στο RX baseband interface, οι προσομοιώσεις έδειξαν ότι το ADA4927 μπορεί να οδηγήσει την είσοδο του Red Pitaya με κατάλληλο πλάτος, χωρίς να φορτώνει υπερβολικά την έξοδο του transceiver.

Τα μοντέλα που χρησιμοποιούνται πέρα από το PCB boundary είναι απλοποιημένα. Επομένως, τα αποτελέσματα ελέγχουν κυρίως την εξωτερική διασύνδεση και όχι το πλήρες εσωτερικό μοντέλο του ALTUS28TR ή του Red Pitaya.

Για την EM ανάλυση επιλέχθηκαν δύο περιοχές της πλακέτας. Η πρώτη ήταν το σημείο της διαδρομής LO όπου γίνεται αλλαγή στρώματος, επειδή αυτή είναι η σημαντικότερη high-frequency layer transition που διατηρήθηκε στο layout. Η δεύτερη ήταν η περιοχή RF/TDD γύρω από το SPDT switch και τις RF συνδέσεις του ALTUS28TR. Οι δύο περιοχές περιλαμβάνουν bends, pads, via environments, ground features και compact multi-port γεωμετρίες. Για τέτοια σημεία, μια απλή transmission-line προσέγγιση δεν αρκεί.

Η διαδρομή LO είχε καλή συμπεριφορά στην προσομοίωση. Στο σημείο αλλαγής στρώματος, το S_{11} βρέθηκε κοντά στα −30 dB, ενώ η απώλεια διέλευσης ήταν μικρότερη από περίπου 0.1 dB στην περιοχή από 8 GHz έως 11 GHz. Με βάση αυτά τα αποτελέσματα, η συγκεκριμένη αλλαγή στρώματος δεν φαίνεται να περιορίζει τη συμπεριφορά της LO διαδρομής.

Η περιοχή RF/TDD γύρω από το SPDT switch ήταν πιο απαιτητική. Εκεί οι απώλειες ήταν μεγαλύτερες και η απόκριση είχε πιο έντονη εξάρτηση από τη συχνότητα. Το S_{21} κυμάνθηκε περίπου από −1 dB έως −3 dB στην περιοχή από 28 GHz έως 31 GHz, ενώ οι όροι σύζευξης παρέμειναν περίπου από −35 dB έως −30 dB. Από τις δομές

που εξετάστηκαν, η περιοχή αυτή έχει το μεγαλύτερο περιθώριο για μελλοντική βελτίωση. Ο βασικός λόγος είναι η πυκνή γεωμετρία γύρω από το switch, τα pad transitions και οι κοντινές RF διαδρομές.

7. Περιορισμοί

Η εργασία σταματά πριν από το στάδιο της κατασκευής. Η πλακέτα δεν έχει συναρμολογηθεί και δεν υπάρχουν ακόμη μετρήσεις από πραγματικό hardware. Για αυτό τα αποτελέσματα πρέπει να διαβαστούν ως αποτελέσματα σχεδιασμού και προσομοίωσης.

Η EM ανάλυση δεν έγινε σε πλήρες μοντέλο όλης της συναρμολογημένης πλακέτας. Επιλέχθηκαν δύο σημεία που ήταν πιο πιθανό να επηρεάσουν τη συμπεριφορά στις υψηλές συχνότητες. Το πρώτο είναι η αλλαγή στρώματος στη διαδρομή LO. Το δεύτερο είναι η περιοχή RF/TDD γύρω από το SPDT switch. Αυτά τα σημεία έχουν bends, pads, vias και κοντινές RF διαδρομές, οπότε η γεωμετρία τους έχει μεγαλύτερη σημασία από ό,τι σε μια απλή ευθεία γραμμή.

Η τοπική EM ανάλυση βοηθά να φανεί πώς συμπεριφέρονται αυτά τα σημεία του layout. Δεν δείχνει όμως από μόνη της τη συνολική συμπεριφορά της πλακέτας στο εργαστήριο. Αυτό θα φανεί μόνο μετά την κατασκευή, το bring-up και τις S-parameter μετρήσεις.

Υπάρχουν επίσης απλοποιήσεις στα κυκλωματικά μοντέλα. Τα ferrite beads του PDN προσομοιώθηκαν με RLC ισοδύναμα και όχι με πλήρη vendor broadband models. Στα baseband simulations χρησιμοποιήθηκαν απλοποιημένα φορτία πέρα από το PCB boundary, ώστε να υπάρχει πιο ρεαλιστικό περιβάλλον στην είσοδο ή στην έξοδο του interface. Οι απλοποιήσεις αυτές είναι αποδεκτές για έλεγχο σχεδίασης, αλλά δεν αντικαθιστούν μοντέλα βασισμένα σε μετρήσεις.

Ένας πρακτικός περιορισμός αφορά και το εύρος του onboard LO. Το NEXUS28Z PLL δεν καλύπτει μόνο του ολόκληρη την περιοχή από 28 GHz έως 31 GHz. Μετά τον εσωτερικό $\times 3$ πολλαπλασιασμό του ALTUS28TR, η onboard LO λύση φτάνει περίπου έως 29.4 GHz. Για το άνω τμήμα της περιοχής χρειάζεται εξωτερικό LO μέσω του αντίστοιχου SMA connector, δυνατότητα που έχει ήδη προβλεφθεί στη σχεδίαση.

8. Συμπεράσματα και Μελλοντική Εργασία

Το τελικό layout φτάνει τη dual-polarization TDD mmWave πλατφόρμα στην περιοχή από 28 GHz έως 31 GHz σε επίπεδο PCB design. Η δυσκολία μοιράζεται σε πολλά σημεία της πλακέτας. Προκύπτει από τον συνδυασμό του stackup, της LO διανομής, των RF διαδρομών, του PDN, των baseband interfaces και του ψηφιακού ελέγχου.

Το αποτέλεσμα της εργασίας είναι ένα τεκμηριωμένο PCB design package. Περι-

λαμβάνει ολοκληρωμένο layout, κυκλωματικές προσομοιώσεις και EM ελέγχους στα σημεία που κρίθηκαν πιο ευαίσθητα. Είναι ένα σχέδιο που μπορεί να περάσει στο επόμενο στάδιο, δηλαδή στην κατασκευή, στο bring-up και στην εργαστηριακή αξιολόγηση.

Το επόμενο βήμα είναι η κατασκευή και το bring-up της πλακέτας. Αρχικά θα πρέπει να ελεγχθούν οι τροφοδοσίες, τα jumper-selected power domains, τα rail-status indicators, η SPI επικοινωνία, η LO αλυσίδα και τα TX/RX baseband interfaces. Έπειτα, οι κρίσιμες RF και LO διαδρομές θα πρέπει να μετρηθούν με S-parameters και να συγκριθούν με τις RFPro εξαγωγές.

Η περιοχή RF/TDD γύρω από το SPDT switch είναι το βασικό σημείο για μελλοντική βελτίωση του layout. Η EM ανάλυση έδειξε ότι η περιοχή αυτή είναι πιο ευαίσθητη από τη διαδρομή LO και συνεισφέρει περισσότερο στις frequency-dependent απώλειες. Σε επόμενη έκδοση της πλακέτας θα μπορούσαν να εξεταστούν διαφορετική γεωμετρία γύρω από το SPDT switch, πιο ομαλές pad transitions, διαφορετική τοπική γείωση και βελτιωμένο via stitching.

Μετά το bring-up, η πλακέτα μπορεί να δοκιμαστεί με κατάλληλη κεραία ή antenna array. Σε αυτό το στάδιο θα φανεί πώς λειτουργούν οι δύο κλάδοι πόλωσης και πώς συμπεριφέρεται το receive path. Μπορούν επίσης να γίνουν channel-sounding πειράματα και δοκιμές TDD με πραγματικά guard intervals. Αυτό είναι το πρακτικό επόμενο βήμα μετά το παρόν PCB design.

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1 Introduction

1.1 Background and motivation

Wireless systems are being pushed toward higher data rates, shorter delay, and operation in denser networks. One way to support these requirements is to move part of the radio link to higher carrier frequencies, where wider channel bandwidths can be used. The 28 GHz to 31 GHz region is relevant here because it overlaps parts of the 5G NR FR2 spectrum and is also used in fixed wireless access and point-to-point microwave links [1].

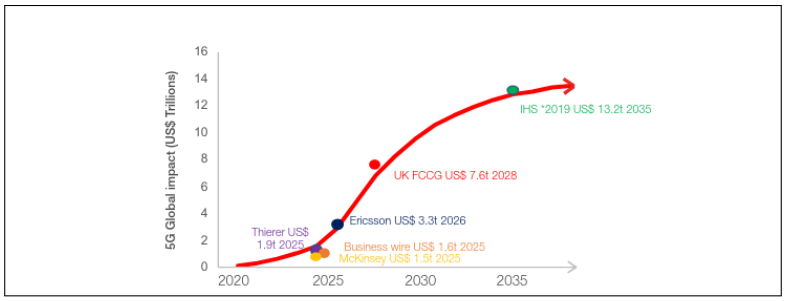


Fig. 1. Cumulative global 5G impact, 2020–2035 [2].

At these frequencies, the PCB becomes part of the RF design. A short trace, a via transition, a connector launch, or a component pad can change the impedance and loss of a route. Coupling between nearby lines also becomes easier to excite. Around 30 GHz, these effects cannot be left as final layout details, because they affect the electrical behavior of the RF and LO paths.

Commercial mmWave ICs reduce the amount of discrete RF circuitry, but they do not remove the PCB problem. The surrounding layout still needs controlled-impedance routing, close return paths, short transitions, and dense grounding around sensitive structures. For that reason, the most important high-frequency interconnects in this thesis are checked with full-wave electromagnetic simulation rather than only with transmission-line estimates.

1.2 Objective of the thesis

The objective of this diploma thesis is to design and simulate a dual-polarization mmWave transceiver PCB platform for operation from 28 GHz to 31 GHz. The platform uses Time Division Duplex (TDD) operation, so transmission and reception share the same RF channel but occur in different time slots.

The design has two polarization branches, vertical and horizontal. Each branch uses one ALTUS28TR mmWave transceiver IC from AKRONIC. A NEXUS28Z PLL/synthesizer, also from AKRONIC, provides the onboard LO source. The shared LO path gives the two branches a common frequency reference. The final amplitude and phase balance still depend on the splitter, routing, package transitions, and board-level

parasitics.

The selected range is not one continuous 5G NR operating band. It overlaps parts of FR2 bands such as n257 and n261 and extends toward spectrum used for fixed-service and point-to-point links. In this thesis, it is treated as a mmWave region of interest for telecommunication hardware rather than as a single 5G band.

Two Red Pitaya STEMlab 125-14 boards are used for external baseband access. They generate and acquire the analog I/Q signals for the TX and RX paths. Since the Red Pitaya interface is single-ended and the ALTUS28TR baseband ports are differential, the PCB includes analog conditioning stages between the two.

The work stops before fabrication. It covers the PCB design, the supporting circuit simulations, and selected EM checks of the routed high-frequency structures. The work produces a layout-complete PCB design package for a future hardware build, bring-up, and measurement campaign.

1.3 Scope of work

The scope is limited to board-level design and pre-fabrication verification. The work includes system partitioning, selection of the main ICs, interface circuits, power distribution, digital control, and PCB layout. The RF, LO, baseband, control, and supply sections are treated as parts of one mixed-signal board.

The commercial ICs are checked against the required operating range, signal levels, supply rails, and control interfaces. The PCB is developed in KiCad. The RF and LO paths are routed as controlled-impedance GCPW structures on the selected stackup. LTspice is used for the lower-frequency circuits, mainly the PDN and the analog baseband interfaces. RFPro is used for selected mmWave interconnects where the routed geometry affects the result.

The thesis also includes a Python host-side control framework for the Red Pitaya boards. The software defines the basic TDD sequence, sets the timing parameters, and controls the RF-switch GPIO states. It is intended for bring-up and functional testing. It is not a deterministic hardware timing controller.

1.4 Methodology

The design process starts from the system architecture and then moves to circuit design and PCB implementation. Component selection is based on frequency coverage, function, supply requirements, interface needs, and availability. The baseband and digital-control interfaces are built around the electrical limits of the Red Pitaya boards and the ALTUS28TR transceivers.

The layout is handled as a mixed-signal mmWave PCB. The stackup, RF and LO routing, grounding, power distribution, and baseband placement are not independent choices. A change in one area can affect return paths, coupling, routing length, or available space in another. This is especially important near the RF front-end, the

LO distribution path, and the fully differential amplifier stages.

The simulation work follows the physical behavior of each part of the board. LTspice is used where lumped circuit models are appropriate, such as supply rails, ripple filtering, and analog baseband interfaces. RFPro is used where the local PCB geometry needs to be included. The extracted structures include the retained LO layer transition, the RF/TDD interconnect around the SPDT switch, and selected GCPW transitions on the final stackup.

The Python framework defines the TDD time slots, inserts guard intervals, and changes the RF-switch GPIO states during the host-side frame sequence. This gives a practical control flow for initial testing. The timing remains software-based, so tighter TDD synchronization would require FPGA or hardware-timed control in a later implementation.

1.5 Contribution of the thesis

The thesis produces a routed PCB design package for a dual-polarization TDD mmWave transceiver platform operating from 28 GHz to 31 GHz. The implementation places the two ALTUS28TR branches, the NEXUS28Z LO source, the Red Pitaya baseband interface, the FDA stages, the multirail PDN, and the digital-control interface on the same PCB.

At board level, the work covers the hybrid stackup, controlled-impedance RF and LO routing, power delivery, Red Pitaya-to-ALTUS28TR baseband conditioning, and selected EM checks of routed high-frequency structures.

The design is not a measured prototype. It is a layout-complete PCB design checked through circuit simulation and selected EM extraction. These checks focus on the parts of the board that carry the main pre-fabrication risk, especially the PDN, the baseband interface, the LO routing, and the RF/TDD switching region.

1.6 Structure of the thesis

Chapter 2 gives the theoretical background used in the design, including mmWave communication, TDD operation, transceiver architecture, PLL operation, high-frequency PCB behavior, and fully differential amplifiers. Chapter 3 presents the system architecture and the main devices used in the platform.

Chapter 4 describes the PCB implementation and verification work. It covers the PDN, baseband interfaces, RF and LO sections, digital control, stackup, layout strategy, software control framework, LTspice simulations, and RFPro extractions. Chapter 5 summarizes the conclusions, limitations, and future work.

Appendix A documents the host-side software framework. Appendix B describes the TDD frame sequencing. Appendix C presents supplementary simulated baseband results. Appendix D summarizes the KiCad-based PCB design workflow.

2 Theoretical Background

2.1 5G and mmWave Communication Systems

The transition to fifth-generation (5G) networks was driven by the need for higher data rates and larger network capacity. To achieve this, telecom infrastructure had to reach into the higher frequency bands of the electromagnetic spectrum where wider bandwidths were available. The 3GPP New Radio (NR) standard defines the Frequency Range 2 (FR2) as a frequency band spanning from 24.25 GHz to 52.6 GHz, which contains the operating range targeted by this work [3], [4].

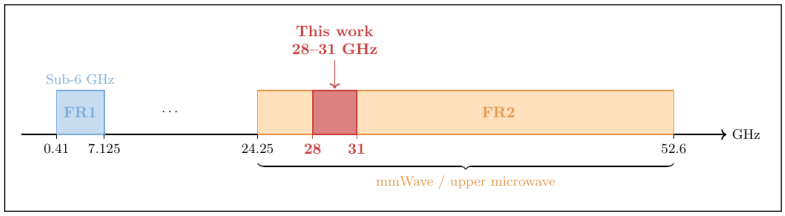


Fig. 2. 3GPP NR frequency ranges. The 28 GHz to 31 GHz band addressed in this work is highlighted within FR2.

The 28 GHz to 31 GHz region is attractive for high-capacity wireless links and 5G-related applications. Frequencies in this range provide access to wider bandwidths, but they also impose stricter design constraints at both the system and printed circuit board (PCB) levels. The main characteristics relevant to this work are:

- wider available bandwidths than sub-6 GHz systems, supporting higher throughput,
- higher propagation losses, which constrain link budgets,
- pronounced sensitivity to parasitics, impedance discontinuities, and layout-dependent effects,
- the need for controlled RF routing and electromagnetic verification.

The 28 GHz to 31 GHz target overlaps with the 3GPP NR mmWave bands n257 (26.5 GHz to 29.5 GHz) and n261 (27.5 GHz to 28.35 GHz). The upper portion of the targeted range, between 29.5 GHz and 31 GHz, lies outside the currently defined 3GPP NR mmWave bands but is allocated in several regions for fixed-satellite-service uplinks and for fixed wireless access under regulatory frameworks such as the FCC Upper Microwave Flexible Use Service (UMFUS) [5], [6] and the Ofcom 27.5 GHz to 30 GHz allocation [1].

2.2 Time Division Duplex Operation and Timing Considerations

A wireless communication link needs a mechanism for the transmitter and the receiver to share the available spectrum. A common method for doing so is to

use Frequency Division Duplex (FDD), where TX and RX simultaneously operate on two different frequency channels. An alternative to FDD is Time Division Duplex (TDD), where there is only one frequency channel and the system switches between transmitting and receiving at different times.

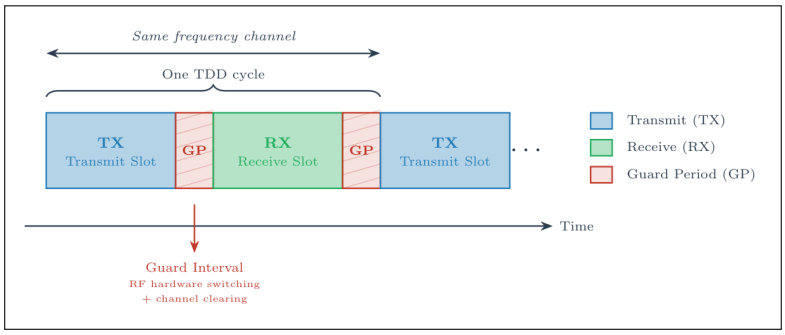


Fig. 3. Basic TDD operation. TX and RX alternate on the same frequency channel, separated by guard intervals that allow the RF hardware to switch state.

The 3GPP NR mmWave bands relevant to the 28 GHz to 31 GHz target (n257 and n261) are both defined for TDD operation [7]. Designing a TDD link therefore requires precise control of the TX-to-RX transition timing. If the receiver turns on too early, it can be desensitized or overloaded by residual energy remaining from the previous transmit time slot. Similarly, if the receiver turns on too late, a portion of the received signal will be lost. Guard intervals are therefore inserted between time slots to allow the RF hardware to switch state and to allow residual signals in the channel to decay.

2.3 Microwave Transceiver System Architecture

The primary function of a microwave transceiver is to perform frequency translation between baseband and RF, for both transmit and receive modes. In the transmit path, the baseband data is up-converted to the appropriate RF band and amplified before being delivered to the antenna. When receiving, the incoming RF signal is amplified and then down-converted to baseband and processed by the Red Pitaya platform.

Both conversions use a mixer driven by a local oscillator (LO), which sets the frequency relationship between baseband and RF.

Two main architectures are used: superheterodyne and direct conversion (Zero-IF). A superheterodyne receiver translates the RF signal to one or more intermediate frequencies before reaching baseband, which adds image-reject filters and IF amplifiers along the chain. A Zero-IF receiver collapses these stages into a single down-conversion step from RF directly to baseband, removing the IF filtering at the cost of new requirements on DC offset and I/Q imbalance. The lower component count and tighter integration make Zero-IF well suited to modern mmWave transceiver ICs, including the device used here.

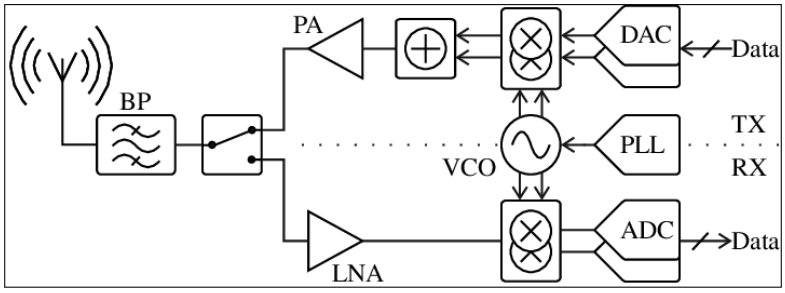


Fig. 4. Simplified block diagram of a Zero-IF transceiver [8].

On the receive side, the first active block is usually the low-noise amplifier (LNA). Because it is placed at the input of the receive chain, its noise figure and gain have a strong influence on the cascaded noise figure of the receiver. The LNA must therefore provide sufficient gain while adding as little noise as possible. After the LNA, the RF signal is split into in-phase and quadrature branches and multiplied by the LO and a $90°$-shifted version of the LO. The two outputs together form a complex baseband signal centered at

$$f_{BB} = f_{RF} - f_{LO}, \tag{1}$$

which preserves the sign of the offset and therefore distinguishes positive from negative baseband frequencies. This signed representation is essential for I/Q demodulation, since otherwise spectra above and below the LO would fold on top of each other.

On the transmit side, the baseband signal is up-converted by the mixer, and then it is amplified by the power amplifier (PA) to the level required for transmission through the antenna. The key issue here is PA linearity: the amplifier must produce the required output power without distorting the signal, particularly since modern 5G waveforms exhibit high peak-to-average power ratios. The PA must therefore operate in its linear region. The detailed PA design lies inside the transceiver IC and is not addressed in this thesis.

Both sides of the transceiver depend on the quality of the LO signal. Any phase noise or frequency instability on the LO transfers directly onto the up- or down-converted signal and degrades the modulation accuracy. At mmWave LO frequencies, maintaining this spectral purity becomes more demanding than at lower frequencies.

2.4 Local Oscillator Generation and PLL Fundamentals

The LO is the reference that drives every frequency conversion in the transceiver, so it must be stable and accurate. In a 28 GHz to 31 GHz system, even small frequency errors or short-term variations in the LO degrade both transmitted and received signal quality. A simple free-running oscillator typically does not provide sufficient reliability for these systems.

A common method for generating a stable LO is the phase-locked loop (PLL), a feedback system that uses a tunable oscillator to track a stable reference clock,

combining the tunability of the oscillator with the stability of the reference.

The PLL has four major components:

- the phase-frequency detector (PFD),
- the charge pump and loop filter,
- the voltage-controlled oscillator (VCO),
- the frequency divider.

The PFD measures the phase difference between a stable reference clock and the output of the VCO after it has been divided by the frequency divider, and generates an error signal based on the measured phase difference.

The charge pump and loop filter convert the error into a smooth control voltage, which controls the VCO until the output of the VCO has locked to the reference.

The feedback divider allows the VCO to oscillate at a higher frequency than the reference, so a low-frequency crystal can be used to generate a stable mmWave LO.

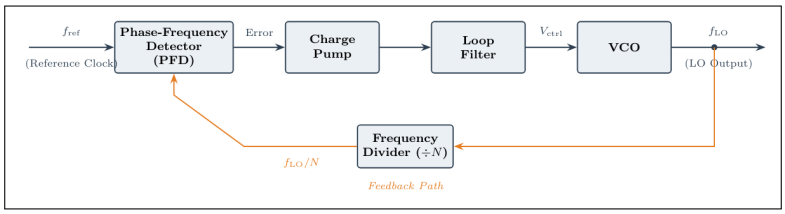


Fig. 5. Block diagram of a PLL.

A critical PLL parameter for LO use is phase noise, which describes short-term random fluctuations in the carrier phase. These fluctuations spread the signal in frequency, disturb the constellation points, and reduce link robustness, particularly at higher modulation orders.

2.5 High-Frequency PCB Design at mmWave Frequencies

The previous subsections treated the transceiver, LO, and TDD timing as system-level concerns. At 28 GHz to 31 GHz, however, the PCB itself becomes part of the RF design: trace geometry, substrate properties, and layer transitions begin to set the matching, loss, and isolation that the rest of the system relies on. The remainder of this chapter develops the transmission-line theory, the S-parameter description used to characterise high-frequency interconnects, and the stackup and routing choices needed to make those interconnects predictable in this band.

2.5.1 Transmission Line Fundamentals

At low frequencies, a short PCB trace behaves as a lumped connection because the physical length of the structure is negligible compared to the signal wavelength. Under these conditions, the voltage and current on the structure are assumed constant. As a result, the interconnect can be represented using lumped resistance, inductance, and capacitance. However, this model is applicable only for interconnects that have very short electrical lengths, which satisfy

$$l < \frac{\lambda}{10},$$

where l is the physical length of the interconnect and λ is the wavelength of the signal.

At microwave frequencies and above, this model breaks down: a structure of just a few millimeters becomes electrically significant compared with the guided wavelength, and the voltage and current along it now vary in both amplitude and phase.

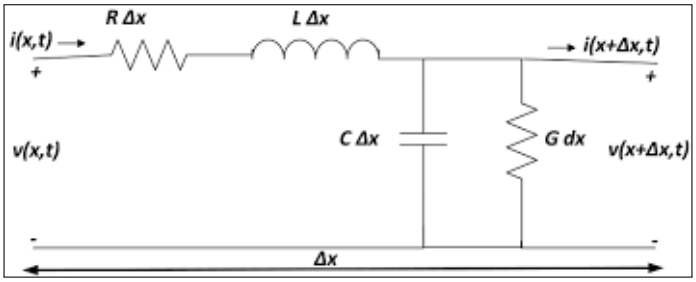


Fig. 6. Distributed RLGC model of a slice of a transmission-line segment of length Δx [9].

The standard model for transmission-line behavior is the distributed RLGC equivalent circuit, in which the line is divided into infinitesimal segments of length Δx. Each segment captures four physical effects: conductor resistance $R'\Delta x$, series inductance $L'\Delta x$ from the magnetic field around the trace, shunt conductance $G'\Delta x$ from dielectric leakage, and shunt capacitance $C'\Delta x$ from charge storage between the conductors. The primed quantities are the per-unit-length parameters of the line.

The characteristic impedance of the line is determined by these distributed parameters. In the general case,

$$Z_0 = \sqrt{\frac{R' + j\omega L'}{G' + j\omega C'}},$$

while for practical low-loss PCB transmission lines this is often approximated by

$$Z_0 \approx \sqrt{\frac{L'}{C'}}.$$

In practice, this balance between the inductive and capacitive properties is fixed by the trace width, the dielectric thickness and constant, and the configuration of the return path.

The electrical length of a line is measured relative to the guided wavelength,

$$\lambda_g = \frac{c}{f\sqrt{\varepsilon_{\text{eff}}}},$$

where c is the speed of light in vacuum, f is the signal frequency, and ε_{eff} is the effective dielectric constant. At high frequencies, a line may become electrically long even when its physical length is only a few millimeters. For example, at 30 GHz on a substrate with $\varepsilon_{\text{eff}} \approx 3.5$, the guided wavelength is about

$$\lambda_g \approx \frac{3 \times 10^8}{30 \times 10^9\sqrt{3.5}} \approx 5.3\,\text{mm}.$$

This means that a trace of only 0.53 mm already approaches the $\lambda_g/10$ limit.

The impedance seen at the input of the line depends on its electrical length. For a lossless transmission line of length l, terminated by a load Z_L, the input impedance is

$$Z_{\text{in}} = Z_0\frac{Z_L + jZ_0\tan(\beta l)}{Z_0 + jZ_L\tan(\beta l)},$$

where

$$\beta = \frac{2\pi}{\lambda_g}.$$

A special case occurs when the line length is $\lambda_g/4$, for which the line acts as an impedance transformer:

$$Z_{\text{in}} = \frac{Z_0^2}{Z_L}.$$

For a line of length $\lambda_g/2$, the input impedance repeats the load impedance:

$$Z_{\text{in}} = Z_L.$$

These relations show why PCB trace lengths cannot be treated as arbitrary interconnects at mmWave frequencies. Routing decisions directly set the impedance presented to each component.

If the load differs from the characteristic impedance, part of the signal is reflected. The reflection coefficient is

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}.$$

When $Z_L = Z_0$, the line is matched and no reflection occurs. In practice, RF systems commonly use 50 Ω transmission lines, since this value provides a widely adopted compromise between attenuation, power handling, and compatibility with RF instruments, connectors, and components.

In PCB implementations, transmission-line behavior is realized through structures such as microstrip lines and grounded coplanar waveguides (GCPW). A microstrip consists of a conductive trace routed above a reference ground plane, with a dielectric substrate in between. A GCPW uses a signal trace with adjacent ground conductors on the same layer, usually together with an underlying reference plane. These structures are discussed in the following subsection, since their geometry directly affects the characteristic impedance and the distribution of the electromagnetic field.

2.5.2 S-Parameters and Reflection Behavior

At microwave and mmWave frequencies, it is often more convenient to describe circuits in terms of incident and reflected waves instead of voltages and currents. Signals at these frequencies propagate as waves along transmission lines, and reflections due to impedance mismatch become important. This approach treats the device as a black box characterized by its input and output behavior.

For a two-port network, a_1 and a_2 denote the waves incident on each port, while b_1 and b_2 denote the waves emerging from each port. The relationship between them is captured by the S-parameter matrix:

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix}.$$

Each S-parameter represents the reflection or transmission coefficient associated with each port and frequency. The two S-parameters of interest here are S_{11} and S_{21}. The parameter S_{11} represents the reflection coefficient at the input port and therefore quantifies the input match. A low value of S_{11} means little of the incident wave is reflected back to the source. The parameter S_{21} represents the transmission coefficient of the forward traveling wave from port 1 to port 2, and quantifies how much signal passes through the structure.

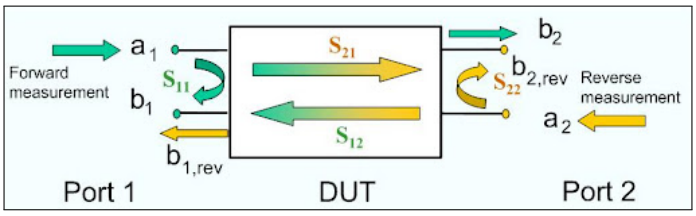


Fig. 7. Two-port interpretation of the main S-parameters [10].

Typically, these S-parameters are presented in log format. The return loss (RL) is defined as

$$\text{RL} = -20 \log_{10} |S_{11}|,$$

and the insertion loss (IL) is defined as

$$\text{IL} = -20 \log_{10} |S_{21}|.$$

The RL indicates how well matched the input is to the system; the IL measures how much of the signal is lost during the signal's travel through the system. Return loss above 10 dB is generally considered acceptable, with values between 15 dB and 20 dB regarded as good.

2.5.3 PCB Stackup and Controlled-Impedance Design

The characteristic impedance of an RF trace is determined by both the conductor geometry and the layer stackup. Dielectric thickness, copper thickness, dielectric permittivity, trace width, and reference-plane location all affect the impedance and

the electromagnetic field distribution. For this reason, the PCB stackup must be fixed before the controlled-impedance routing dimensions are selected.

Dielectric materials are commonly characterized by their relative permittivity ε_r. However, since the electromagnetic field is often distributed throughout the dielectric and not confined within it, the transmission line is more accurately represented by an effective dielectric constant ε_{eff}, which is related to the relative permittivity of the dielectric material by the equation:

$$1 < \varepsilon_{\text{eff}} < \varepsilon_r.$$

This effective value of the dielectric constant determines the speed of propagation of the signal through the transmission line, the guided wavelength, and therefore the electrical length of the trace.

The characteristic impedance of a microstrip trace is primarily a function of the trace width (w), the dielectric height (h), and the effective dielectric constant of the material. The ratio of the trace width to the dielectric height (w/h) directly influences the characteristic impedance. Increasing the w/h ratio reduces Z_0, while decreasing it raises Z_0.

For a grounded coplanar waveguide (GCPW), the characteristic impedance is a function of the trace width (w), the dielectric height (h), and the lateral spacing (s) between the center trace and the two adjacent ground traces. A smaller gap between the center trace and the adjacent ground traces results in increased electric-field coupling to the ground traces, which lowers the characteristic impedance.

Controlled-impedance design involves maintaining a specified characteristic impedance (Z_0), usually 50 Ω, over the intended frequency range of operation. That requires the stackup and routing geometry to be defined simultaneously, not independently. That is, the choice of ε_r, the dielectric thickness, the conductor width, and the conductor spacing must all be decided together.

Reference-path continuity is also critical. A controlled-impedance trace relies on a continuous reference path provided by one or more reference planes or ground conductors. If the reference path is broken, the impedance at that point will change and parasitic effects may be introduced. This is particularly important when designing mmWave layouts, as even minor interruptions in the reference path can reduce the matching performance and increase the loss associated with the trace.

The specific stackup configuration, substrate type, and controlled impedance dimensions is documented in Chapter 4, where the application of these principles to the 28 GHz to 31 GHz implementation is demonstrated.

2.5.4 Microstrip and Grounded Coplanar Waveguide (GCPW) Structures

Two common controlled-impedance structures are the microstrip line and the grounded coplanar waveguide (GCPW). Both are widely used in microwave and mmWave

PCBs because they can be designed to provide a target characteristic impedance and integrate naturally into multilayer stackups.

A microstrip line consists of a signal trace routed above a reference ground plane, with a dielectric layer between them. Because the electromagnetic field is split between the dielectric below the trace and the air above it, the effective dielectric constant of the line is lower than the bulk relative permittivity of the substrate.

A grounded coplanar waveguide (GCPW) has a signal trace flanked by ground regions on the same layer, typically with an additional reference plane underneath. Because the field is more tightly confined between the signal trace and the surrounding ground regions, the impedance depends both on the trace width and on the lateral gap to the side grounds.

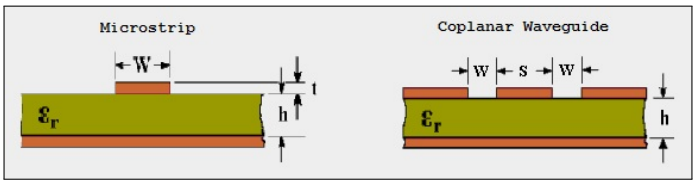


Fig. 8. Cross-sectional comparison of microstrip and grounded coplanar waveguide PCB transmission-line structures.

Microstrip lines are generally simpler to manufacture and are therefore common in PCB layouts. GCPW is preferred when stronger field confinement is needed, for example for better isolation between adjacent traces or a more clearly defined return path. These properties make GCPW useful in dense RF and mmWave layouts where unwanted coupling is a concern.

2.5.5 Electromagnetic Effects and the Need for EM Simulation

At microwave and mmWave frequencies, a real PCB layout no longer follows the assumptions of an ideal transmission line. Small geometric features that are negligible at lower frequencies can affect matching, insertion loss, and signal quality. Examples include pads, vias, bends, width changes, and ground discontinuities.

The most important layout-dependent effects include:

- impedance discontinuities caused by pads, bends, vias, and trace-width changes,
- parasitic capacitances and inductances introduced by component pads, interconnect transitions, and return-path interruptions,
- conductor losses associated with finite metal conductivity and skin-effect behavior,
- dielectric losses inside the substrate material,
- unwanted coupling and partial radiation from closely spaced high-frequency structures.

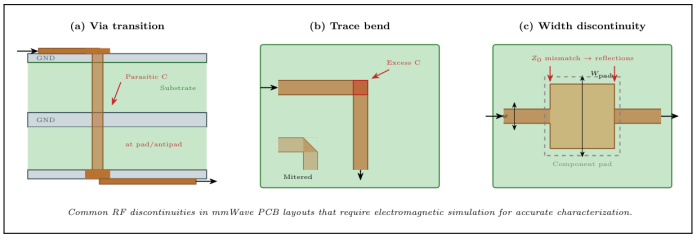


Fig. 9. Typical PCB discontinuities in mmWave routing that can introduce parasitic effects and additional reflections.

The extent of these effects depends on the geometric configuration of the PCB and the field distribution around it. Even when a circuit-level model captures the nominal characteristic impedance of a trace, it cannot represent the discontinuities along the layout, which add reflections and loss.

Electromagnetic simulation is used to verify mmWave PCB structures before fabrication. Full-wave EM tools predict the reflection and transmission behavior of an actual layout, capture the contribution of bends, vias, and pads, and reveal mismatch or excess loss that lumped models cannot. The high-frequency interconnects of this work are verified by EM simulation in Chapter 4.

2.6 Amplifiers and Fully Differential Amplifiers (FDAs)

Electronic amplifiers are circuits that increase the amplitude of a signal or condition it so that it can be processed at the next stage. In the simplest voltage-mode case, the gain of an amplifier is given by

$$A_v = \frac{V_{\text{out}}}{V_{\text{in}}},$$

where V_{in} is the input voltage and V_{out} is the output voltage. In practical analog interfaces, amplifiers are also used for buffering, impedance transformation, and level adjustment between successive stages.

Single-ended circuits measure the signal with respect to ground, while differential circuits measure it as the potential difference between two nodes:

$$V_{\text{id}} = V_{i+} - V_{i-}.$$

The corresponding differential output voltage is

$$V_{\text{od}} = V_{o+} - V_{o-}.$$

In addition to this differential component, the signal also carries a common-mode component, defined as the average of the two node voltages:

$$V_{\text{icm}} = \frac{V_{i+} + V_{i-}}{2}, \quad V_{\text{ocm}} = \frac{V_{o+} + V_{o-}}{2}.$$

A fully differential amplifier accepts a differential input and produces a differential output, with the output common-mode voltage held at a defined level. The differential gain is defined as

$$A_{vd} = \frac{V_{\text{od}}}{V_{\text{id}}}.$$

In practice, most FDA implementations use an external feedback network to establish the closed-loop gain. For the standard resistive feedback configuration, the differential gain is given by:

$$|A_{vd}| \approx \frac{R_F}{R_G},$$

while the exact sign depends on the selected topology and signal polarity convention.

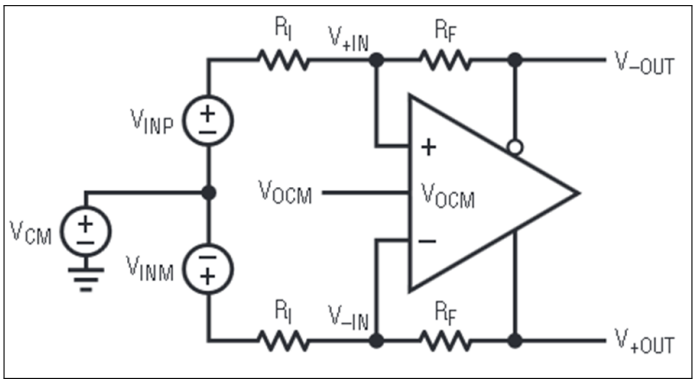


Fig. 10. Typical fully differential amplifier (FDA) configuration with symmetric feedback and output common-mode control [11].

The output common-mode control of an FDA is useful when the FDA drives a high-speed ADC, mixer, or another analog block that requires both a controlled signal swing and a specific common-mode voltage at its input.

Compared with single-ended amplifiers, FDAs fit into differential signal chains and help reject common-mode interference. They integrate naturally with differential downstream blocks such as ADCs and mixers, and are widely used as differential gain stages, single-ended-to-differential converters, and level-shifting interfaces between mismatched signal formats. Here, they serve this role between the Red Pitaya analog ports and the ALTUS28TR baseband ports.

3 System Architecture

3.1 Overall System Architecture

The platform consists of two transceiver chains operating in the 28 GHz to 31 GHz band: one for vertical polarization (V-pol) and one for horizontal polarization (H-pol). The two chains share a common LO source, which keeps the polarization branches frequency-coherent and avoids independent PLL frequency errors between them. The architecture is built from commercially available ICs and modules and is developed as a PCB-level mmWave transceiver design.

Each polarization chain contains identical functional blocks. The main RF IC in each chain is the ALTUS28TR transceiver IC [12] implemented in a SiGe BiCMOS process that uses a Zero-IF, or direct-conversion, architecture. Each ALTUS28TR includes a power amplifier (PA), a low-noise amplifier (LNA), frequency conversion mixers with LO leakage compensation, and fully differential I/Q baseband low-pass filters with programmable cutoff frequencies. The ALTUS28TR requires an input LO signal in the 8 GHz to 11 GHz range and then multiplies this signal internally to produce the millimeter-wave carrier signal. In this platform, the onboard PLL covers part of the target RF range after the internal LO multiplication of the ALTUS28TR. An external LO input is also provided to support extended frequency coverage. An SPDT switch at the RF front end connects the antenna-side RF port to either the TX path or the RX path according to the TDD state. The external RF interface of each polarization chain is realized using a 2.92 mm RF connector.

One NEXUS28Z PLL synthesizer chip is used to supply the LO signal for both ALTUS28TR chips [13]. The NEXUS28Z is a CMOS fractional-N PLL with internal VCO and loop filter that produces an LO signal in the range of 7.9 GHz to 9.8 GHz. It operates using a 40 MHz crystal oscillator reference clock and is configured through SPI.

Each polarization chain is also connected to a separate Red Pitaya STEMlab 125-14 board, which handles the external baseband generation and acquisition interface. One Red Pitaya board supplies the two DAC and two ADC channels required for one I/Q polarization branch, so two boards are used for the dual-polarization platform.

The baseband interface between the Red Pitaya and ALTUS28TR chips is realized using fully differential amplifiers (FDAs). For transmission, the I and Q signals from the Red Pitaya are conditioned by two LTC6409 FDA ICs before being sent to the transmitter input of the ALTUS28TR. For reception, the differential I and Q signals from the ALTUS28TR receiver outputs are conditioned by two ADA4927 FDA ICs before being delivered to the ADC inputs of the Red Pitaya.

Each polarization chain uses two LTC6409 FDAs [14] for transmission and two ADA4927 FDAs [15] for reception. Across the platform, this gives four LTC6409 devices and four ADA4927 devices. These FDA stages provide the required interface adaptation between the single-ended Red Pitaya ports and the differential baseband ports of the transceivers. Their role is to condition the baseband signal between the single-ended Red Pitaya ports and the differential ALTUS28TR ports.

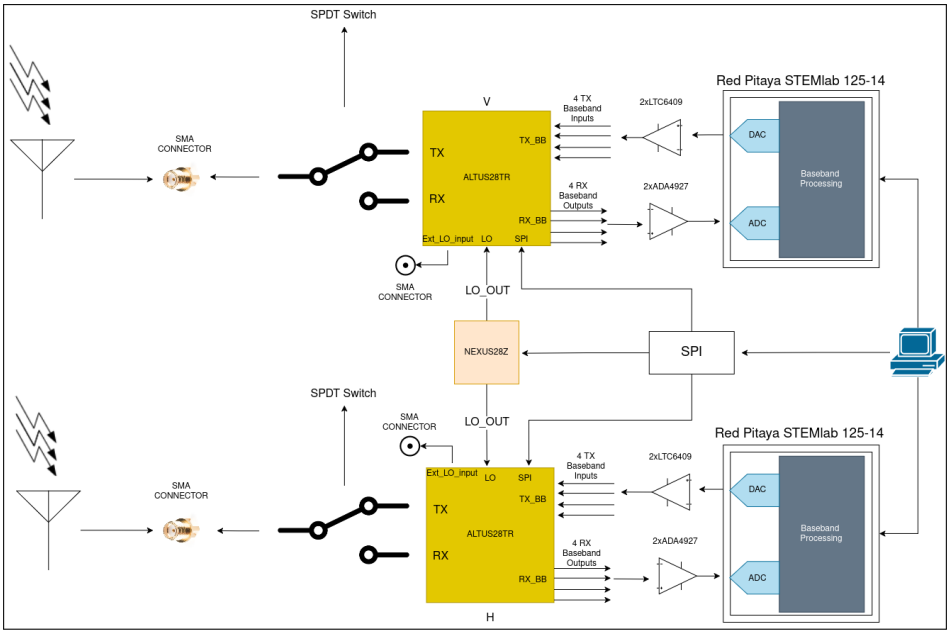


Fig. 11. Overall system block diagram of the dual-polarization 28 GHz to 31 GHz TDD transceiver platform.

The ALTUS28TR transceivers and the NEXUS28Z synthesizer are configured from the host PC through SPI. The Red Pitaya boards are connected to the host PC through Ethernet. The RF signal paths in the 28 GHz to 31 GHz range are designed as 50 Ω controlled-impedance interconnects.

3.2 Transceiver IC: ALTUS28TR

Each polarization channel contains the ALTUS28TR, which acts as the main RF transceiver in that branch. The ALTUS28TR is a highly integrated Zero-IF millimeter-wave transceiver implemented in SiGe BiCMOS technology. The transceiver covers all key RF functions, as well as an internal LO processing path and differential I/Q interface with the baseband block.

The architecture of the ALTUS28TR consists of three main parts: the transmit path, the receive path, and the LO path. The schematic view of these subsystems is shown in Figure 12. The figure highlights the main signal paths between the differential I/Q baseband interface and the external RF ports.

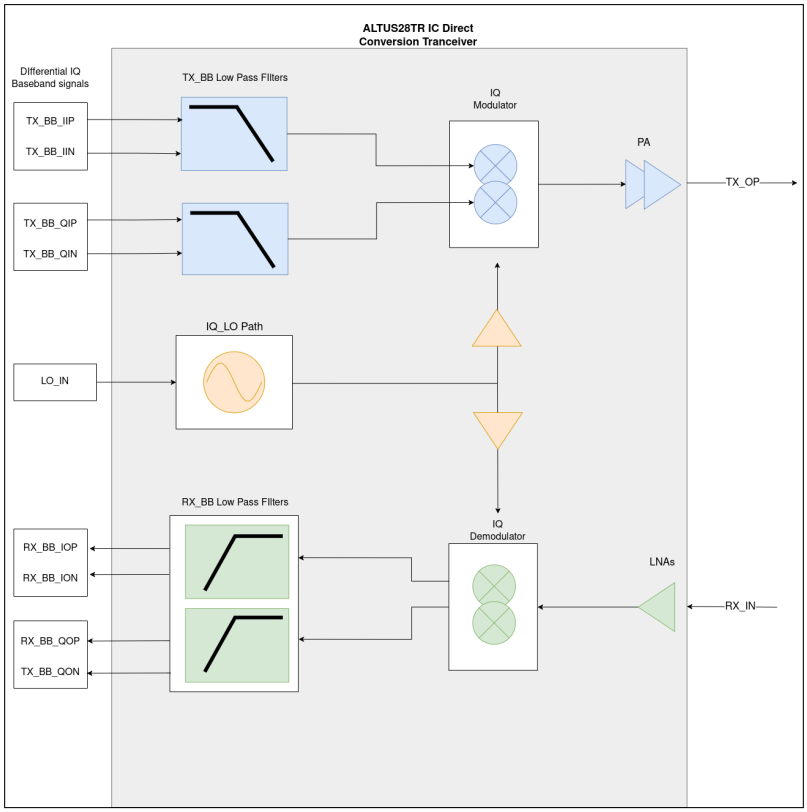


Fig. 12. Schematic view of the main functional modules in the ALTUS28TR with focus on the signal processing paths between the differential I/Q interface and the RF ports.

The internal signal flow is clearer when the transmit, receive, and LO paths are described separately: the transmit chain starts at the differential I/Q baseband input ports and the receive chain ends with the output of the differential I/Q baseband ports.

3.2.1 Transmit Path

During transmit operation, the I/Q input signal arrives on the differential baseband pins of the ALTUS28TR. The input signals then pass through the internal TX low-pass filters, which define the actual baseband bandwidth for the upconversion process. The transmit LPF response is programmable through the digital configuration interface, allowing the baseband bandwidth to be adjusted for the selected operating mode.

After low-pass filtering, the I and Q signals are applied to the internal I/Q modulator. Within the modulator stage, the input signals undergo mixing with the internally distributed LO signal and translation to the RF band directly from the baseband. Because the transceiver uses a Zero-IF architecture, no intermediate-frequency stage is used during this translation process.

The transmitter circuitry also includes LO leakage compensation (LLC) circuitry related to the I/Q modulator stage. LLC reduces LO feedthrough at the output port,

which is a common impairment in direct-conversion transmitters. After modulation and compensation, the signal is amplified by the internal power amplifier (PA), after which the signal appears at the RF output port.

3.2.2 Receive Path

On reception, the RF signal enters through the receive RF input and is amplified by the integrated low-noise amplifier (LNA). As the first active block in the receive chain, the LNA supplies front-end gain and has a strong influence on the receiver noise figure and sensitivity.

The amplified RF signal next proceeds to the internal I/Q demodulator, where it is directly converted to baseband using the internal LO signal path. As in the transmit path, the direct-conversion topology avoids an intermediate-frequency stage, but it is sensitive to impairments such as DC offset caused by LO self-mixing.

To reduce this effect, the receive path includes the DC offset cancellation circuitry at the receive low-pass filter stage input. Finally, after DC-offset compensation, the I/Q receive signals pass through the internal low-pass filter stage of the receiver, which sets the receive bandwidth through the digital configuration interface.

3.2.3 LO Path

The LO path forms the third principal internal subsystem of the ALTUS28TR. The device accepts an external LO signal in the approximate range of 8 GHz to 11 GHz through the `LO_IN` pin. This signal is further multiplied internally, and then used for the transmit and receive conversion stages.

Internal frequency multiplication is performed by a frequency tripler, followed by a limiting amplifier and an active splitter. This allows the external LO to remain at a lower frequency than the final RF carrier, with the transceiver generating the higher-frequency LO needed for the 24 GHz to 33 GHz mmWave band internally. The external LO source therefore only needs to cover a lower frequency range. This lower-frequency LO requirement allows one common LO distribution path to serve both polarization branches.

The interaction among the internal transmit, receive, LO, and control blocks is shown in more detail in Fig. 13. This functional diagram highlights the complete internal organization of the device, including the SPI interface, calibration-related paths, the LO multiplication stage, and the internal signal routing between the RF and baseband subsystems.

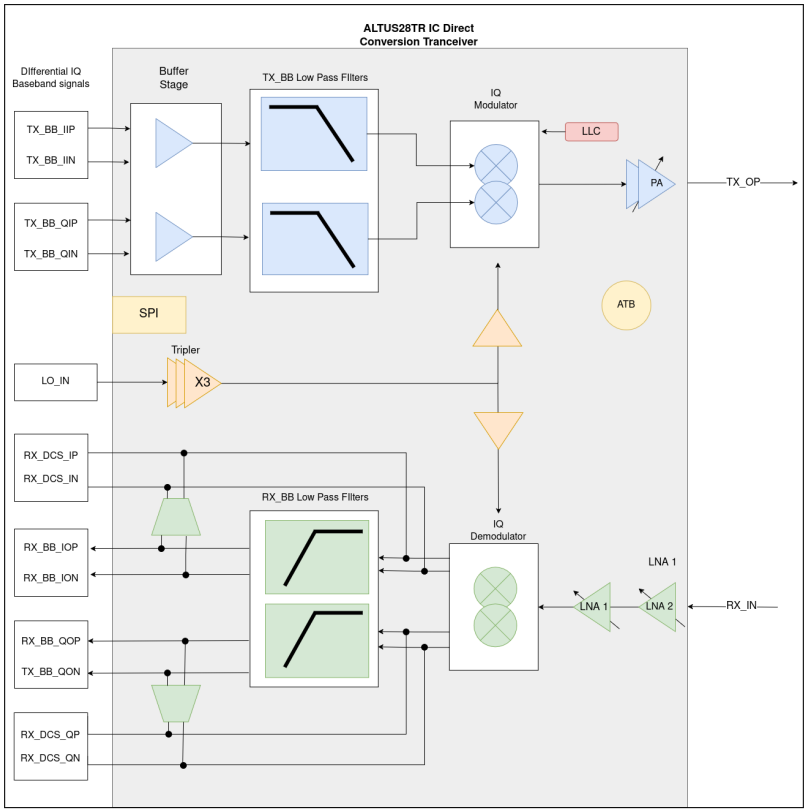


Fig. 13. Functional block diagram of the ALTUS28TR, showing the internal transmit chain, receive chain, LO path, and SPI control interface.

The ALTUS28TR integrates the RF transmit chain, RF receive chain, LO processing path, differential I/Q baseband interface, and digital configuration interface. This integration reduces the amount of RF circuitry required on the PCB, so the board-level design can focus on interconnects, LO distribution, power delivery, and baseband signal conditioning rather than discrete RF gain and frequency-conversion stages, which is why the ALTUS28TR was selected as the RF core of each polarization branch.

3.2.4 Electrical Specifications

Table 1 is based on manufacturer-provided design information available during the thesis work. Only board-level interface parameters required for the present design are included [12].

Tab. 1. Key specifications of the ALTUS28TR transceiver IC.

Parameter	**Value**
Transmitter	
RF frequency range	24 GHz to 33 GHz
Typical TX gain	14 dB
TX gain control range	45 dB (typ.)
5 dB output compression ($OP_{5\text{dB}}$)	12.8 dBm (typ.)
1 dB output compression ($OP_{1\text{dB}}$)	10 dBm (typ.)
Output IP3 (OIP3)	21 dBm (typ.)
Baseband filter cut-off frequency (4 bands)	90 MHz, 180 MHz, 360 MHz and 720 MHz
LO leakage (uncalibrated)	−20 dBc
Image rejection (uncalibrated)	−25 dBc
Receiver	
Max. RX voltage gain	55 dB (ref. 600 Ω diff.)
RX gain control range	90 dB
Noise figure	5.5 dB (typ.)
1 dB input compression point	−10 dBm
Baseband filter cut-off frequency (4 bands)	30 MHz, 60 MHz, 120 MHz and 240 MHz
Input reflection coefficient (S_{11})	−15 dB
LO Path	
Input frequency range	8 GHz to 11 GHz
Input power	−10 dBm to 3 dBm
Internal multiplier	$\times 3$
General	
Supply voltage	2.6 V
DC power (RX / TX)	540 mW / 500 mW
SPI interface	4-wire
Package	2.7 mm $\times$ 2.7 mm CSP, 34-ball

3.3 PLL/Synthesizer: NEXUS28Z

The proposed platform uses a single *NEXUS28Z* PLL synthesizer as the common LO source for the two ALTUS28TR transceivers. This avoids the need for individual frequency generation hardware for each polarization branch. The NEXUS28Z synthesizer generates an LO over a typical range of 7.9 GHz to 9.8 GHz and accepts an external reference in the 40 MHz to 320 MHz range. Here, a 40 MHz crystal oscillator is used as the reference clock. This output range corresponds to an RF carrier coverage of 23.7 GHz to 29.4 GHz after the internal $\times 3$ multiplication of the ALTUS28TR. Thus, the onboard PLL covers the lower part of the target range, while operation above this range relies on the external LO option.

As shown in Fig. 14, the reference signal enters the reference input path and is compared with the divided VCO output by the phase-frequency detector (PFD). The charge pump (CP) and the integrated loop filter (LPF) generate the control signal that tunes the integrated VCO, and the LO driver buffers the synthesized signal to the external `LO` output.

The frequency synthesis function can operate in either integer-N or fractional-N

mode. In the first case, the PLL limits the output frequency to integer multiples of the reference frequency, while fractional-N operation allows for finer frequency resolution. When operated in fractional-N mode, the multi-modulus divider (MMD) is controlled by a MASH sigma-delta modulator, which varies the instantaneous division ratio to achieve a fractional average value while shaping quantization noise out of band.

Loop dynamics are programmable: the bandwidth can be selected to trade settling speed against integrated noise, and additional SPI parameters such as charge-pump current and internal bias settings allow further tuning of lock behavior, stability margin, and LO output level. The VCO has automatic sub-band calibration that selects the correct VCO band for a given output frequency, ensuring reliable operation across process and temperature variation. The PLL is configured using a 4-wire SPI interface, and it also exposes status information such as lock indication and calibration status.

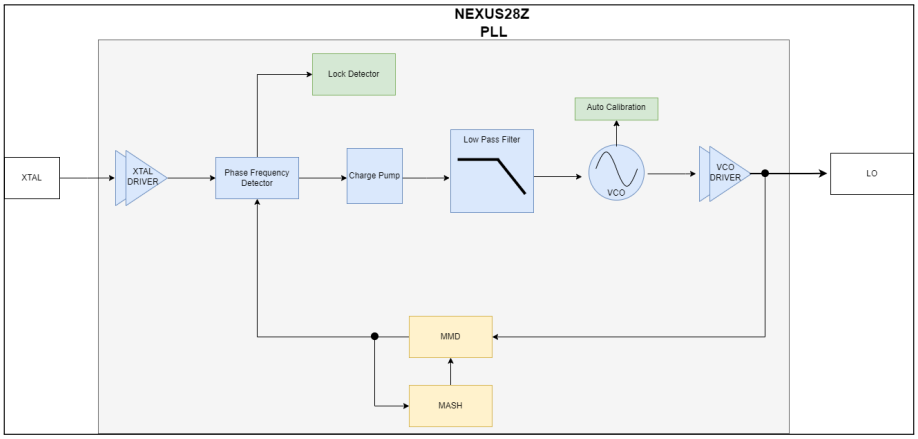


Fig. 14. Functional block diagram of the NEXUS28Z PLL synthesizer, showing the reference path, PFD/CP, loop filter, integrated VCO, fractional-N divider path, LO output driver, and SPI control interface.

3.3.1 Electrical Specifications

The key electrical parameters of the NEXUS28Z [13] are summarized in Table 2.

Tab. 2. Key specifications of the NEXUS28Z PLL synthesizer.

Parameter	**Value**
Output frequency range	7.9 GHz to 9.8 GHz
Reference input range	40 MHz to 320 MHz
Fractional frequency resolution	38 Hz
Divider range (MMD)	16–511
MMD input frequency range	4 GHz to 13.5 GHz
RMS jitter @ 9.7 GHz (integrated, 10 kHz–80 MHz)	0.52° (typ.)
DSB phase noise @ 9.7 GHz, 100 kHz offset	−111 dBc/Hz to −93 dBc/Hz
LO output power	−3 dBm to 3 dBm
LO output impedance	50 Ω
Loop filter bandwidth (programmable)	150 kHz to 1000 kHz
VCO sub-bands	64
SPI interface	4-wire
VCO calibration time	7 µs to 56 µs
Typical lock time (excl. calibration)	4 µs to 32 µs

3.4 Baseband Interface: Red Pitaya STEMlab 125-14

The transceiver platform uses the Red Pitaya *STEMlab 125-14* as the baseband data-conversion platform. In this implementation, it is used to generate and acquire the analog in-phase (I) and quadrature (Q) baseband signals exchanged with the RF front-end. A board was needed that exposes the raw analog baseband, since all RF conversion happens inside the ALTUS28TR. Most SDR platforms hide the baseband behind their own RF front-end and present the user with an IF or RF port. The Red Pitaya does the opposite: its DACs and ADCs are wired straight to SMA connectors, which is exactly what this architecture needs. The external platform therefore operates only at baseband and handles the generation and acquisition of the I/Q signals without additional mixing or filtering stages.

The Red Pitaya is based on the Xilinx Zynq-7010 SoC, which combines a dual-core ARM Cortex-A9 processor with FPGA fabric. The CPU handles configuration, networking, and data flow, while the FPGA manages the high-speed path between the ADC/DAC converters and the Ethernet interface. The board includes 512 MB of DDR3 memory for software execution and data buffering.

The STEMlab 125-14 is equipped with two analog input channels and two analog output channels, each accessible via SMA connectors. On the transmit side, dual 14-bit DACs running at 125 MS/s generate single-ended analog baseband waveforms with a full-scale range of ± 1 V into a 50 Ω load [16]. These outputs are then routed to the transmit FDA stages for signal conditioning. Specifically, these stages provide amplitude scaling, DC-level translation, and single-ended-to-differential conversion before the ALTUS28TR transmitter inputs.

On the receive side, the dual 14-bit ADCs sample at 125 MS/s and digitize the conditioned baseband I/Q signals delivered by the receive FDA stages. These stages provide the required gain and common-mode translation before the Red Pitaya ADC inputs.

3.4.1 Firmware and Software Control

Here, the Red Pitaya STEMlab 125-14 is operated using the *Wideband SDR Transceiver* design developed by Pavel Demin [17]. This design implements an FPGA-based transmit/receive chain for the dual-channel DAC/ADC converters and exposes a control and data interface to a host PC over Ethernet. This makes the Red Pitaya suitable as a configurable baseband I/Q waveform generator and acquisition platform for controlled mmWave transceiver evaluation.

In streaming mode, I/Q samples flow continuously over the Ethernet link in both directions: transmit samples are interpolated and converted to analog by the DACs, while received samples are captured by the ADCs and sent back to the host for processing. Burst mode uses finite-length I/Q blocks within defined time intervals, which is useful when the same waveform and timing sequence must be replayed.

The achievable real-time I/Q bandwidth in streaming operation is determined by the selected FPGA configuration (including interpolation/decimation settings) and the Ethernet transport. In the configurations used here, practical real-time streaming bandwidths are on the order of a few MHz, which is sufficient for the intended baseband control and waveform-generation framework.

3.4.2 Electrical Specifications

The specifications of the Red Pitaya STEMlab 125-14 platform are presented in Table 3.

Tab. 3. Key specifications of the Red Pitaya STEMlab 125-14 baseband platform.

Parameter	Value
SoC	Xilinx Zynq-7010 (ARM + FPGA)
System memory	512 MB DDR3
DAC channels / resolution	2 channels, 14-bit
DAC sample rate	125 MS/s
DAC full-scale output	± 1 V (typ.), 50 Ω load
ADC channels / resolution	2 channels, 14-bit
ADC sample rate	125 MS/s
ADC coupling	DC-coupled inputs
Analog I/O interface	SMA connectors
Host interface	Ethernet

3.5 Signal Conditioning Chain (FDA Interface)

The baseband interface between the Red Pitaya STEMlab 125-14 modules and the ALTUS28TR transceivers is built around fully differential amplifier (FDA) stages. The Red Pitaya exposes single-ended DAC and ADC channels, while the ALTUS28TR expects differential I/Q signals at its baseband ports, and a direct connection between the two is not possible. Beyond the single-ended to differential conversion, the transceiver ports also require specific common-mode levels and bounded volt-

age swings, which are set by the FDA stages across the operating range. In this role, the FDA chain acts as the analog voltage translation layer between the Red Pitaya baseband platform and the mmWave front end. Fig. 15 gives a high-level view of the resulting interface.

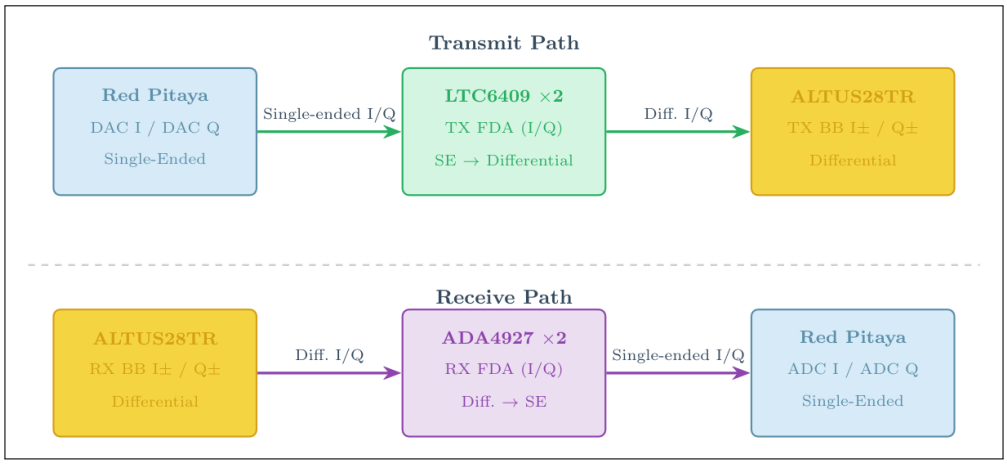


Fig. 15. High-level FDA-based baseband interface between the Red Pitaya and the ALTUS28TR for one polarization branch.

On the transmit side, each polarization branch uses two LTC6409 devices, one for I and one for Q. They receive the baseband single-ended waveforms from the Red Pitaya DACs and provide the differential I/Q signals that drive the ALTUS28TR transmitter inputs. The same stages set the required amplitude scaling and common-mode level so that the transceiver baseband inputs remain within their intended linear range.

On the receive side, each branch uses two ADA4927 devices, again one for I and one for Q. They take the differential baseband outputs of the ALTUS28TR, apply the level scaling and common-mode adaptation required by the Red Pitaya ADCs, and provide enough headroom to handle the receiver gain range without saturating the ADC inputs.

Detailed gain and attenuation values, resistor network calculations, and the resulting common-mode voltages for both the transmit and receive FDA stages are given in Chapter 4.

3.6 TDD Control and Timing

The proposed transceiver platform operates in time-division duplexing (TDD) mode, where transmission and reception take place in separate time intervals while sharing the same frequency resources.

A key characteristic of the selected transceiver architecture is that the ALTUS28TR has separate RF ports for the transmit and receive paths. Since these two ports cannot be connected to the same antenna at the same time, an external switching element is required in each polarization branch.

A MACOM *MASW-011145* absorptive SPDT switch is used at the RF front-end of each polarization branch [18]. The device is specified for operation in the 27 GHz to 31 GHz frequency range and provides typical insertion loss of 1.3 dB, typical isolation of 45 dB, and return loss of approximately 18 dB. These characteristics make the device suitable for the intended TDD front-end of this work.

During transmission, isolation from the receive path is needed to reduce leakage into the LNA input and to protect the receive front end. During reception, the transmit path must be disconnected so that it does not load the receive path or degrade receiver noise performance.

The MASW-011145 is controlled through a single digital control line which is compatible with 1.8 V, 2.5 V, and 3.3 V CMOS logic. The switch state is selected through a GPIO-driven digital control signal provided by either the Red Pitaya or the onboard FTDI. Depending on the logic level applied to the control pin, the switch connects the antenna to the active path according to the current TDD state of the branch. The switch also requires a 3.3 V power supply and supports switching times of 135 ns, which is compatible with the timing requirements of this TDD implementation [18].

At system level, TDD timing requires coordination between the baseband waveform timing, the ALTUS28TR transmit/receive state, and the RF switch state, with the switch connecting the antenna-side port to the active path during each interval. The implementation details are discussed in Chapter 4.

3.6.1 RF Switch Specifications

Key specifications of the MASW-011145 RF switch are summarized in Table 4.

Tab. 4. Key specifications of the MASW-011145 SPDT RF switch.

Parameter	**Value**
Frequency range	27 GHz to 31 GHz
Insertion loss	1.3 dB (typ.)
Isolation	45 dB (typ.)
Return loss	18 dB (typ.)
Logic compatibility	1.8 V, 2.5 V and 3.3 V CMOS
Supply voltage	3.3 V
Switching time	135 ns (typ.)

The system architecture presented in this chapter establishes the functional organization of the platform and the role of its main subsystems. The next chapter examines the circuit-level design and PCB layout of these subsystems.

4 Circuit Design and PCB Implementation

4.1 Power Distribution Network

The power-distribution network (PDN) generates all required supply rails from an external 5 V input. The transceivers, synthesizer, analog signal-conditioning stages, and digital control circuitry operate at different voltage levels and exhibit different noise sensitivities, so the board uses a multirail architecture combining switching converters, linear regulators, and local filtering. The PDN also limits coupling between supply domains and maintains supply quality appropriate for each load domain.

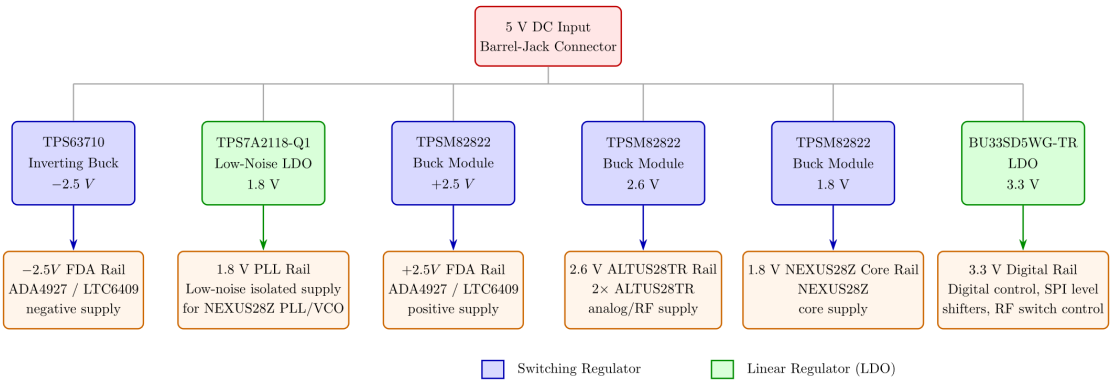


Fig. 16. High-level architecture of the power-distribution network, showing the 5 V input, the regulator stages, and the principal load domains supplied by each rail.

From the common 5 V input, the PDN generates separate rails for the ALTUS28TR transceivers, the NEXUS28Z core and PLL domains, the FDA-based signal-conditioning circuitry, and the digital-control subsystem.

4.1.1 Voltage Rail Requirements

Table 5 summarizes the main supply rails of the platform, together with the corresponding load domains, regulator types, and estimated current demand used during PDN design.

Tab. 5. Voltage rails, loads, regulators, and estimated current demand.

Rail (V)	Load Domain	Regulator	Current (mA)
2.6	ALTUS28TR analog/RF	Buck converter	750
1.8	NEXUS28Z core	Buck converter	200
1.8	NEXUS28Z PLL (isolated)	LDO	150
3.3	Digital control / interface	LDO	20
2.5	FDA positive supply	Buck converter	350
−2.5	FDA negative supply	Inverting buck	400

The partitioning reflects the mixed-signal nature of the platform: mmWave transceivers and synthesizer circuits demand tighter noise specifications than the digital control domain, while the FDA blocks require dual-polarity supplies to cover their analog swing range. The current values were used both for regulator selection and for sizing the PDN layout.

4.1.2 Input Protection and Initial Decoupling

The PDN is fed from an external 5 V source through a barrel-jack connector. Since this node supplies all downstream regulators, a protection and decoupling stage is placed directly at the power entry (Fig. 17).

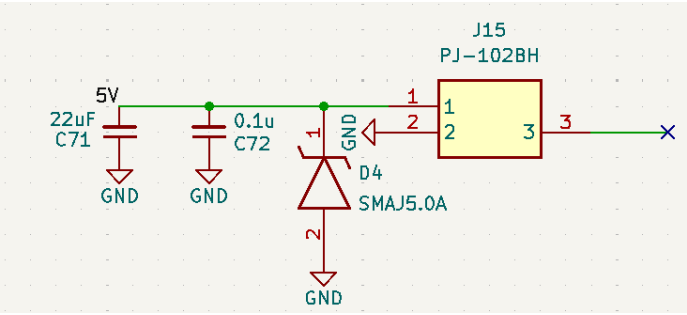


Fig. 17. Input protection and initial decoupling stage of the 5 V power-entry network.

The input line is protected by an SMAJ5.0A TVS diode. Under normal operation the diode is off; during a surge or hot-plug event it clamps the input and shunts the transient to ground, limiting the voltage stress on the downstream regulators.

Initial decoupling is provided by two capacitors at the power entry. The 22 µF capacitor acts as a bulk reservoir during slower input variations and during pulsed current demand from the downstream switching stages. The 100 nF capacitor provides a low-impedance path for higher-frequency components that the bulk capacitor cannot handle alone. Together they keep the input impedance low across a wide frequency range.

4.1.3 Buck Converters for the Main IC Supply Rails

The main positive supply rails are generated with switching converters to limit power dissipation. The 2.6 V ALTUS28TR rail and the 1.8 V NEXUS28Z core rail use TPSM82822 power-module buck converters (Fig. 18), which integrate the inductor within the package and deliver enough current for these loads in a compact footprint [19].

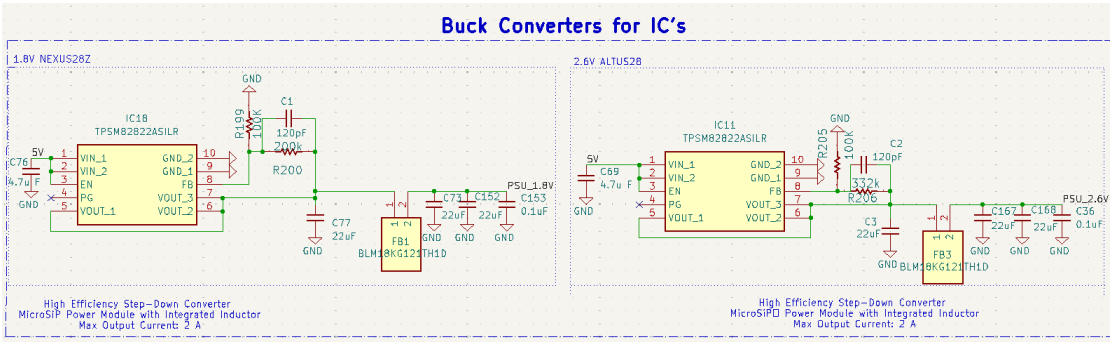


Fig. 18. Buck-converter implementation of the main positive IC supply rails for the NEXUS28Z core and ALTUS28TR analog/RF domains.

Linear regulation from 5 V to these rails would dissipate

$$P_{\mathrm{LDO}} \approx (V_{\mathrm{IN}} - V_{\mathrm{OUT}})\, I_{\mathrm{OUT}} \tag{2}$$

at each load, which for the 750 mA ALTUS28TR rail alone amounts to roughly 1.8 W of unnecessary heat dissipation. Switching conversion is therefore the appropriate choice for these main IC supplies. For the adjustable-output TPSM82822 stages, the output voltage is set by an external feedback divider,

$$V_{\mathrm{OUT}} = V_{\mathrm{REF}} \left(1 + \frac{R_{\mathrm{TOP}}}{R_{\mathrm{BOT}}}\right), \qquad V_{\mathrm{REF}} = 0.6\,\mathrm{V}. \tag{3}$$

For the 2.6 V ALTUS28TR rail, the divider uses $R_{\mathrm{TOP}} = 332\,\mathrm{k\Omega}$ and $R_{\mathrm{BOT}} = 100\,\mathrm{k\Omega}$, yielding

$$V_{\mathrm{OUT}} = 0.6\,\mathrm{V}\left(1 + \frac{332}{100}\right) = 2.592\,\mathrm{V} \approx 2.6\,\mathrm{V}. \tag{4}$$

For the 1.8 V NEXUS28Z core rail, the values $R_{\mathrm{TOP}} = 200\,\mathrm{k\Omega}$ and $R_{\mathrm{BOT}} = 100\,\mathrm{k\Omega}$ give

$$V_{\mathrm{OUT}} = 0.6\,\mathrm{V}\left(1 + \frac{200}{100}\right) = 1.8\,\mathrm{V}. \tag{5}$$

Beyond the DC operating point, the ripple behavior is also of interest. The first-order inductor ripple current is

$$\Delta I_L \approx \frac{(V_{\mathrm{IN}} - V_{\mathrm{OUT}})\, D}{L f_{\mathrm{SW}}}, \tag{6}$$

where $D = V_{\mathrm{OUT}}/V_{\mathrm{IN}}$ is the duty cycle, L is the inductance, and f_{SW} is the switching frequency. Substituting $V_{\mathrm{OUT}} = D\,V_{\mathrm{IN}}$ rewrites the ripple as

$$\Delta I_L \approx \frac{V_{\mathrm{IN}}\, D\,(1 - D)}{L f_{\mathrm{SW}}}, \tag{7}$$

which is parabolic in duty cycle and reaches its maximum at $D = 0.5$. The two rails of this design operate at $D \approx 0.36$ for the 1.8 V rail and $D \approx 0.52$ for the 2.6 V rail, both close to the worst-case point of the curve. The dominant levers for ripple reduction are therefore the inductance and the switching frequency rather than the duty cycle, which justifies the use of a high switching frequency and the integrated inductor solution. Each buck output is followed by a ferrite bead and a local decoupling network before reaching the IC supply pins, using the post-regulator filtering scheme described in Section 4.1.6. This filtering attenuates switching components from the raw converter output before they reach the noise-sensitive ALTUS28TR analog/RF and NEXUS28Z core supply domains.

4.1.4 Low-Noise and Auxiliary LDO Supply Rails

Switching conversion is not the best choice for every rail. For lower-power, noise-sensitive domains, linear regulation gives a cleaner output spectrum with a simpler implementation. The 1.8 V PLL rail of the NEXUS28Z and the 3.3 V digital rail therefore use LDOs (Fig. 19).

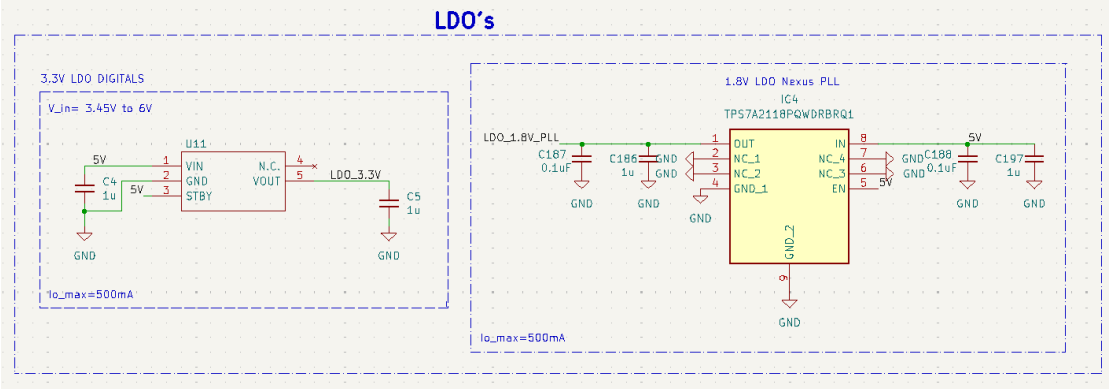


Fig. 19. LDO-based implementation of the auxiliary 3.3 V digital rail and the isolated low-noise 1.8 V PLL rail.

The 1.8 V PLL rail is produced by a TPS7A2118-Q1 ultra-low-noise LDO from TI [20]. This rail is kept isolated from the rest of the distribution network because the frequency-synthesis path is sensitive to switching transients and broadband noise that would otherwise couple through the supply. PLL and VCO circuits are more sensitive to supply variation than the surrounding blocks, so meeting the DC voltage requirement alone is not sufficient; supply noise directly degrades LO spectral purity.

The 3.3 V digital rail is provided by a BU33SD5WG-TR LDO from ROHM [21]. It feeds the digital control subsystem, the SPI level shifters used with the ALTUS28TR and NEXUS28Z, and the RF-switch driver. Phase-noise requirements are relaxed here compared to the PLL and RF domains, but a clean regulated output is still valuable since this rail fans out across several logic and control lines on the board.

Both LDOs include small bypass capacitors near their output pins, which support loop stability and provide local decoupling. Neither rail carries significant current,

which made linear regulation the natural choice: at these load levels the efficiency penalty is minor, while the output noise stays low with a simple implementation.

4.1.5 FDA Supply Rails

The fully differential amplifier stages in the baseband interface run from bipolar supplies, so the board generates a $+2.5\,\mathrm{V}$ rail and a $-2.5\,\mathrm{V}$ rail (Fig. 20). These allow the FDA interface to support the required analog swing range and keep the transmit and receive conditioning stages properly biased.

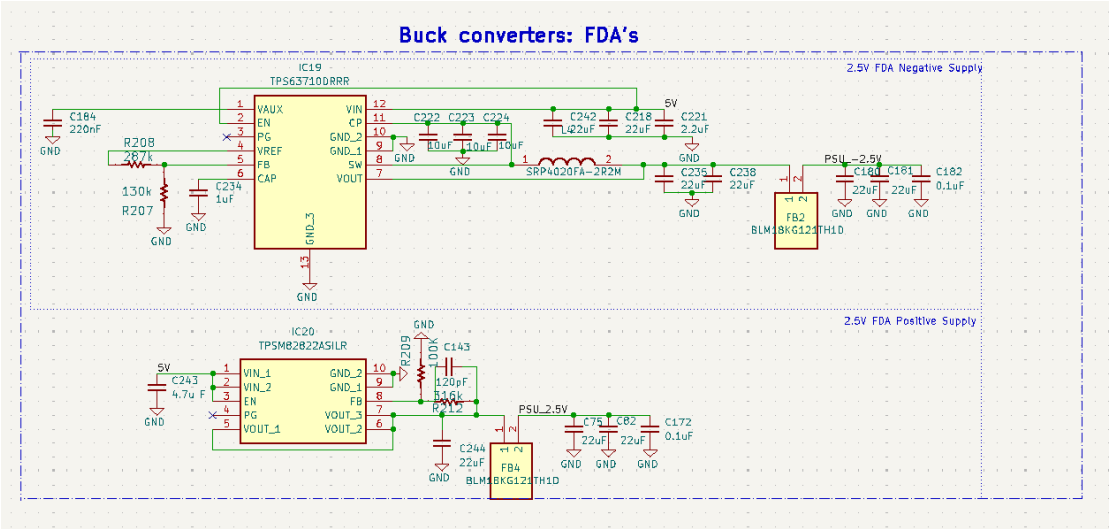


Fig. 20. Positive and negative supply-rail generation for the FDA-based analog signal-conditioning stages.

The positive rail uses another TPSM82822 buck. With $R_{\mathrm{TOP}} = 316\,\mathrm{k\Omega}$ and $R_{\mathrm{BOT}} = 100\,\mathrm{k\Omega}$,

$$V_{\mathrm{OUT}} = 0.6\,\mathrm{V}\left(1 + \frac{316}{100}\right) = 2.496\,\mathrm{V} \approx 2.5\,\mathrm{V}. \tag{8}$$

The negative rail uses the TPS63710 synchronous inverting buck converter [22]. Its feedback reference is

$$V_{\mathrm{FB,SET}} = -0.778\,\mathrm{V}, \tag{9}$$

and the output voltage is set by

$$V_{\mathrm{OUT}} = V_{\mathrm{FB,SET}}\left(1 + \frac{R_1}{R_2}\right). \tag{10}$$

For $R_1 = 287\,\mathrm{k\Omega}$ and $R_2 = 130\,\mathrm{k\Omega}$,

$$V_{\mathrm{OUT}} = -0.778\,\mathrm{V}\left(1 + \frac{287}{130}\right) \approx -2.50\,\mathrm{V}. \tag{11}$$

Both FDA rails use the same post-regulator filtering approach as the other buck outputs before reaching the analog stages. At layout level, both require careful

routing toward the FDA region to avoid unnecessary series resistance and to keep return-current paths well controlled; they are therefore handled as dedicated analog supply branches.

4.1.6 Filtering and Decoupling Strategy

Supply quality depends on both the nominal output voltage and the frequency-dependent impedance between the regulator and the load. The PDN uses a cascaded filtering and decoupling scheme (Fig. 21) to keep this impedance low across the relevant frequency range.

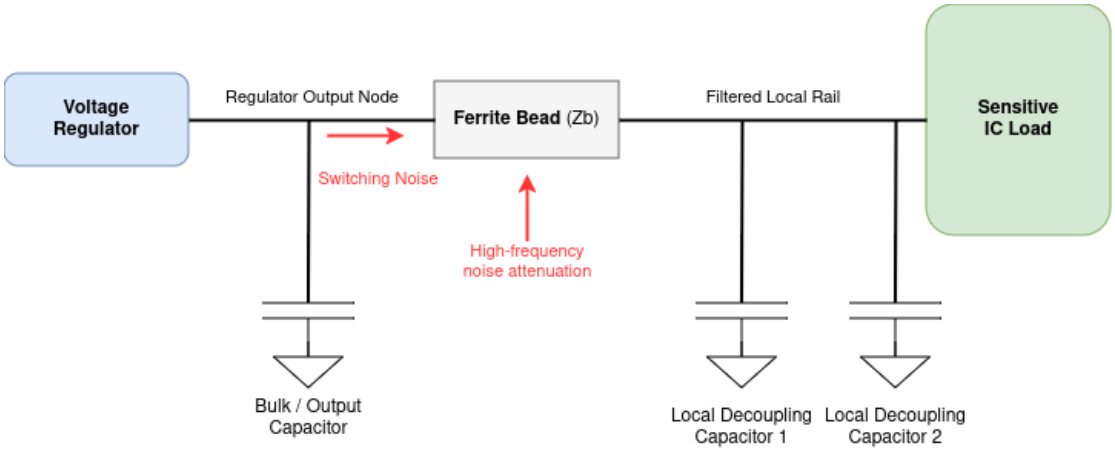


Fig. 21. Conceptual illustration of the local filtering and decoupling approach used in the PDN, combining regulator-side output capacitance, ferrite-bead isolation, and local decoupling close to the load.

The regulator output capacitors provide local energy storage and reduce low-frequency fluctuations, but the regulator output node is not always the right final supply node for a sensitive RF, analog, or PLL load. Several rails are therefore followed by a ferrite bead and a second local decoupling network before reaching the IC supply pins. The bead raises the impedance presented to higher-frequency conducted noise, while the local capacitors shunt those components to ground and supply transient current close to the load. This effectively decouples the regulator-side and load-side nodes at high frequency, reducing conducted noise propagation toward the load.

The impedance of a capacitor falls with frequency,

$$Z_C(j\omega) = \frac{1}{j\omega C}, \tag{12}$$

whereas parasitic inductance from traces, vias, and interconnects rises,

$$Z_L(j\omega) = j\omega L. \tag{13}$$

The local decoupling must therefore sit close to the IC supply pins: even ample capacitance loses effectiveness at high frequency if the inductive path to the load is long.

Placement is hierarchical. Smaller capacitors sit close to the IC pins to handle high-frequency transient current, while larger capacitors sit near the regulator output or at the start of the local branch to provide lower-frequency storage. At layout level, short supply traces, low-inductance returns, and minimal shared impedance between noisy and sensitive nodes all help to suppress conducted coupling through common parasitics. Because the mmWave transceivers, PLL, analog conditioning, and digital control all share the same board, filtering and decoupling were treated as an integral part of the PDN layout, with decoupling capacitors placed as close as practical to the corresponding supply pins.

4.1.7 Digital Supply and Logic-Level Considerations

The 3.3 V rail is the main digital control domain of the board because this is the logic level used by the FTDI interface and the Cheetah SPI host adapter. The host-side SPI and control signals are therefore all referenced to 3.3 V.

The controlled ICs, however, run at different logic levels: the ALTUS28TR SPI is referenced to 2.6 V and the NEXUS28Z SPI to 1.8 V. Dedicated level-shifting stages translate the 3.3 V host-side signals to each device's domain. Beyond SPI, the 3.3 V rail also supplies the RF-switch control lines and a few auxiliary peripherals. Current demand is low, but the functional scope of this rail is wide, since it coordinates configuration across all the main transceiver subsystems.

4.1.8 PDN Layout

The layout of the PDN was treated as part of the supply-network design, not as a routing step after the schematic was finalized. The regulator stages are grouped in a dedicated PDN region so that current paths to the main load regions stay compact, with low loop area and minimal overlap with unrelated functional domains. Each rail is organized locally around its load, not distributed through a shared supply structure, which keeps common-impedance coupling between unrelated domains low. Switching elements are kept physically separated from the feedback nodes and from the lower-noise branches to limit field coupling into the regulation loop and the nearby analog circuitry. The resulting PDN layout is shown in Fig. 22.

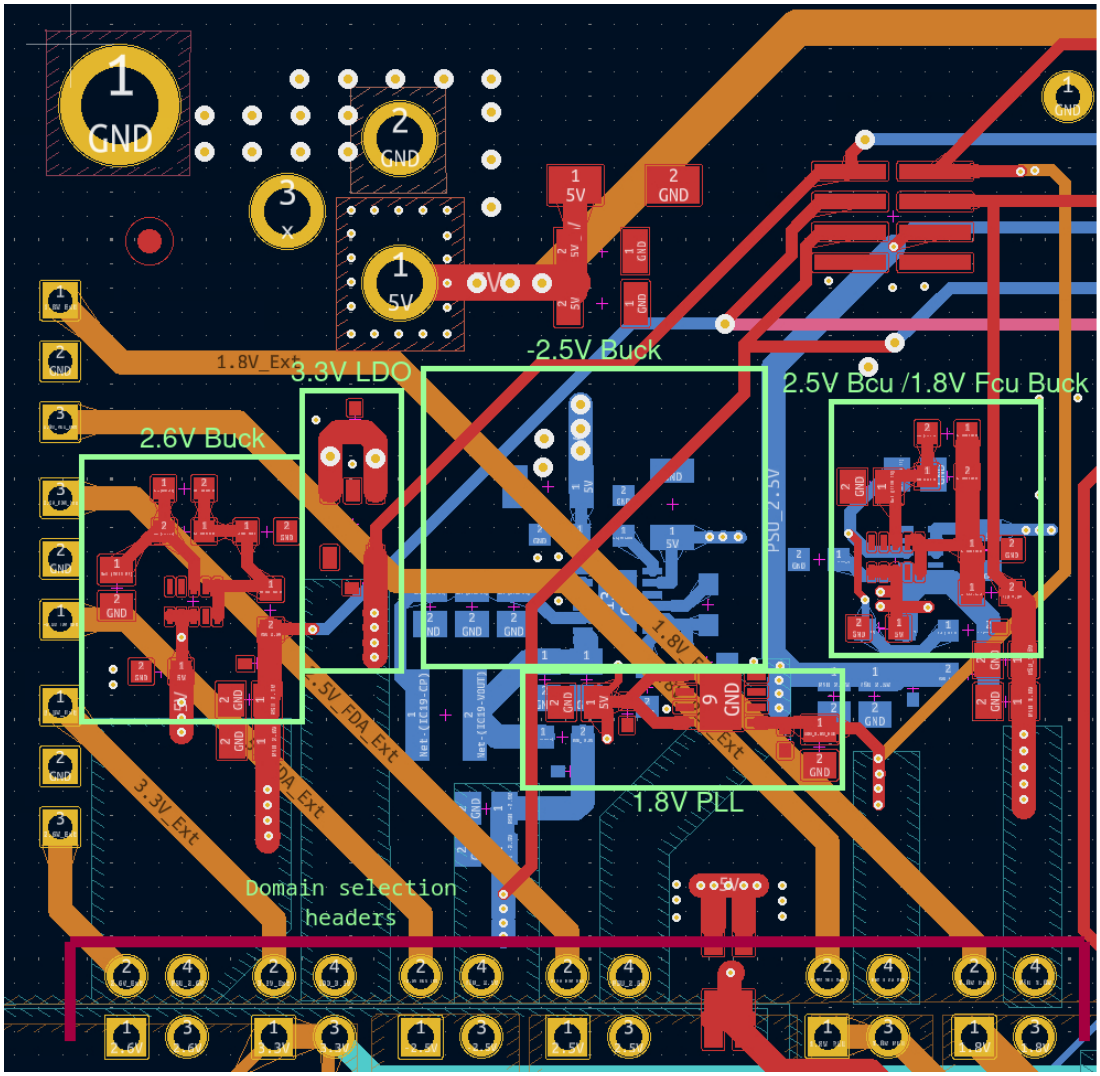


Fig. 22. PCB layout overview of the PDN, showing the regulator stages, the local organization of each power domain, and the domain-selection header region.

The higher-current rails (the 2.6 V ALTUS28TR, the 1.8 V NEXUS28Z core, and the ± 2.5 V FDA rails) are routed as localized copper pours or widened sections rather than narrow traces, which keeps series resistance and voltage drop low. There is no single default width across the board; each rail was sized to its own expected current and route length. The lower-current auxiliary branches stay on narrower, more conventional routing.

The 2.6 V ALTUS28TR rail illustrates the approach. It is implemented as a localized copper region feeding both transceiver devices (Fig. 23). The corresponding feed paths toward the load pins are shown in Fig. 24, alongside the equivalent detail for the $+2.5$ V FDA branch; both follow the same principle of supplying the local decoupling network through a widened pour before reaching the device pins.

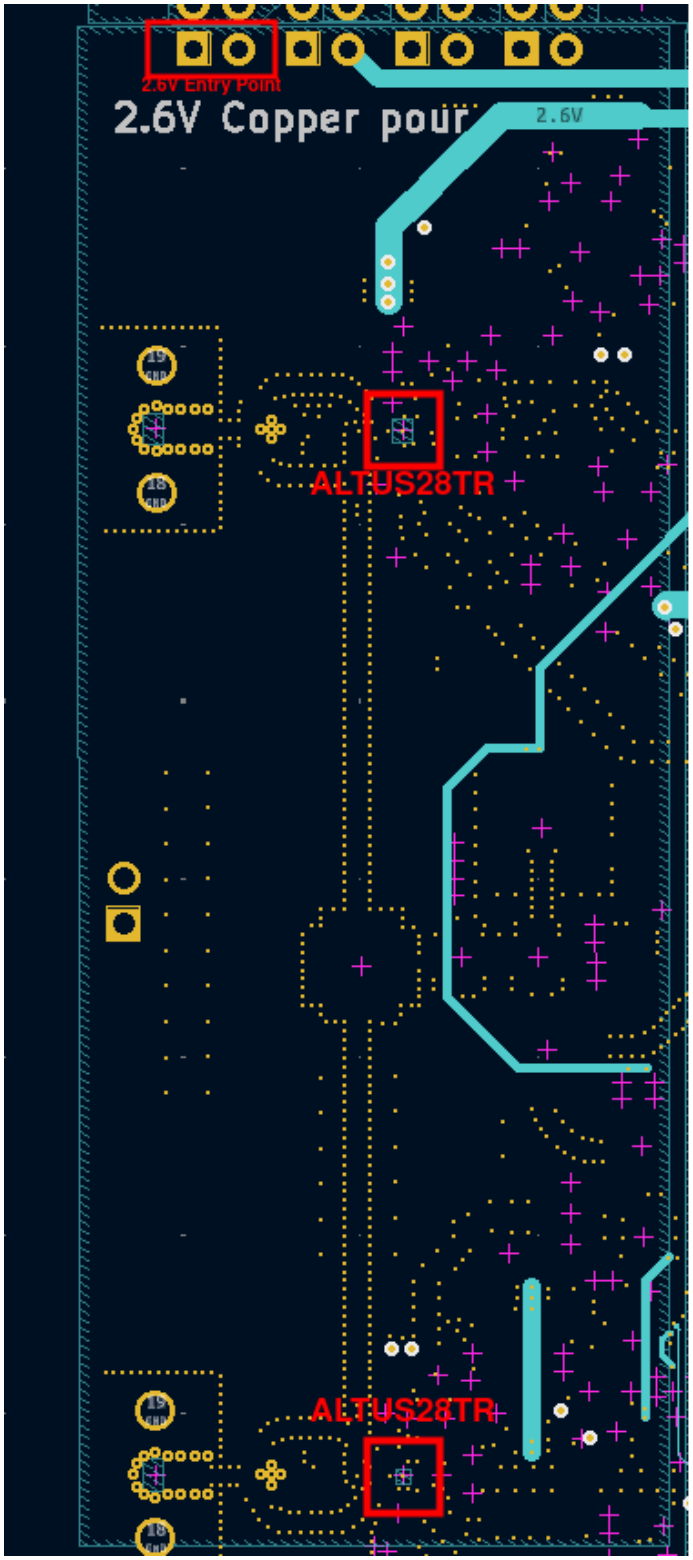
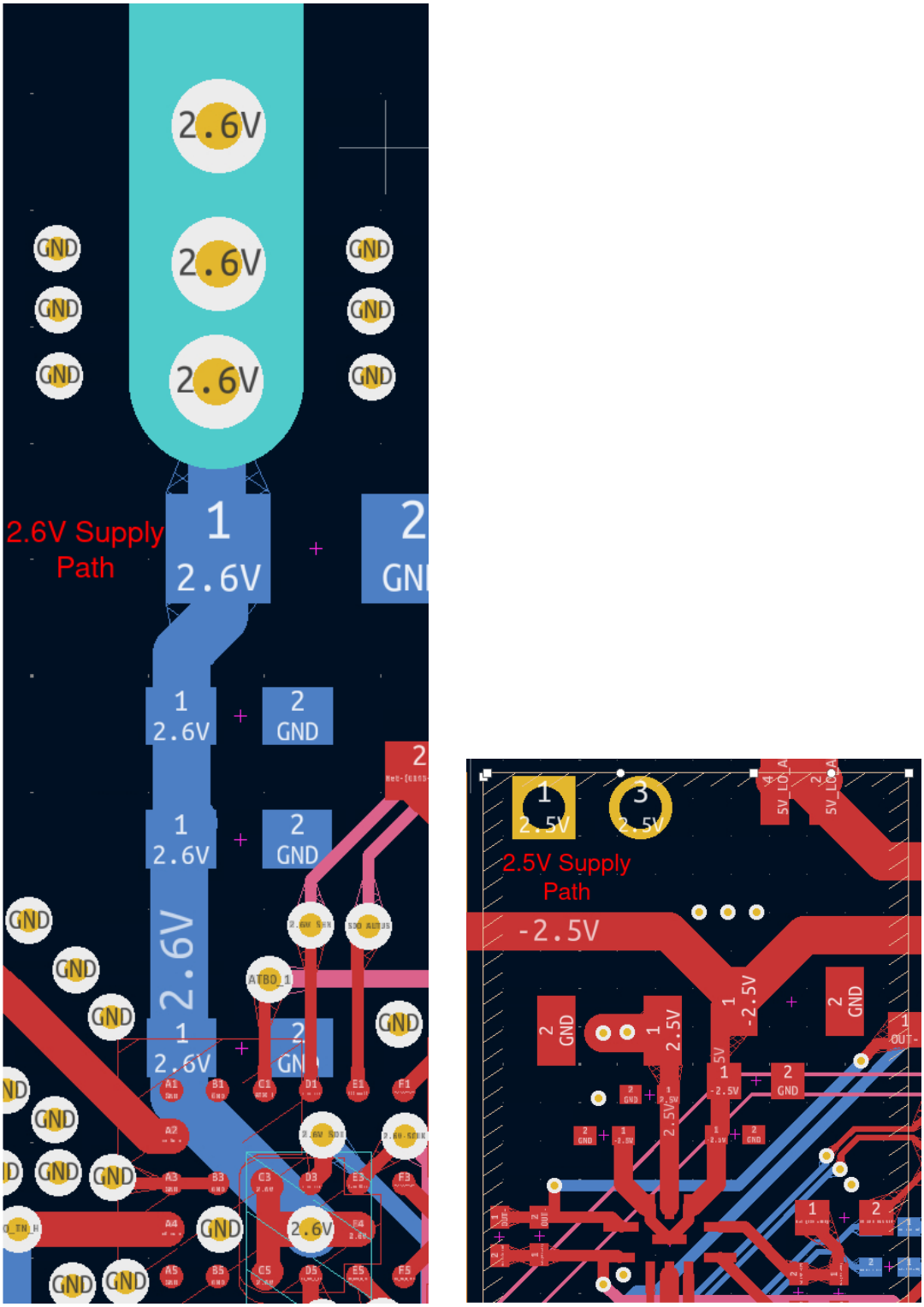


Fig. 23. Localized 2.6 V copper region supplying both ALTUS28TR devices. The rail is realized as an extended pour to reduce series resistance and distribute current across the load region.



(a) 2.6 V ALTUS28TR supply path.

(b) 2.5 V FDA supply branch.

Fig. 24. Layout detail of two representative high-current supply paths. In both cases, the rail is distributed through a widened copper path that reaches the local decoupling network before the device pins.

For a rectangular copper path, the DC resistance is estimated as

$$R = \rho \frac{l}{wt}, \tag{14}$$

where ρ is the resistivity of copper, l is the path length, w is the conductor width, and t is the copper thickness. The voltage drop is then $\Delta V = IR$. For the higher-current rails, the layout reduces this drop by using wider copper where space allows and by keeping the main supply paths short.

Layer transitions in the PDN use multiple parallel vias instead of a single via. The parallel vias lower the effective series resistance and inductance of the transition, share current between layers, and avoid forcing the full rail current through one vertical interconnect. This is used on the higher-current branches and at copper-pour transitions between layers.

4.1.9 Bring-Up, Monitoring, and External Power Access

The PDN also supports staged bring-up and rail-by-rail debugging. Each major rail includes an external power-injection pad, so that one supply domain can be powered, checked, or diagnosed without energizing the whole board. A jumper selects whether the rail is fed by its on-board regulator or by the external pad (Fig. 25).

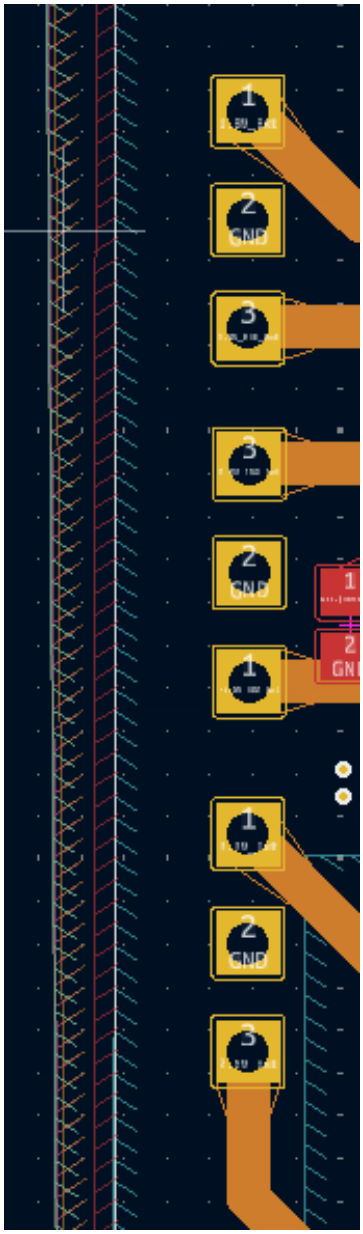


Fig. 25. External power-injection pads and jumper selection for the main supply rails. Each rail can be fed either from its on-board regulator or from an external source during bring-up, isolated validation, or fault diagnosis.

The selection headers use TE Connectivity AMPMODU 826656-2 through-hole parts on a 2.54 mm pitch, rated at 5 A per contact [23]. Through-hole mounting was preferred over surface-mount because the jumpers may be inserted and removed repeatedly during bring-up, and plated through-hole pads tolerate this mechanical stress better.

Each major rail also has a dedicated status LED. The LEDs provide a quick visual check during bring-up and help catch simple faults such as a missing rail or an incorrect jumper setting before probing starts. They are not a substitute for voltage measurement.

4.2 Baseband Signal Conditioning

The Red Pitaya and the ALTUS28TR cannot be connected directly at baseband. Their signal amplitudes, common-mode levels, and interface formats are different: the Red Pitaya uses single-ended analog I/O, while the ALTUS28TR baseband interface is differential. Separate FDA stages are therefore used in the TX and RX directions.

In the TX path, the single-ended Red Pitaya output is scaled, level shifted, and converted to a differential signal for the ALTUS28TR input. In the RX path, the differential ALTUS28TR baseband output is conditioned and converted to a single-ended signal for the Red Pitaya ADC.

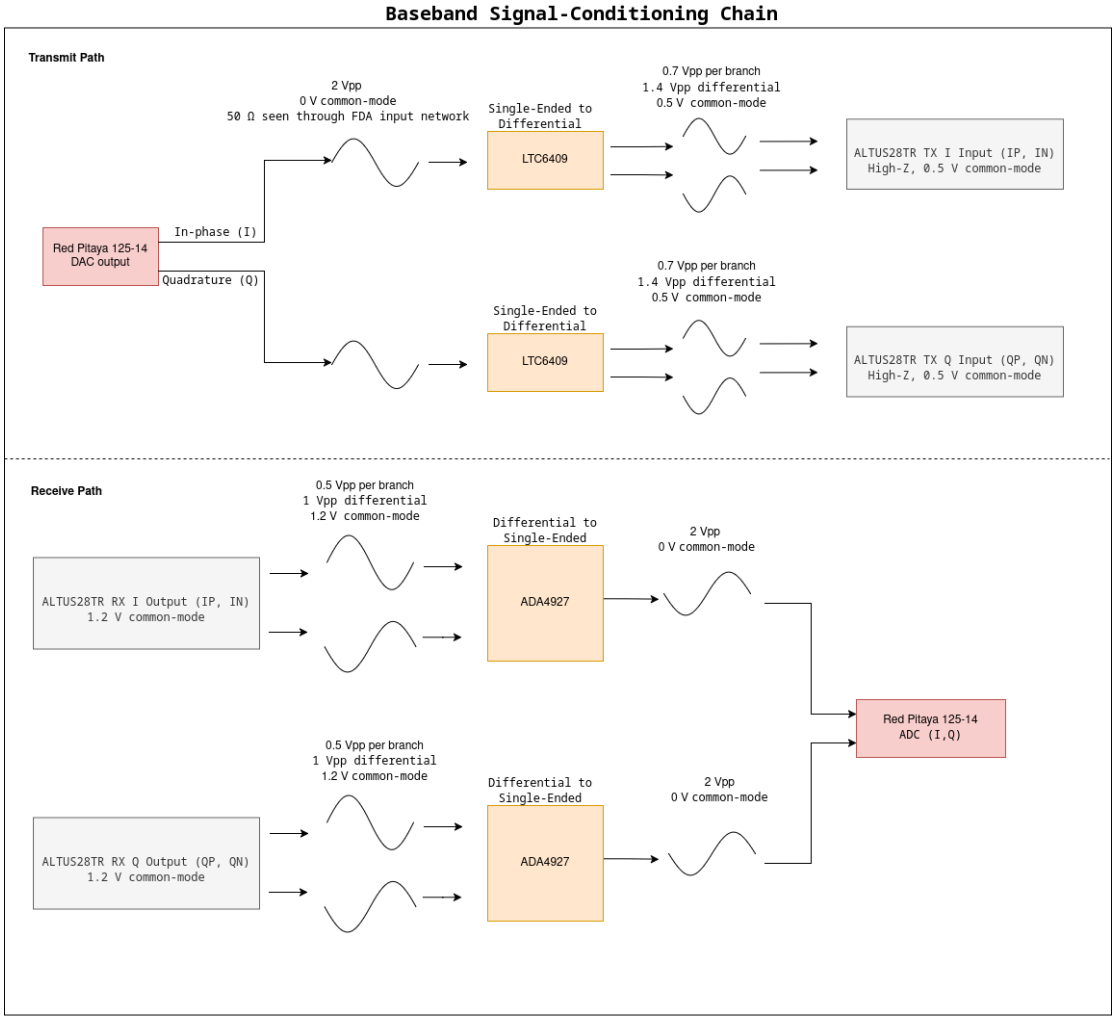


Fig. 26. High-level overview of the baseband signal-conditioning chain between the Red Pitaya platform and the ALTUS28TR transceiver, showing the distinct transmit and receive adaptation paths for the in-phase and quadrature channels.

The FDA-based interface translates voltage levels and common-mode conditions between the Red Pitaya and the ALTUS28TR, and lets the converters operate near full scale. This matters for signal integrity: with N-bit converters, the signal-to-quantization-noise ratio improves by 6 dB for every doubling of the signal swing

relative to full scale, while the quantization and thermal noise floor stay fixed. Driving the converters well below full scale therefore wastes effective bits on the desired signal. Setting the swing in the analog conditioning stages, rather than constraining the source directly, keeps the signal as far above the noise floor as the 14-bit converters allow.

4.2.1 Transmit Path Design

On the transceiver side, the ALTUS28TR transmit baseband input expects a differential signal centered around a specific common-mode voltage. The device uses a Zero-IF I/Q modulator stage operating from a 2.6 V supply, so a target differential swing of 1.4 Vpp centered near 0.5 V keeps the modulator within its linear region while leaving headroom for I/Q modulation peaks. The FDA stage targets a delivered swing of 1.6 Vpp at the transceiver input, with a small margin above the nominal target to absorb routing losses and component tolerances between the FDA output and the ALTUS28TR input.

The transmit conditioning stage therefore performs three functions: amplitude scaling of the DAC signal, translation of the common-mode level, and single-ended-to-differential conversion. The LTC6409 implements these functions and the topology is shown in Fig. 27.

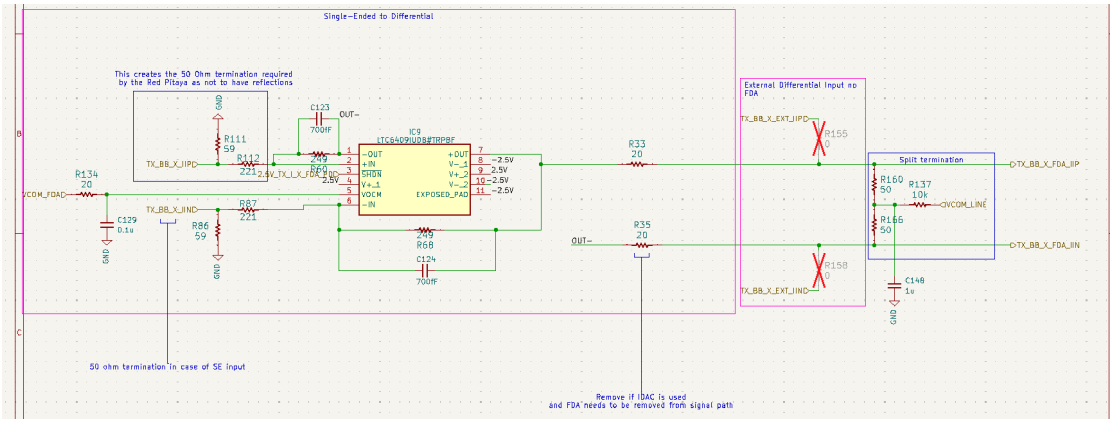


Fig. 27. Schematic implementation of the transmit-path signal-conditioning stage based on the LTC6409, including input termination, single-ended-to-differential conversion, common-mode generation, and split termination toward the ALTUS28TR transmit baseband input.

In this configuration, the LTC6409 is used as a single-ended-to-differential driver for the ALTUS28TR transmit baseband input. The selected topology also preserves the option for future differential-to-differential operation, should the external baseband source be upgraded to a differential interface.

The LTC6409 was selected for the transmit path because it supports the common-mode range required by the ALTUS28TR baseband interface and offers sufficient gain-bandwidth product for the intended baseband bandwidth. The ALTUS28TR transmit baseband input supports bandwidths up to 720 MHz, so the FDA stage must not be the dominant bandwidth limit of the transmit path.

On the source side, the Red Pitaya DAC drives an effective load close to 50 Ω. This requirement comes from two considerations. The DAC is specified for a 50 Ω load [16], with an output range of ± 1 V, or 2 Vpp. An incorrect load would shift the delivered DC level and reduce the available swing, since the nominal 2 Vpp output is specified at 50 Ω loading. In addition, the SMA cable between the Red Pitaya and the FDA input is a 50 Ω transmission line, and a matched load at the FDA input reduces reflections that could distort fast baseband transitions.

This termination is implemented by the FDA input resistor network itself, not by a separate discrete resistor across the input. When the amplifier is driven single-ended, the effective input impedance differs from R_G. The non-driven input also moves slightly with the signal through the feedback network, which partially bootstraps the driven input and raises its effective impedance. The resulting single-ended input impedance is

$$R_{\mathrm{IN,SE}} = \frac{R_G}{1 - \dfrac{R_F}{2(R_G + R_F)}}. \tag{15}$$

Using the selected values $R_G = 221\,\Omega$ and $R_F = 249\,\Omega$,

$$R_{\mathrm{IN,SE}} = \frac{221\,\Omega}{1 - \dfrac{249}{2(221 + 249)}} \approx 300.6\,\Omega. \tag{16}$$

This is higher than the 50 Ω target, so an additional shunt resistor R_T is placed from the input to ground. The required shunt value follows from the parallel-combination condition $R_T \parallel R_{\mathrm{IN,SE}} = 50\,\Omega$, which gives

$$R_T = \frac{R_{\mathrm{IN,SE}} \cdot 50}{R_{\mathrm{IN,SE}} - 50} \approx 60.0\,\Omega. \tag{17}$$

The closest selected standard value is 59 Ω, which yields

$$R_{\mathrm{SEEN}} = 59\,\Omega \parallel 300.6\,\Omega \approx 49.3\,\Omega. \tag{18}$$

The 50 Ω-class source load is therefore set by the combination of the FDA input network and the shunt resistor, while the same feedback network sets the closed-loop gain of the stage.

In contrast, no 50 Ω matching is required at the FDA output, since the ALTUS28TR baseband input is high-impedance. The output network was therefore designed for controlled signal delivery and biasing rather than for transmission-line matching.

The same input network sets the differential closed-loop gain. With $R_G = 221\,\Omega$ and $R_F = 249\,\Omega$, the gain of the LTC6409 stage is

$$A_{vd} \approx \frac{R_F}{R_G} = \frac{249}{221} \approx 1.13. \tag{19}$$

The Red Pitaya DAC is operated at full scale, 2 Vpp single-ended, so that the available DAC dynamic range is fully used at the source. With the gain above, the FDA produces 2.26 Vpp differential at its output. The signal is then attenuated by the split-termination network at the ALTUS28TR side, described below, to deliver approximately 1.6 Vpp differential at the transceiver baseband input.

A direct external input path through SMA connectors is also provided so that the FDA stage can be bypassed for testing or alternative drive scenarios. Removable 20 Ω series resistors are placed in the signal path to allow manual path reconfiguration during bring-up. The 0603 package was chosen specifically because it can be handled and re-soldered with standard hand tools, which is not practical with smaller package sizes.

At the transceiver-side interface, a split-termination network is used to provide a defined differential load for the FDA stage and to establish the common-mode reference. The ALTUS28TR baseband input itself is a high-impedance node, so this network is not used for transceiver-side matching. It is included to give the FDA a well-defined symmetric load over frequency, which supports stable wideband operation, and to leave room for the current-mode excitation option. With two equal termination resistors R_{TERM} connected from the two signal branches to a common midpoint, the differential termination is

$$Z_{\mathrm{diff}} = 2R_{\mathrm{TERM}}. \tag{20}$$

For $R_{\mathrm{TERM}} = 50\,\Omega$, the resulting differential load becomes

$$Z_{\mathrm{diff}} = 2 \times 50\,\Omega = 100\,\Omega. \tag{21}$$

The center of the split termination is biased via the `VCOM_LINE` node, which carries the buffered 0.5 V common-mode reference described below, so that the differential signal, denoted by v_d, is centered around the intended common-mode level V_{COM}. The two terminal voltages of the split termination are given by

$$v_+ = V_{\mathrm{COM}} + \frac{v_d}{2}, \qquad v_- = V_{\mathrm{COM}} - \frac{v_d}{2}. \tag{22}$$

The split termination therefore defines the differential load, sets the DC operating point, and preserves symmetry at the transceiver-side interface.

The same network also preserves the option of direct current-mode excitation. If the FDA stage is removed and the transmit path is driven by a differential current-output DAC, then the two termination resistors convert the differential branch currents into corresponding branch voltages with respect to the common-mode reference. When the two branches carry equal and opposite currents $\pm I$, the resulting branch voltages become

$$v_+ = V_{\mathrm{COM}} + IR_{\mathrm{TERM}}, \qquad v_- = V_{\mathrm{COM}} - IR_{\mathrm{TERM}}, \tag{23}$$

and the resulting differential signal is

$$v_d = v_+ - v_- = 2IR_{\mathrm{TERM}}. \tag{24}$$

Thus, for $R_{\mathrm{TERM}} = 50\,\Omega$,

$$v_d = 2I \times 50\,\Omega = I \cdot 100\,\Omega, \tag{25}$$

showing that the network can also act as a passive differential current-to-voltage conversion stage.

The required common-mode voltages are generated by resistor dividers and buffered to provide low-impedance bias references for the surrounding analog circuitry. For a divider driven from V_{DD}, the tap voltage is

$$V_{\mathrm{tap}} = V_{\mathrm{DD}} \frac{R_{\mathrm{below}}}{R_{\mathrm{total}}}, \tag{26}$$

where R_{below} is the effective resistance from the selected tap to ground, and R_{total} is the overall resistance of the ladder. Auxiliary bias levels are derived from the 3.3 V divider network and then buffered before being distributed to the surrounding signal-conditioning circuitry. The 0.5 V common-mode reference used by the transmit FDA path is generated separately from the 5 V rail. The tap voltage resulting from resistor values of 750 kΩ, 150 kΩ, and 100 kΩ is given by

$$V_{0.5} = 5\,\mathrm{V} \cdot \frac{100}{750 + 150 + 100} = 0.5\,\mathrm{V}. \tag{27}$$

The feedback capacitors of the transmit FDA were not chosen as an aggressive low-pass reconstruction filter. They shape the high-frequency response of the amplifier network and help stabilize the closed loop near the upper end of the useful baseband range. The corner frequency associated with this first-order response can be approximated as

$$f_c \approx \frac{1}{2\pi R_F C_F}. \tag{28}$$

For $R_F = 249\,\Omega$ and $C_F = 700\,\mathrm{fF}$, this gives

$$f_c \approx \frac{1}{2\pi \cdot 249\,\Omega \cdot 700\,\mathrm{fF}} \approx 913\,\mathrm{MHz}. \tag{29}$$

At 913 MHz this pole sits near the top of the intended baseband range, so it trims the high-frequency response without limiting the in-band signal. The value of C_F was chosen iteratively, by sweeping a range of feedback capacitor values in LTspice and selecting the value that gave the widest flat passband without compromising stability. This option is available because the LTC6409 is a voltage-feedback amplifier; the same approach is not used in the receive path, where the ADA4927 is a current-feedback amplifier.

4.2.2 Receive Path Design

The receive baseband path conditions the differential output of the ALTUS28TR for connection to the Red Pitaya ADC. The Red Pitaya ADC is a bipolar DC-coupled converter with a ± 1 V input range [16], so the target signal at the ADC input is 2 Vpp single-ended, centered at 0 V. The receive stage must therefore provide gain, DC-level translation, and differential-to-single-ended conversion in a single block.

The receive conditioning stage is shown in Fig. 28 and uses the ADA4927 fully differential amplifier.

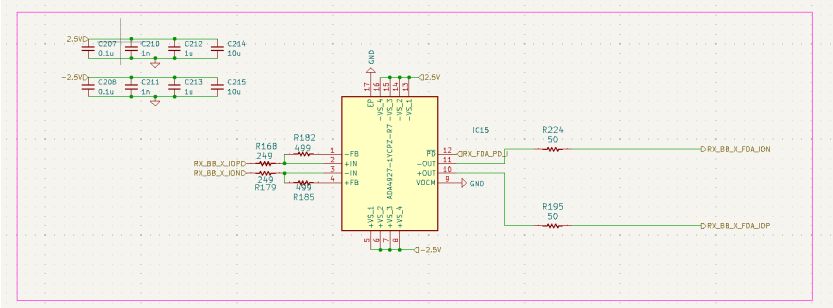


Fig. 28. Receive-path signal-conditioning stage based on the ADA4927, providing gain, DC-level translation, and differential-to-single-ended conversion between the ALTUS28TR receive baseband output and the Red Pitaya ADC input.

The ADA4927 takes the differential baseband output of the ALTUS28TR and processes both branches together, so the differential signal is amplified while the common-mode component at the input is rejected. The single-ended output to the ADC is taken from one of the two FDA output pins. The complementary output is not used as a signal output in this interface and is handled according to the stability and loading guidance of the ADA4927 datasheet. A useful property of the device is that it supports relatively large gain-setting resistors, which reduces loading on the ALTUS28TR baseband output.

The receive baseband output is internally driven by the I/Q demodulator stage of the ALTUS28TR, which operates from the same 2.6 V supply. The output therefore delivers 1.0 Vpp on each branch (2.0 Vpp differential) with a common-mode level near 1.2 V, set by the internal demodulator bias. With $R_G = 249\,\Omega$ and $R_F = 499\,\Omega$, the closed-loop differential gain of the FDA is

$$A_v \approx \frac{R_F}{R_G} = \frac{499}{249} \approx 2.00. \tag{30}$$

Since only one of the two FDA outputs drives the ADC, the relevant transfer function is the differential-to-single-ended gain,

$$A_{v,\text{diff}\to\text{SE}} = \frac{R_F}{2R_G} \approx 1.0, \tag{31}$$

which scales the 2.0 Vpp differential signal from the transceiver to a 2.0 Vpp single-ended waveform at the ADC input.

The ADA4927 also performs the common-mode level translation between the two sides. The device includes a dedicated `VOCM` input that sets the output common-mode voltage independently of the input common-mode level, so the output is centered wherever the `VOCM` pin is biased. In this design, the `VOCM` input is tied to system ground, which centers the FDA output at 0 V and matches the bipolar input range of the Red Pitaya ADC. The transceiver-side common-mode of approximately 1.2 V is therefore absorbed by the FDA itself, and no AC coupling is required on the signal path. This preserves the zero-IF content near DC, which would otherwise be removed by series capacitors.

The choice of R_G also sets the differential input resistance of the FDA. To first order,

$$R_{\text{IN,diff}} \approx 2R_G = 498\,\Omega, \tag{32}$$

which is high enough to limit loading on the transceiver output while preserving the target gain setting.

Unlike the transmit path, the receive-side feedback network carries no explicit capacitor across R_F. This is deliberate: the ADA4927 is a current-feedback amplifier, whose stability depends on the feedback impedance being primarily resistive. A capacitor in parallel with R_F would not produce a well-behaved low-pass response and could introduce peaking or ringing, as the datasheet warns [15]. Bandwidth limitation on the receive side therefore comes from the amplifier itself and the surrounding signal chain, not from a dedicated RC pole in the feedback network.

4.3 RF Front-End and TDD Switching Interface

Chapter 3 introduced the system-level role of the MASW-011145 SPDT switch. This subsection focuses on its circuit- and PCB-level integration within each polarization branch, covering the RF interconnect topology and the layout choices around it.

4.3.1 RF Path Implementation

Each branch routes three RF signals to the switch: the ALTUS28TR TX output, the ALTUS28TR RX input, and the external connector at the common RF port of the SPDT. Since the platform operates with direct mmWave signal paths and the ALTUS28TR RF ports are specified for direct 50 Ω interfacing, no series DC-blocking capacitors were inserted in the RF lines. The switch control pin is driven directly from a 3.3 V GPIO line, which is within the CMOS logic range accepted by the MASW-011145, so no level-shifting stage is needed.

The external RF interface uses a 50 Ω 2.92 mm (K-type) end-launch connector [24]. The 2.92 mm standard was selected because its usable bandwidth extends to 40 GHz, well above the 28 GHz to 31 GHz operating range. This margin is useful for connector-launch nonidealities, parasitic resonances near the transition, calibration bandwidth, and possible future extension of the operating range.

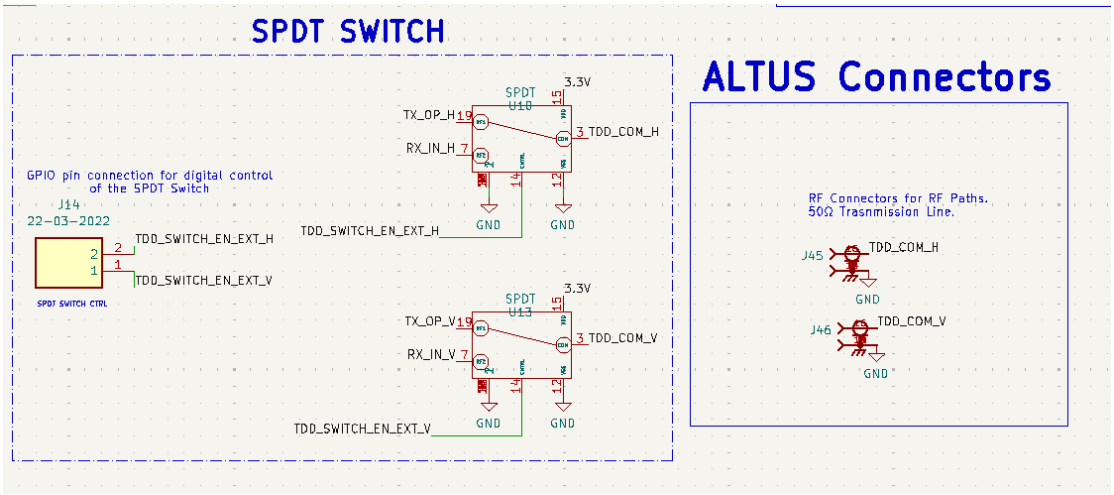


Fig. 29. Schematic of the RF front-end for one polarization branch, showing the MASW-011145 SPDT, the 2.92 mm connector interface, and the connection points to the ALTUS28TR.

4.3.2 RF Layout Considerations

At 28 GHz to 31 GHz, the guided wavelength is about 5 mm on this substrate, so geometric features of 0.1 mm or larger represent a non-negligible fraction of λ_g and translate into measurable impedance mismatch and radiation. The switch is placed close to both the connector and the transceiver, which keeps all three RF segments short; each segment is routed as GCPW with top-layer via stitching along both edges of the signal trace to preserve the reference and suppress surface-wave coupling.

The most sensitive region is the connector launch and the short section of GCPW that follows it. The launch footprint follows the connector manufacturer's recommended geometry, and the surrounding ground is stitched with a dense via fence so that the transition from the coaxial reference to the PCB reference stays clean. Trace widths are set for 50 Ω on the chosen stackup, and trace bends in the switching region are avoided where possible or implemented as mitered corners to limit localized mismatch. The extracted electromagnetic behavior of this region, including the asymmetry between the TX and RX paths, is discussed in Section 4.9.

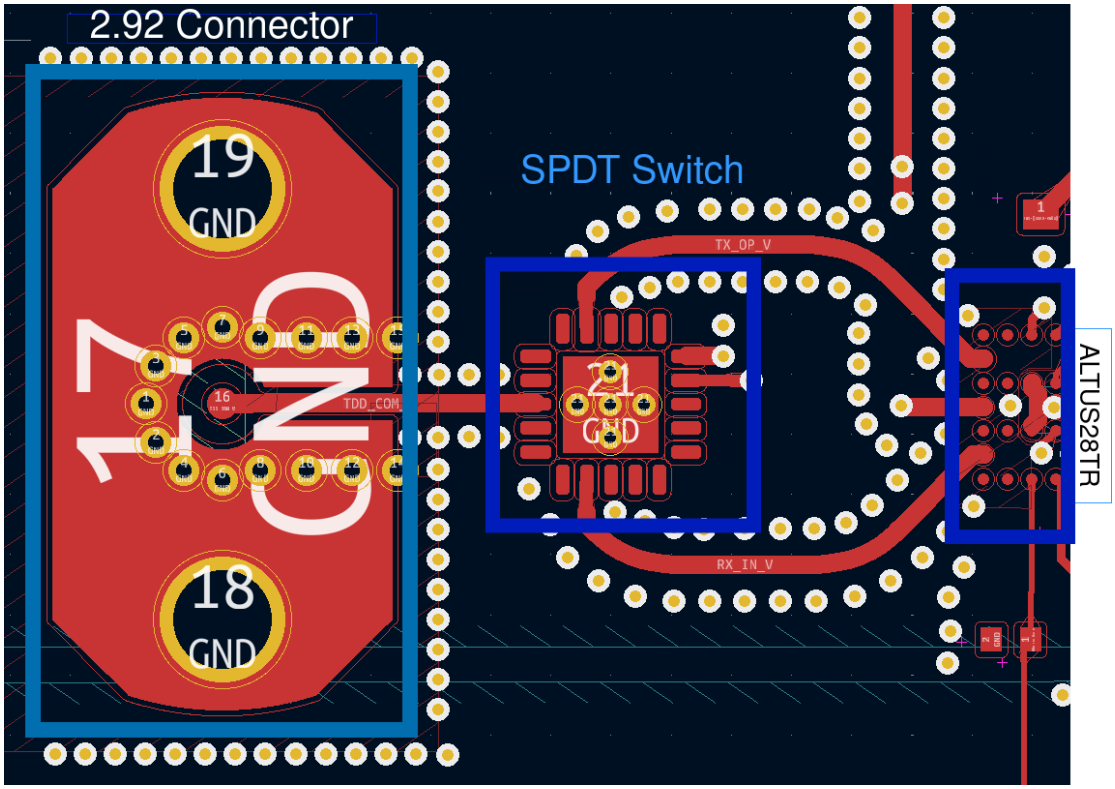


Fig. 30. PCB layout of the RF front-end, showing the 2.92 mm connector launch, the MASW-011145 SPDT, and the short GCPW segments toward the ALTUS28TR RF ports.

4.4 Local Oscillator Generation and Distribution

The system-level role of the LO subsystem was covered in Chapter 3. This subsection presents its circuit- and PCB-level implementation within the platform. The LO chain handles four functions: LO generation by the on-board PLL, provision for an alternative external LO input, intermediate amplification, and final fan-out to the two ALTUS28TR devices. These are implemented, respectively, by the PLL subsystem together with its auxiliary output, a PE42520 SPDT absorptive RF switch for source selection [25], a CMX90G702QF gain block for amplification [26], and an EP2RKU+ two-way resistive splitter [27] for the final distribution.

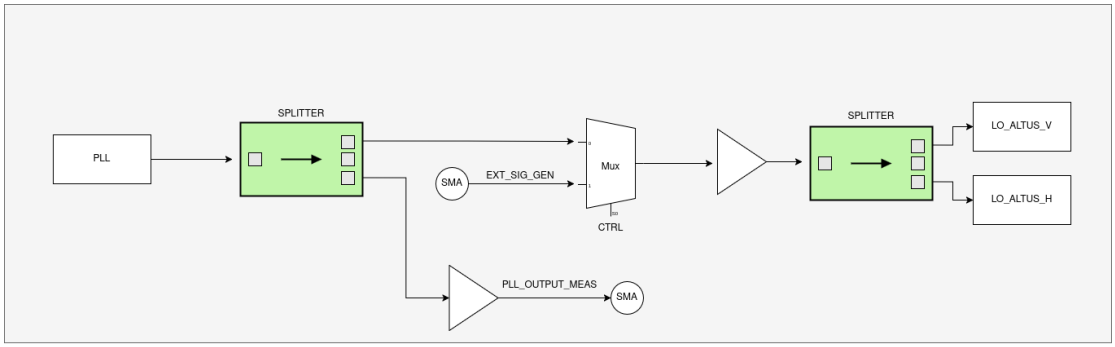


Fig. 31. Block-level view of the LO chain, showing the PLL path, the external LO input, the source-selection and amplification stages, and the fan-out toward the two ALTUS28TR devices.

4.4.1 On-Board LO Generation and Auxiliary Output

The primary LO source is the on-board PLL subsystem. Its output is not routed directly to the source-selection switch. Instead, a small two-way resistive splitter divides the PLL signal between the source-selection switch (the main LO path) and an auxiliary SMA port. The auxiliary port carries a copy of the PLL output for laboratory measurement or for driving external equipment, while the main signal continues to the rest of the distribution chain. The use of a resistive splitter here introduces 6 dB of loss on each branch, which is acceptable on the auxiliary side and is compensated downstream on the main path by the gain block discussed below.

4.4.2 External LO Provision and Source Selection

An SMA input is provided so that an external LO can be used in place of the PLL output. Selection between the two sources is performed by the PE42520 SPDT switch, whose output feeds the rest of the distribution chain. The downstream amplification and splitting stages therefore operate regardless of which source is selected.

4.4.3 LO Amplification and Final Distribution

The EP2RKU+ resistive splitter introduces roughly 6 dB of splitting loss per output, (3 dB inherent power splitting plus an additional 3 dB associated with the resistive isolation network), and the cascade of interconnect, switch, and splitter adds further insertion loss along the path. To bring the LO level at each ALTUS28TR input back within the specified −10 dBm to 3 dBm window, a CMX90G702QF gain block is inserted in the chain. The gain block is placed ahead of the splitter so that a single amplifier feeds both branches; placing it after the splitter would have required two gain blocks (one per branch), at higher cost, additional supply current, and the risk of mismatched gain between the two paths. The splitter fan-out drives the vertical- and horizontal-polarization transceivers from the same source, so both chains op-

erate from a shared LO. This keeps the two polarization paths frequency-coherent and reduces relative phase drift between them.

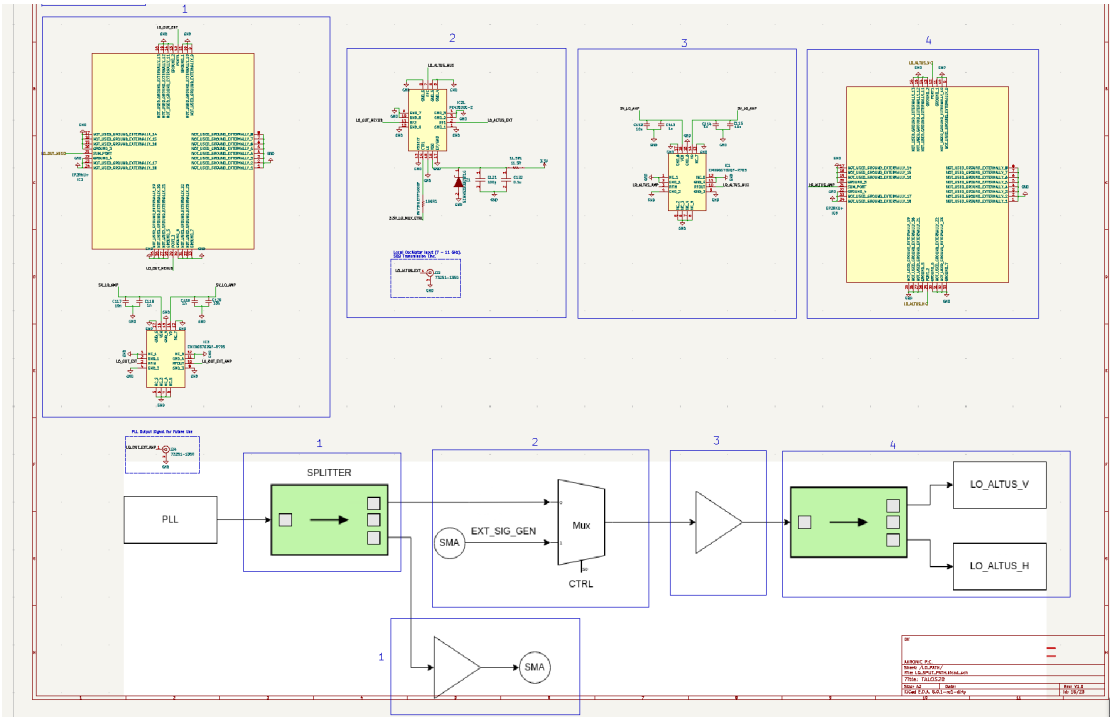


Fig. 32. Schematic of the LO generation and distribution chain, including the PLL path, external LO input, source-selection switch, gain block, and two-way splitter feeding the ALTUS28TR LO inputs.

4.4.4 LO Path Loss Budget

To estimate the LO level reaching each ALTUS28TR, the loss budget of the on-board LO path was tabulated (Table 6). The gain block appears as a positive gain term, while all other elements are losses; the totals at the bottom of the table give the net gain of the complete path.

Tab. 6. LO path budget from the PLL output to one ALTUS28TR LO input.

Path Element	Best (dB)	Typ (dB)	Worst (dB)
Auxiliary tap (resistive splitter)	-5.7	-6.0	-6.5
50 Ω GCPW, 20 mm	-0.15	-0.17	-0.23
PE42520 SPDT switch	-0.8	-1.0	-1.2
50 Ω GCPW, 20 mm	-0.15	-0.17	-0.23
CMX90G702QF gain block	+10.3	+9.8	+9.3
50 Ω GCPW, 20 mm	-0.15	-0.17	-0.23
EP2RKU+ two-way resistive splitter	-5.7	-6.0	-6.5
50 Ω GCPW, 20 mm	-0.15	-0.17	-0.23
Net gain	**-2.50**	**-3.88**	**-5.82**

For the budget calculation, the PLL output is taken at its specified worst case of

−3 dBm; in practice the device may deliver up to 3 dBm, which would simply add margin on top of the values below. Under this conservative assumption, the estimated LO power at each ALTUS28TR input is between −8.82 dBm (worst case) and −5.50 dBm (best case), with a typical value near −6.88 dBm. All three values lie inside the −10 dBm to 3 dBm acceptance window of the ALTUS28TR LO input, with approximately 1.2 dB of margin above the lower limit even in the worst case. Since this margin is calculated against the lowest specified PLL output, any actual PLL level above −3 dBm translates one-to-one into additional margin at the ALTUS28TR inputs. The gain block is essential to the chain. Without it, the path would have a net loss of 13.7 dB (typ.), driving the LO at the ALTUS28TR inputs to about −16.7 dBm for a −3 dBm PLL output, well below the specified range and leaving no margin for component or supply variation.

4.4.5 LO Routing and Layout Considerations

Because the LO network operates in the 8 GHz to 11 GHz range and feeds two coherent transceiver branches from a single source, its physical implementation is as important as the schematic-level topology. The source-selection switch, the gain block, and the splitter are placed close together so that the full chain fits in a compact region of the board, which minimizes interconnect length and simplifies shielding from adjacent digital and control circuitry.

From the splitter onward, the two outputs are routed to the ALTUS28TR LO inputs as GCPW with identical geometry and via fencing, matched in length and bend pattern so that the branch-to-branch phase offset stays small. The corresponding PCB layout is shown in Fig. 33.

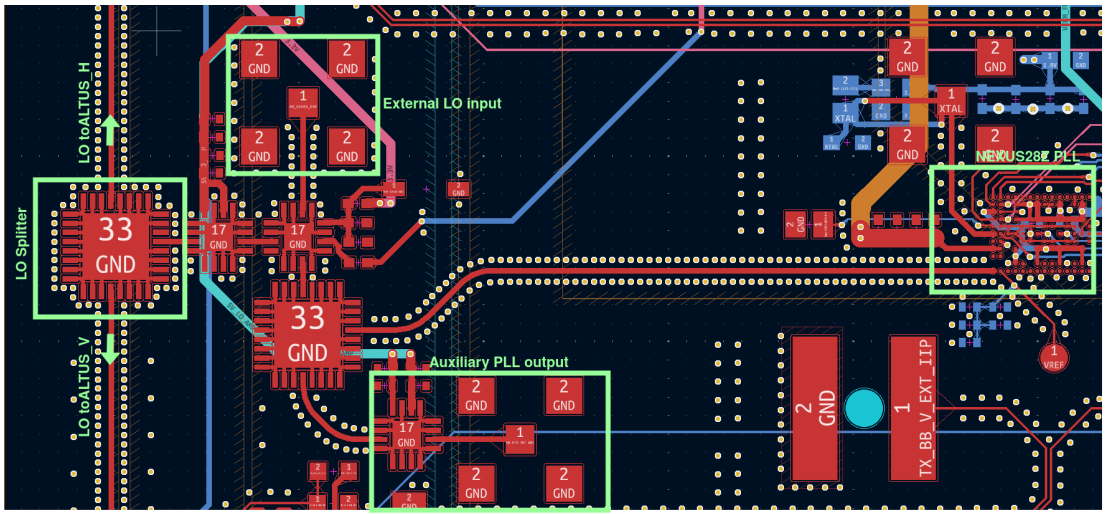


Fig. 33. PCB layout of the LO distribution region, showing the selected-source routing, the gain block, the two-way splitter, and the symmetric fan-out toward the two ALTUS28TR LO inputs.

4.5 Digital Control and Interface Subsystem

The digital control subsystem is the bridge between the host-side controller and the mixed-voltage devices on the board. It handles SPI communication with the ALTUS28TR and NEXUS28Z, carries the GPIO-level control signals for the RF switch and the LO source selection, and adapts logic levels between the host-side domain and each device-specific interface.

The host-side control domain runs at 3.3 V, matching both the FT232H USB controller and the external Cheetah header. The 3.3 V rail therefore acts as the primary digital reference of the board, and all lower-voltage device interfaces are derived from it through level-shifting stages.

4.5.1 Host-Side Control Architecture

Two control paths are provided. The first is a micro-USB interface driving an on-board FT232H, which generates the SPI and GPIO signals used by the board-level control network. The second is an external header that allows a Cheetah SPI host to drive the same host-side SPI lines. An isolation header on the board selects which source is active at a time, preventing contention between the two host devices. Both paths feed into the same downstream network, so the board can be controlled either through USB during normal operation or through the external header for bring-up and debugging.

The FT232H domain also carries reset and enable lines for the controlled devices, as well as a set of debug signals collected on a dedicated debug header for bring-up and probing.

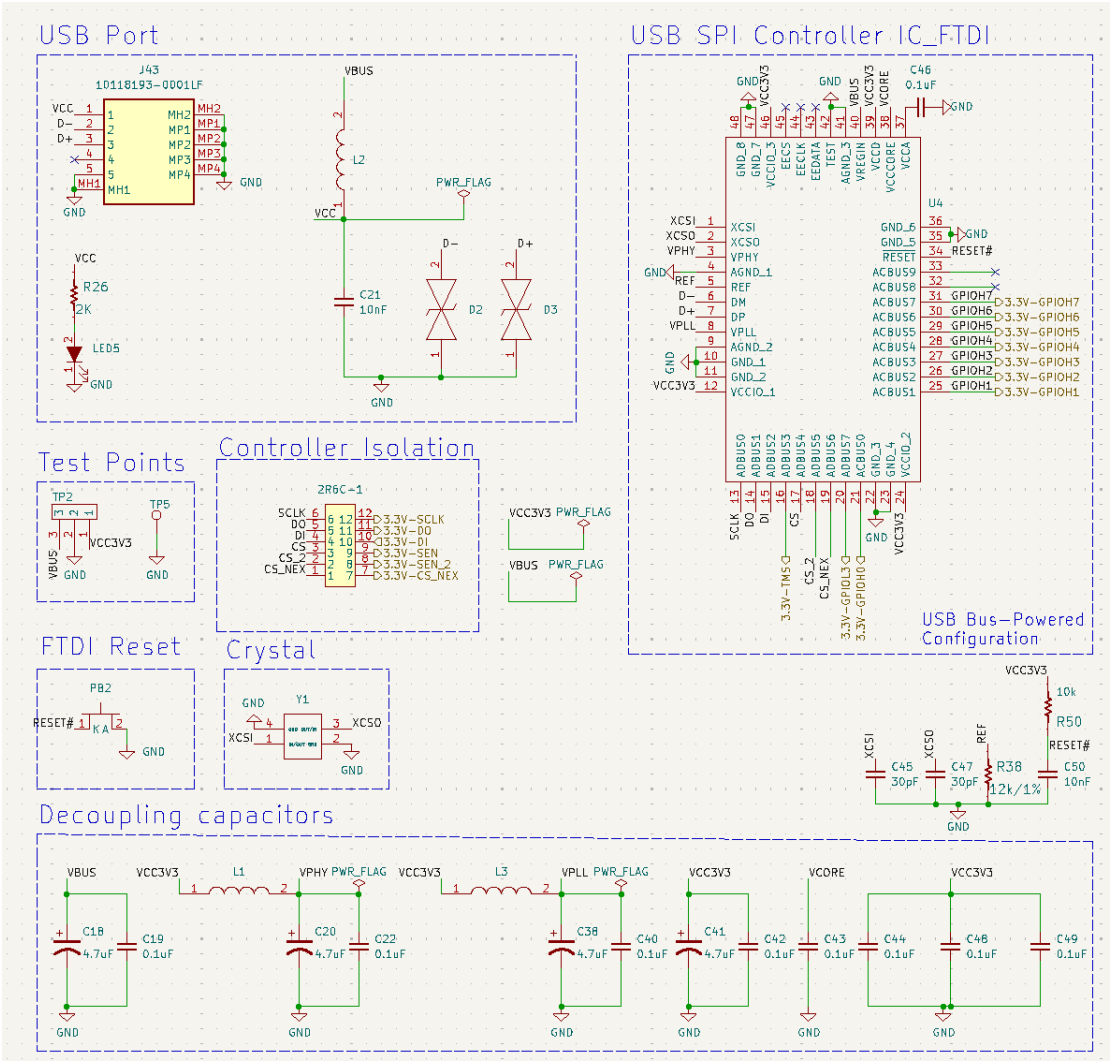


Fig. 34. FT232H-based USB control subsystem, showing the micro-USB interface, the USB-to-SPI controller, the support circuitry, and the 3.3 V host-side signals it generates.

4.5.2 SPI Communication and Logic-Level Translation

The two main controlled device families do not share a common logic level: the ALTUS28TR digital interface operates at 2.6 V, while the NEXUS28Z operates at 1.8 V. Dedicated level-shifting stages are therefore placed between the 3.3 V host-side domain and each device family.

The shifters are built around the TXB0108RGYR, an 8-bit bidirectional voltage-level translator [28]. The TXB0108 detects the signal direction automatically on each line, which removes the need for separate shifters per direction or for an external direction-control signal. This is useful in the SPI interface because the bus includes signals in both directions: SCLK, MOSI, and $\overline{\text{CS}}$ travel from the host to the device, while MISO returns from the device to the host. Two instances are used: one configured for 3.3 V-to-2.6 V translation feeding the two ALTUS28TR devices, and one for 3.3 V-to-1.8 V feeding the NEXUS28Z. Both instances present the same

3.3 V interface to the host side, so the controller sees a uniform SPI environment regardless of which device is addressed.

At protocol level, two separate SPI buses are used. The two ALTUS28TR devices share one bus, with one dedicated chip-select line per device distinguishing between them. The NEXUS28Z is placed on its own SPI bus, since it operates at a different logic level and is fed through a separate level-shifter. This split keeps each shifter dedicated to a single voltage domain and avoids mixing logic levels on the same MISO line.

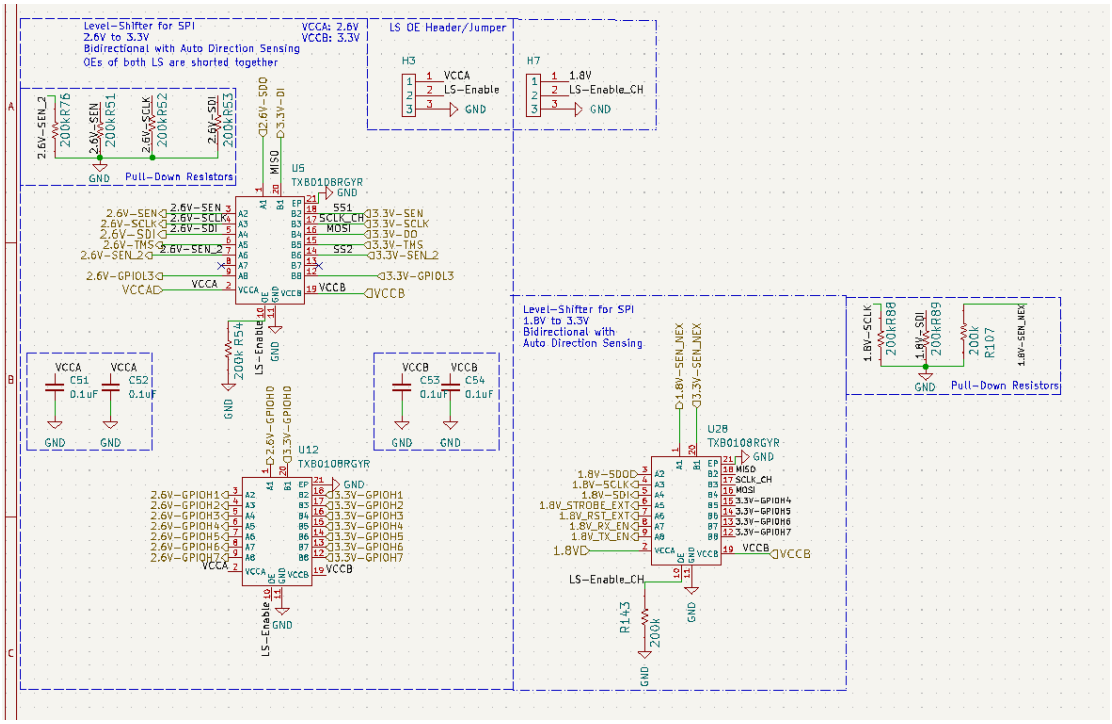


Fig. 35. SPI level-shifting network based on TXB0108RGYR devices, translating 3.3 V host-side signals to 2.6 V for the two ALTUS28TR devices and 1.8 V for the NEXUS28Z.

4.5.3 GPIO and Support Circuitry

Beyond SPI, the digital subsystem also carries the GPIO lines that drive the RF switch in each polarization branch and the LO source-selection switch. These lines run at the same 3.3 V as the host-side SPI, so they can be driven directly from the FT232H or the external header without additional translation.

Pull-down resistors are placed on the four SPI signals (SCLK, MOSI, MISO, $\overline{\text{CS}}$) of both the ALTUS28TR bus and the NEXUS28Z bus, so that all SPI lines settle in a defined low state during power-up and whenever no controller is actively driving them. This keeps the controlled devices in a known idle state before the host enables them, and prevents floating inputs that could otherwise pick up noise and cause spurious SPI transactions. Jumpers on the enable inputs of the TXB0108 instances allow each shifter to be isolated during debugging, which is useful when probing one device at a time or forcing a specific interface into a known state.

On the ALTUS28TR shared bus, the MISO path needs protection against bus contention. The ALTUS28TR does not implement a tri-state MISO output: when the device is not selected, its MISO line remains actively driven instead of going to high impedance. Connecting both ALTUS28TR MISO pins directly to the same net would therefore produce a contention condition whenever one of them is addressed, since the un-addressed device would still be driving its data-out line. To avoid this, an SN74AHC1G126 single-channel tri-state buffer is placed in series with each ALTUS28TR MISO output [29]. The buffer enables follow the corresponding chip-select lines, so only the currently selected ALTUS28TR drives the shared MISO net, while the buffer of the un-addressed device places its output in high impedance. The NEXUS28Z is on its own SPI bus and does not share MISO with any other device, so no buffer is required there. The resulting topology is illustrated in Fig. 36.

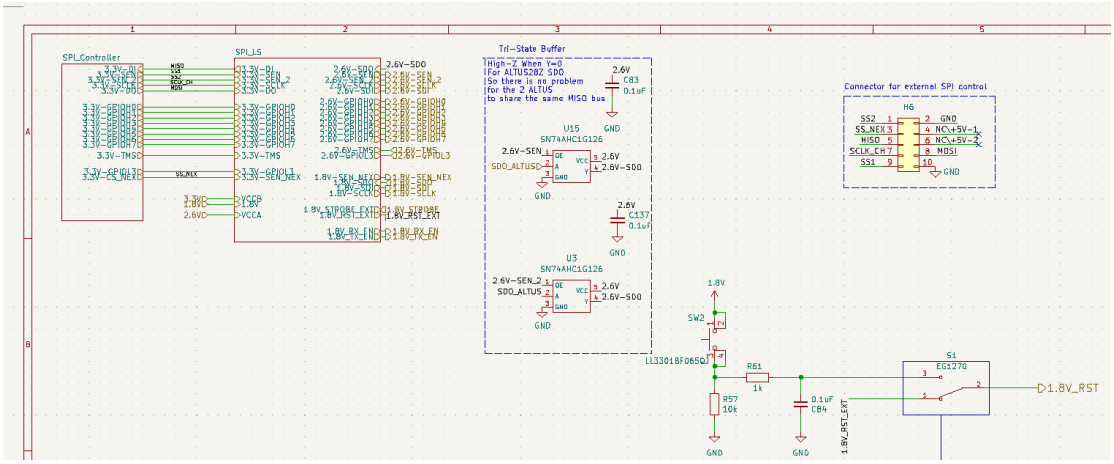


Fig. 36. SPI bus distribution showing the two separate buses (ALTUS28TR shared bus and NEXUS28Z dedicated bus), the external Cheetah header, and the tri-state isolation of the ALTUS28TR MISO paths using SN74AHC1G126 buffers.

4.5.4 Layout and Integration Considerations

The digital control circuitry is grouped in its own region of the PCB, physically separated from the RF and LO routing. Keeping the two domains separated reduces the chance of SPI and GPIO switching edges coupling into the sensitive high-frequency paths, and also simplifies the return-current routing: each domain has a clean local ground reference rather than forcing sensitive RF return currents to share paths with digital switching currents or regulator-current loops. The FT232H, level shifters, tri-state buffers, and associated headers are placed close together so that all SPI and GPIO traces stay short and the digital region remains physically isolated from the analog and RF sections.

4.6 PCB Stackup and Layout Considerations

At 28 GHz to 31 GHz, interconnect behavior is set as much by the PCB stackup as by the schematic. Dielectric loss, layer spacing, trace geometry, and the return-current environment all become first-order design parameters. The stackup was therefore chosen in parallel with the routing strategy, not after it.

The stackup and layout decisions described in this section were implemented in KiCad. Appendix D briefly describes KiCad and summarizes the PCB-design workflow followed here.

Figure 37 shows the complete PCB with its main functional regions marked: RF front-end, LO distribution, baseband interface, digital control, and power delivery. This partitioning is the context for every stackup and routing decision that follows.

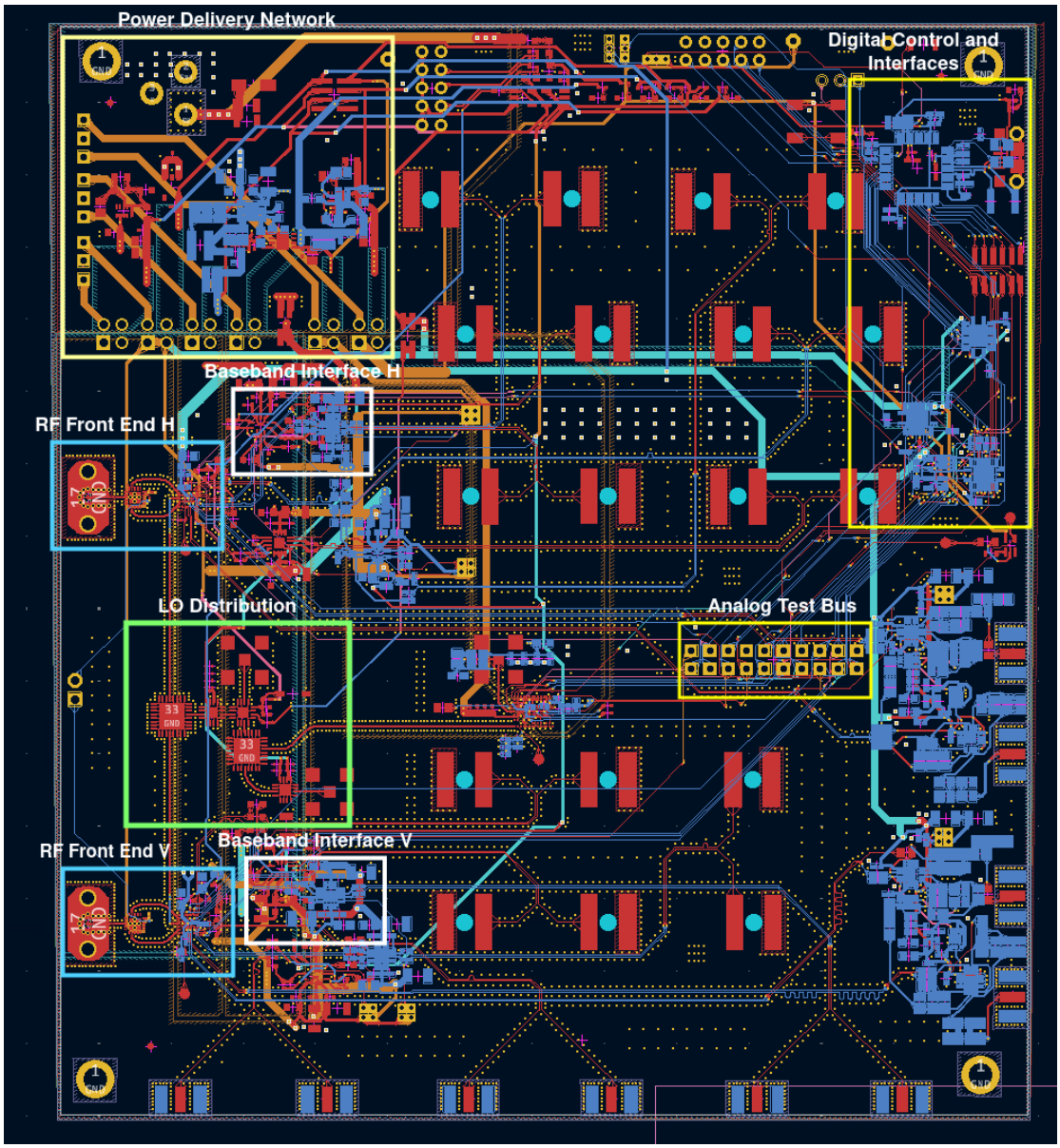
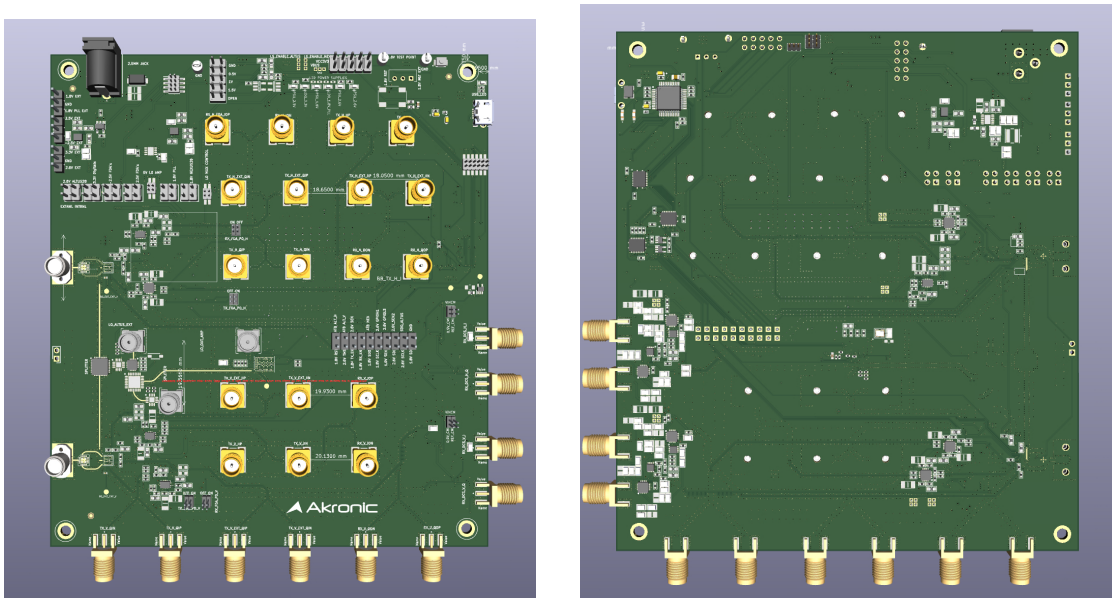


Fig. 37. Annotated full PCB layout of the platform, showing the main functional regions of the implementation, including the RF front-end sections, the LO distribution network, the baseband interfaces, the digital-control area, and the power-delivery subsystem.

A three-dimensional rendering of the same board is shown in Fig. 38. The front and back views illustrate the physical realization of the multilayer platform, the overall placement of the major components, and the relation between the top RF-oriented implementation and the lower support and control regions.



(a) Front-side three-dimensional view.

(b) Back-side three-dimensional view.

Fig. 38. Three-dimensional rendering of the implemented PCB platform, illustrating the physical realization of the multilayer board from the F.Cu (top, RF-side) and B.Cu (bottom, control-side) sides.

The six-layer stackup is shown in Fig. 39. The material selection, layer assignment, and dielectric spacing that define this stackup are discussed in the following subsections.

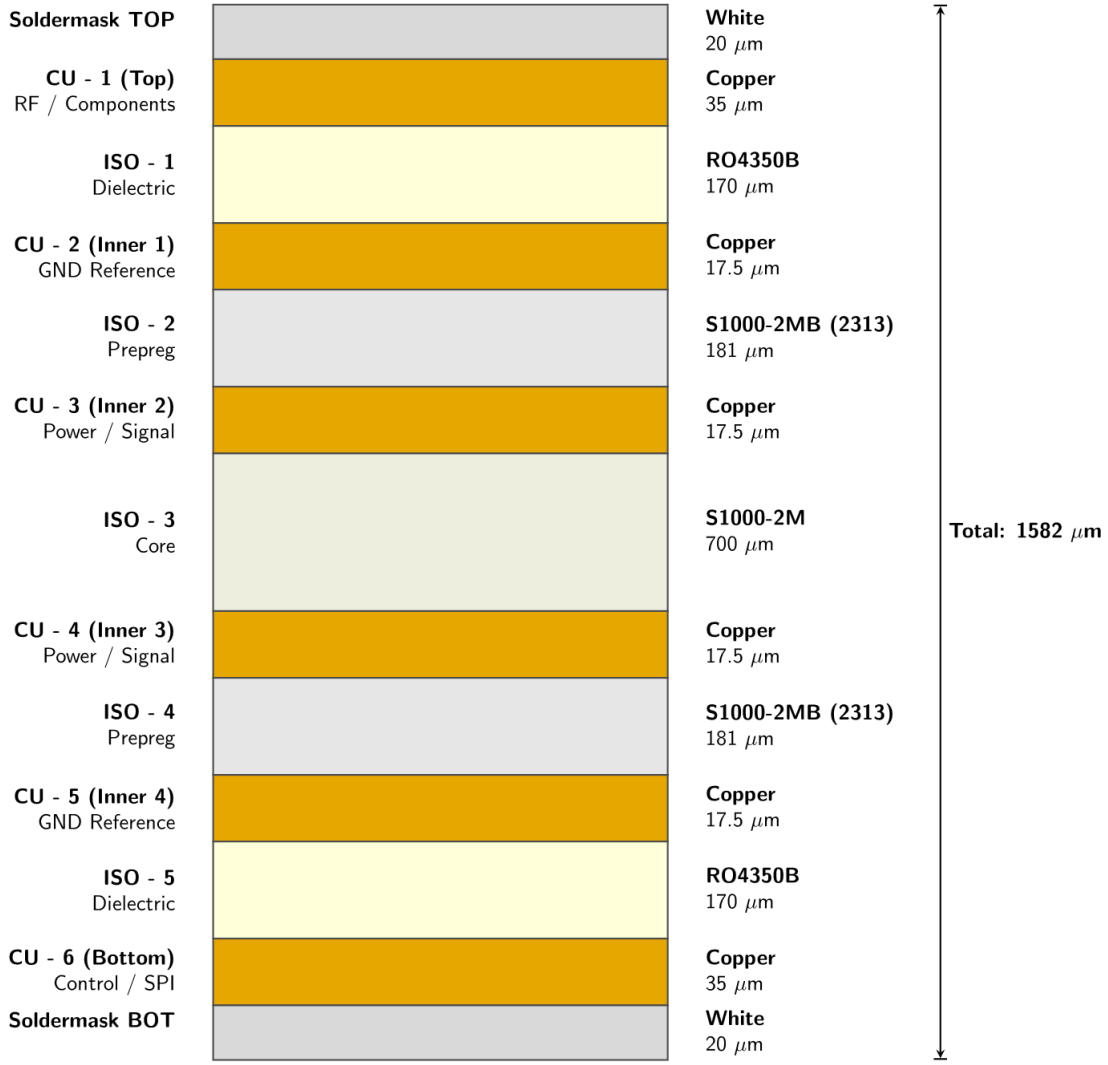


Fig. 39. Implemented six-layer PCB stackup of the platform, showing the layer order, dielectric materials, and the separation between RF, ground, power, and control layers.

4.6.1 Stackup Material Selection

The board uses a six-layer hybrid stackup. The two outer dielectric regions adjacent to the external copper layers are based on RO4350B, a low-loss high-frequency laminate with a typical dielectric constant of $D_k = 3.48 \pm 0.05$ and dissipation factor of approximately 0.0037 at 10 GHz [30]. The internal stackup uses FR-4-class glass-reinforced epoxy materials from the S1000-2M/S1000-2MB family, with S1000-2MB used as the prepreg layers and S1000-2M used as the central core [31]. Both have similar electrical properties at the digital and PDN frequencies of interest, and their combination forms the mechanical and electrical backbone of the multilayer structure. This choice was made because the most critical RF and LO interconnects are routed on the outer layer, where lower dielectric loss and more stable dielectric properties are important for operation in the 28 GHz to 31 GHz range.

Using RO4350B in the outer RF region improves the transmission-line environment of the high-frequency routing and connector-launch areas. In particular, its lower dielectric loss makes it more suitable than conventional FR-4-class materials for microwave and mmWave interconnects. By contrast, the internal layers are used mainly for ground reference, power distribution, and digital/control routing, so implementing the entire board on a uniformly low-loss RF substrate would not provide a proportional benefit.

For this reason, a hybrid stackup was preferred. It provides lower-loss routing conditions where they matter most, while preserving a more practical and cost-effective internal layer structure for PDN, control routing, and mechanical support.

The stackup is also kept thermally and mechanically symmetric: RO4350B is used in both outer dielectric regions, and S1000-2MB and S1000-2M are mirrored on either side of the central core. Symmetric construction is important in hybrid stackups, since the different coefficients of thermal expansion between RO4350B and the FR-4-class internal materials would otherwise produce mechanical warping after reflow soldering.

4.6.2 Layer Assignment and Stackup Rationale

The upper outer copper layer, `F.Cu.RF`, is assigned to the RF- and LO-sensitive routing of the board. Directly beneath it, `In1.Cu.GND` provides an immediate ground-reference plane. This close signal-to-ground arrangement is important for controlled-impedance routing, return-path continuity, and confinement of the electromagnetic fields associated with the high-frequency interconnects.

The internal copper layers `In2.Cu.PWR` and `In3.Cu.PWR` are used primarily for power distribution, whereas `In4.Cu.GND` provides an additional internal ground reference. The lower outer copper layer, `B.Cu.CNTRL`, is used mainly for the digital-control and support circuitry. This layer assignment allows the RF and LO functions to remain on the top high-frequency layer, while the lower layer is used for the lower-speed digital subsystem. In this way, the stackup supports both electrical and functional partitioning of the board.

The dielectric spacing between the top RF layer and the adjacent ground plane was selected so that practical controlled-impedance routing could be achieved using manufacturable trace widths and gaps. In particular, the spacing between `F.Cu.RF` and `In1.Cu.GND` is about 0.17 mm, which supports compact high-frequency routing with a closely coupled return path. The corresponding outer copper thickness is 35 µm, whereas the internal copper layers use 17.5 µm. The internal dielectric sections of 0.181 mm, together with the central 0.70 mm core, provide the required total board thickness and mechanical rigidity while preserving a practical multilayer structure for the internal power-distribution and support layers. The total board thickness is 1.5832 mm.

4.6.3 Controlled-Impedance RF and LO Routing

The RF and LO top-layer interconnects are implemented as grounded coplanar waveguide (GCPW), with coplanar ground on either side of the signal trace and the `In1.Cu.GND` plane as the lower reference. GCPW was chosen over conventional microstrip because it confines the field more tightly and gives a better-controlled return path at mmWave frequencies, where the edge fields of a microstrip line become harder to manage.

On this stackup, the selected geometry used a trace width of 0.30 mm and a lateral ground clearance of 0.10 mm, with dielectric height $H = 0.17$ mm and copper thickness $T = 0.035$ mm. This geometry was selected to realize practical controlled-impedance routing in the 50 Ω class while remaining compatible with the manufacturing limits of the board.

The impedance of the routed geometry was first estimated with the KiCad transmission-line calculator, as shown in Fig. 40.

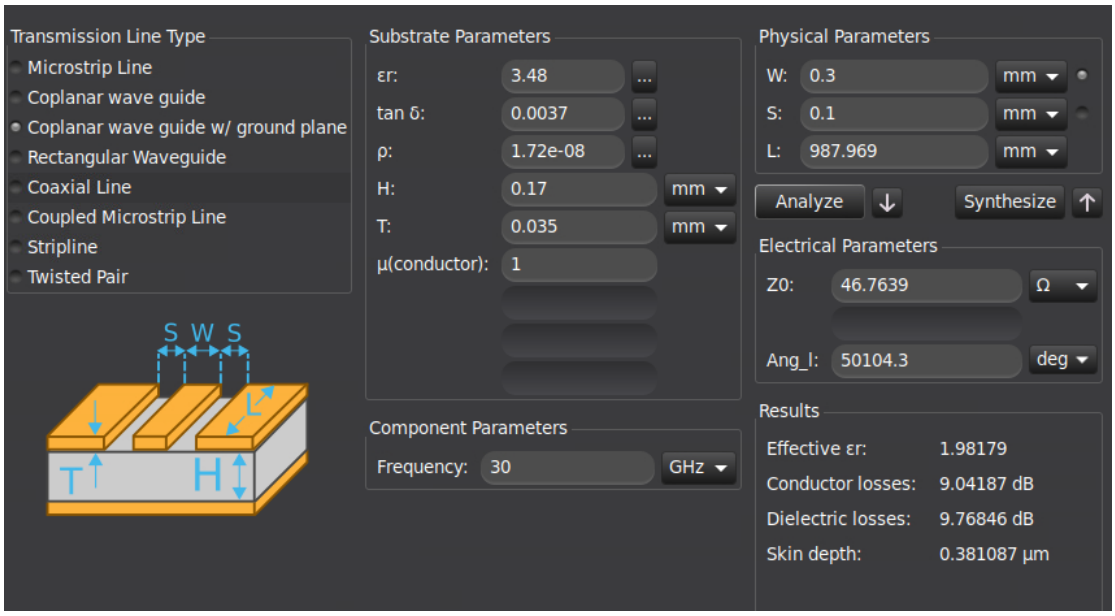


Fig. 40. Estimated characteristic impedance of the implemented top-layer GCPW line geometry.

Over the RF- and LO-critical top-layer traces, the solder mask was locally removed. Solder mask is a lossy dielectric at mmWave, and its thickness and permittivity are poorly controlled compared to the base laminate; opening the mask over these traces keeps the line environment close to that of a bare-copper GCPW with air as the upper dielectric, which improves impedance predictability and reduces dielectric loss in the most critical sections.

The characteristic impedance of a transmission line is given by

$$Z_0 = \sqrt{\frac{L'}{C'}}, \tag{33}$$

where L' and C' are the per-unit-length inductance and capacitance of the implemented structure. Both quantities depend on the layer stackup, trace width, copper thickness, and distance to the reference plane. For this reason, the RF and LO traces cannot be treated independently of the surrounding PCB structure, since their final behavior is determined by the combination of line geometry and stackup.

The guided wavelength of the corresponding high-frequency interconnects is

$$\lambda_g = \frac{c}{f\sqrt{\varepsilon_{\text{eff}}}}, \tag{34}$$

where c is the speed of light, f is the operating frequency, and ε_{eff} is the effective dielectric constant seen by the routed line. At mmWave frequencies, even relatively short discontinuities and path-length mismatches correspond to non-negligible phase shifts. This is one of the main reasons why the relevant RF and LO paths were kept short and compact.

4.6.4 High-Frequency Loss Considerations

A useful first-order approximation for dielectric attenuation in a low-loss transmission-line medium is

$$\alpha_d \approx \frac{\beta \tan\delta}{2} = \frac{\pi f \sqrt{\varepsilon_r}\tan\delta}{c}, \tag{35}$$

where f is the operating frequency, ε_r is the dielectric constant, and $\tan\delta$ is the dielectric loss tangent. This expression is only a first-order approximation, but it is sufficient to show why the choice of the outer dielectric material is important in both the RF and LO sections.

For the RO4350B outer dielectric, with $\varepsilon_r \approx 3.48$ and $\tan\delta \approx 0.0037$ at 10 GHz [30], the corresponding dielectric attenuation is on the order of

$$\alpha_d(10\,\text{GHz}) \approx 0.06\,\text{dB/cm}, \tag{36}$$

and

$$\alpha_d(30\,\text{GHz}) \approx 0.18\,\text{dB/cm}. \tag{37}$$

These values are indicative, not exact, but they illustrate the key design requirement: even at 30 GHz, the dielectric attenuation of the outer RO4350B region stays well below 0.2 dB/cm, which is acceptable for the short top-layer routes used in this design. By contrast, if the RF routing were implemented in the internal S1000-2M/S1000-2MB FR-4-class sections, the higher dissipation factor of these materials

would lead to larger attenuation and would degrade the performance of the interconnects [31].

Conductor loss also becomes increasingly important at these frequencies, since the skin depth decreases according to

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}}, \tag{38}$$

where $\omega = 2\pi f$, μ is the permeability, and σ is the conductivity of copper. For copper, the skin depth is approximately 0.66 µm at 10 GHz and approximately 0.38 µm at 30 GHz. This means that current is confined to a very thin surface region of the conductor, increasing sensitivity to conductor loss and surface-quality effects at the upper end of the operating band. This was taken into account in the layout, where RF/LO trace edges were kept as smooth as possible and conductor-surface effects were considered when evaluating high-frequency loss.

4.6.5 Differential Routing Strategy

The differential analog interconnects of the baseband sections were routed with emphasis on geometric symmetry rather than on the stricter timing rules used for high-speed digital links. The selected geometry uses 0.14 mm traces with 0.10 mm spacing between the two branches of each pair, giving a differential impedance close to 100 Ω on this stackup. This impedance is compatible with the split-termination network at the ALTUS28TR side. The routes are kept short, length-matched, and exposed to a similar surrounding EM environment, to reduce skew, phase imbalance, and differential-to-common-mode conversion.

A path-length mismatch Δl between the two branches produces a time skew

$$\Delta t = \frac{\Delta l}{v_p}, \tag{39}$$

where v_p is the propagation velocity, and a corresponding phase error

$$\Delta\phi = \beta\,\Delta l = \frac{2\pi}{\lambda_g}\,\Delta l. \tag{40}$$

Although these traces do not carry high-speed serial data, the same propagation effects still apply: mismatch turns directly into skew and common-mode conversion. The matching rules were therefore treated as part of the analog layout requirements, not as optional digital-style guidelines.

4.6.6 Grounding, Copper Zones, and Return-Path Control

Ground copper regions were placed alongside the RF and LO traces on the top layer to form the coplanar reference structure of the GCPW routing and to improve field

confinement. These regions were tied to the internal ground reference through stitching vias, which improved return-path continuity and helped maintain a more stable high-frequency reference environment.

The stitching vias were used not only for shielding, but also to maintain return-current control. At high frequency, the return current follows the lowest-impedance path supported by the available reference structure. If discontinuities are introduced in the surrounding ground environment, the return path is disturbed, the effective loop area increases, and the routing environment becomes less controlled. For this reason, the top-layer ground regions and the nearby stitched ground vias were treated as part of the RF/LO routing structure itself, not as a secondary layout feature.

In the power-distribution sections, localized copper regions were also used on the corresponding supply layers to reduce series impedance and improve current delivery. However, large shared copper regions were avoided across unrelated functional domains, particularly near sensitive analog, PLL, RF, and switching-regulator nodes. Instead, copper fills were organized locally according to function, so that the PDN could provide low-impedance delivery without unnecessarily increasing coupling between unrelated domains.

4.6.7 Layer Transitions and Discontinuity Control

As a general design rule, RF and LO layer transitions were minimized so that the most critical high-frequency paths remained on the top RF layer as far as practical. This reduces the number of vertical discontinuities and simplifies preservation of the intended reference environment.

The only high-frequency layer transition of practical significance is in the LO path. To clear an unrelated structure, the LO trace was routed briefly on the `B.Cu.CNTRL` layer to cross under the obstruction, and then returned to `F.Cu.RF`. Each transition between layers was implemented with two parallel vias instead of a single via, which lowers the series inductance of the transition, distributes the current between two paths, and produces a smoother impedance profile across the discontinuity. The complete two-via transition was verified by full-wave simulation, giving S_{11} close to −20 dB and insertion loss below approximately 0.1 dB across the LO band (Section 4.9), and was therefore kept. All other critical RF and LO routes stay on the top layer.

Particular care was also taken at connector launches, bends, and transitions between active devices and transmission-line sections. The corresponding interconnects were kept short and direct to limit discontinuity-induced mismatch and unwanted excess path length.

4.6.8 Functional Layout Partitioning

In addition to the selected stackup, the board layout was organized through functional partitioning of the main subsystems. The RF front-end and the LO distribu-

tion network were placed in the high-frequency region of the board so that the associated interconnects, launches, and splitter-based fan-out paths could remain compact and electrically controlled. This was especially important in the LO network, where unnecessary path-length asymmetry would directly affect branch-to-branch phase consistency.

The PDN was grouped as a localized subsystem, allowing the main regulator stages, ferrite-bead isolation points, and filtered supply branches to be implemented with short interconnects and localized copper regions. This reduced long power-routing paths across unrelated areas of the board and helped keep the switching supply stages away from the more sensitive RF, LO, and low-noise analog regions.

The digital-control circuitry was grouped separately and routed primarily on the lower control layer. In this way, SPI and GPIO switching activity was kept as far as possible from the upper high-frequency region of the board. Although this partitioning cannot eliminate all inter-domain coupling, it reduces unnecessary overlap between RF, LO, PDN, and digital-control routing and leads to a cleaner mixed-signal implementation.

4.7 Simulation and Verification

The PDN of the platform was verified through circuit-level simulations in LTspice. The aim of these simulations was to confirm that the implemented supply architecture behaves consistently with the intended design. The main points examined were nominal rail generation under representative load conditions, startup behavior, steady-state ripple at both the regulator and point-of-load nodes, and the effect of the post-regulation filtering stages introduced to suppress the switching noise of the buck regulators.

The simulation effort focused mainly on the buck-converter-based rails, where switching ripple and post-regulation filtering are the dominant verification concerns. The LDO-based rails were not simulated in detail, since their PSRR and noise behavior are well characterized in the manufacturer datasheets and they feed lower-power or already isolated domains.

A common modeling approach was adopted for the ferrite-bead stages placed after the regulator outputs. No dedicated vendor LTspice model was used for the selected component. Instead, each bead was represented by a simplified equivalent circuit consisting of a small series resistance, a series inductance, and a parasitic capacitance across its terminals. The aim was a first-order evaluation of the bead as a filtering element, not an exact broadband fit of a specific commercial part. This level of detail is sufficient for verifying the post-regulation filtering structure, but does not capture vendor-specific resonances or saturation behavior at high DC bias.

4.7.1 Verification of the 1.8 V Core Supply Rail

The 1.8 V core rail was used to supply the main core circuitry of the NEXUS28Z. This branch therefore provides a useful baseline case for evaluating the effect of the post-regulation filtering strategy under moderate load conditions.

The corresponding LTspice setup is shown in Fig. 41. In addition to the regulator model and the post-regulation filtering network, a pulse current source was applied to emulate changes in load demand and observe the resulting transient and steady-state behavior of the rail.

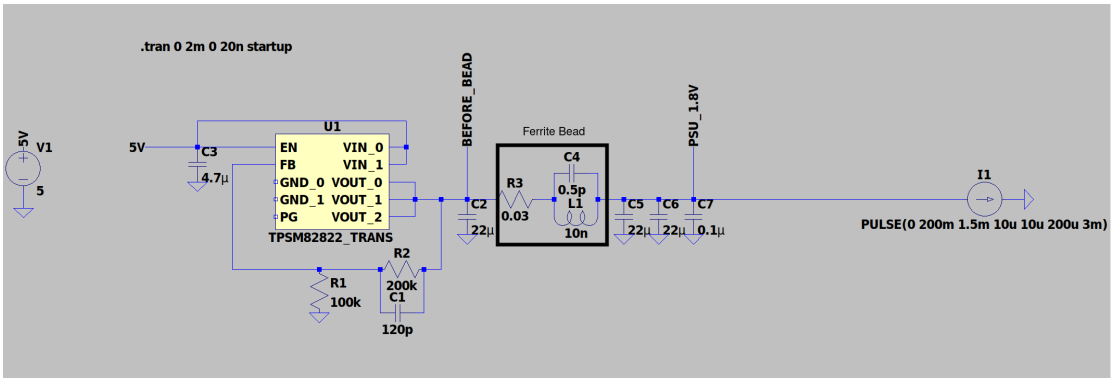


Fig. 41. LTspice simulation setup used for the verification of the 1.8 V core rail.

The transient response is shown in Fig. 42. Both the regulator output node and post filtered load node converge close to the intended operating level.

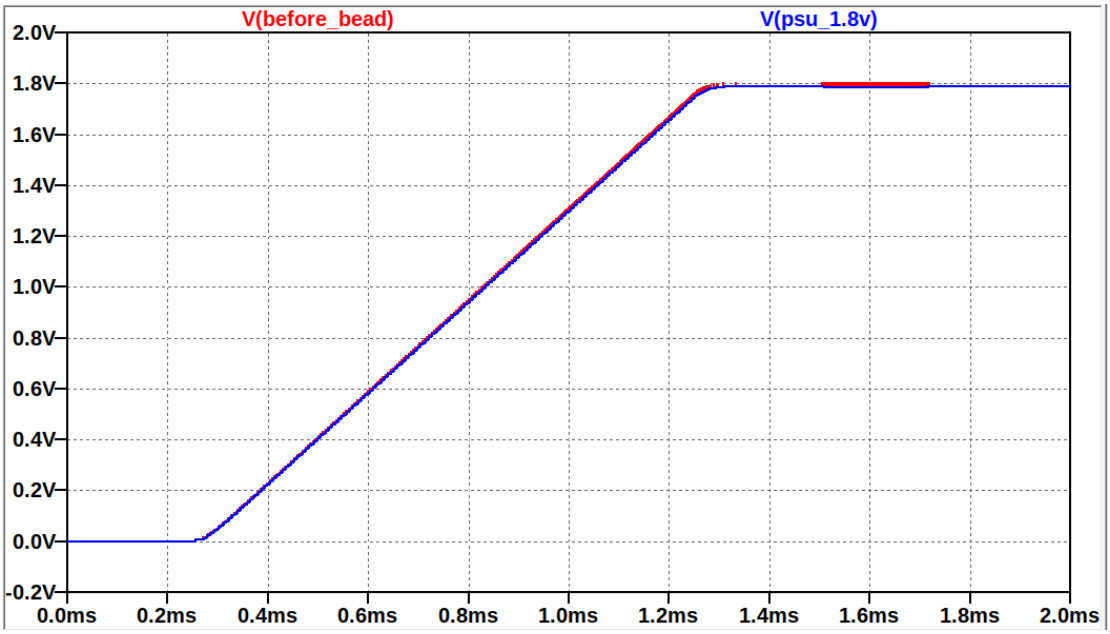


Fig. 42. Transient response of the simulated 1.8 V core rail under the applied load profile.

A steady-state zoom is shown in Fig. 43. The converter-side node carries about

6 mV_{pp} of switching ripple; after the bead this drops to sub-millivolt, with a $\sim$7 mV DC offset across the bead DCR.

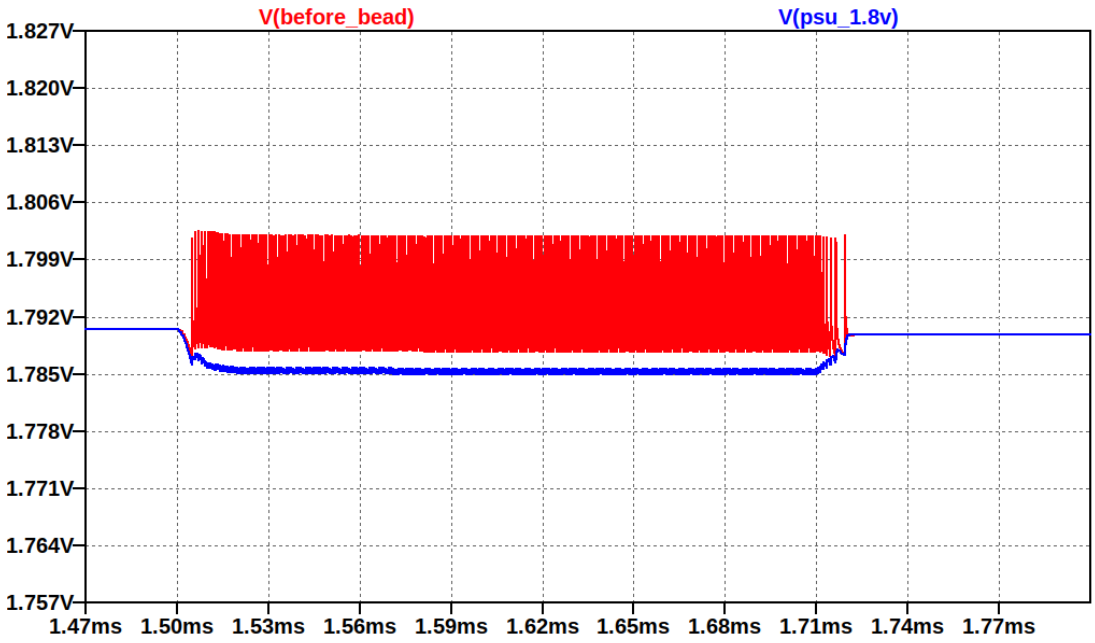


Fig. 43. Steady-state ripple comparison for the 1.8 V rail.

4.7.2 Verification of the +2.5 V FDA Supply Rail

The positive FDA rail was verified next, since it feeds the analog signal-conditioning stages and is therefore more sensitive to supply noise than a purely digital branch.

The corresponding LTspice setup is shown in Fig. 44. As in the previous case, a pulse current source was used to apply representative load variation, here adjusted to the current demands of the $+2.5$ V branch.

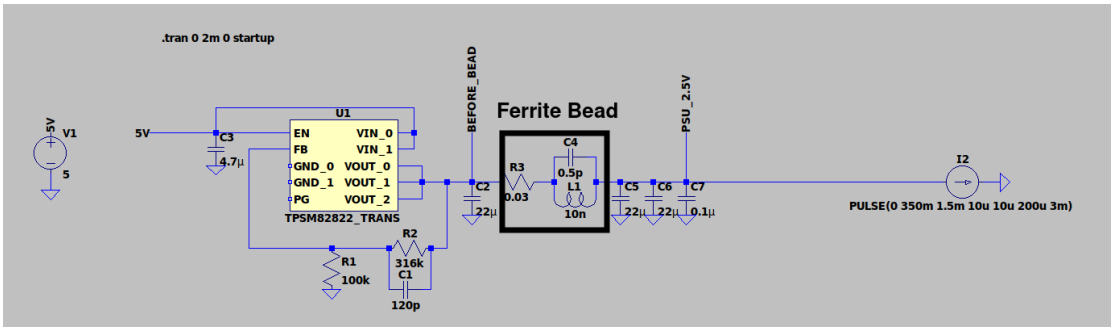


Fig. 44. LTspice simulation setup used for the verification of the $+2.5$ V FDA supply rail.

The transient behavior is shown in Fig. 45. The rail converges close to the required operating value under the applied load step.

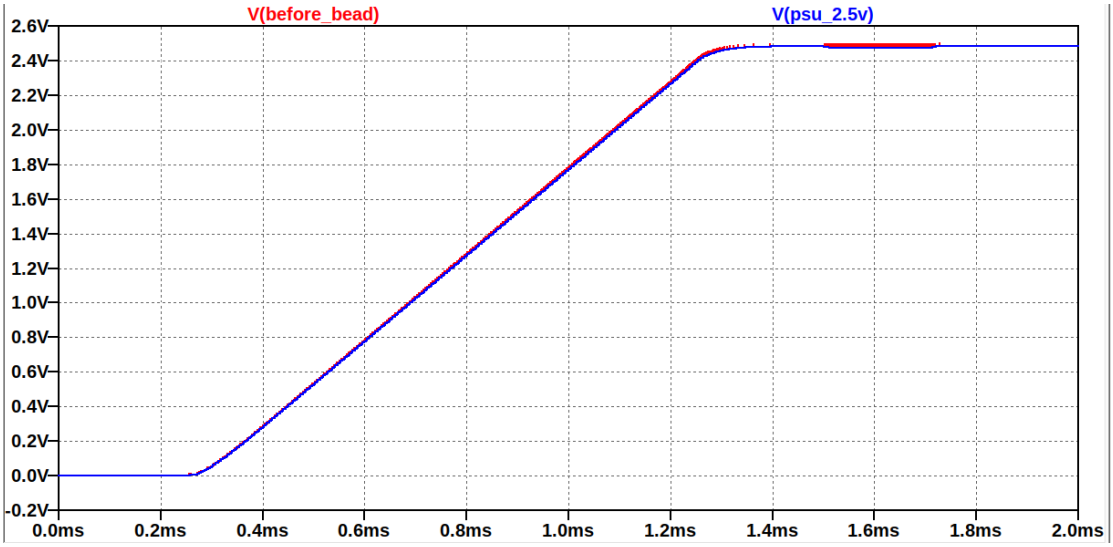


Fig. 45. Transient response of the simulated $+2.5$ V FDA supply rail under the applied load profile.

A steady-state zoom in Fig. 46 shows the same comparison for this rail: $\sim$10 mV_{pp} before the bead, sub-millivolt after it, and a $\sim$9 mV offset. The clean load-side node matters here, since the FDA stage is directly sensitive to local rail noise.

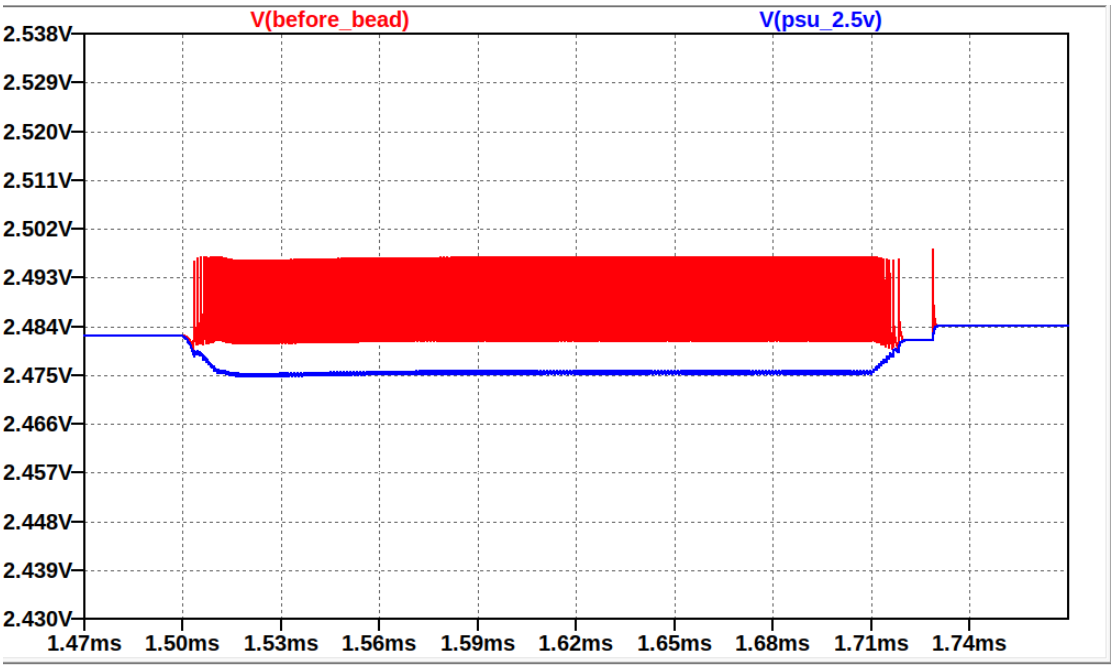


Fig. 46. Steady-state ripple comparison for the $+2.5$ V rail.

4.7.3 Verification of the 2.6 V RF Supply Rail

The 2.6 V rail represents the most demanding positive supply branch of the board, since it feeds the ALTUS28TR transceiver devices and carries the highest current among the main positive rails. For this reason, the rail was simulated under a

representative high-load condition of approximately 0.7 A. This gives a meaningful stress case for the PDN and allows the effect of the local filtering stage to be examined under the most demanding positive-rail operating condition.

The corresponding simulation setup is shown in Fig. 47. It includes the regulator model, the converter-side output capacitor, the ferrite-bead isolation stage, the local decoupling network at the final point-of-load node, and the applied load excitation.

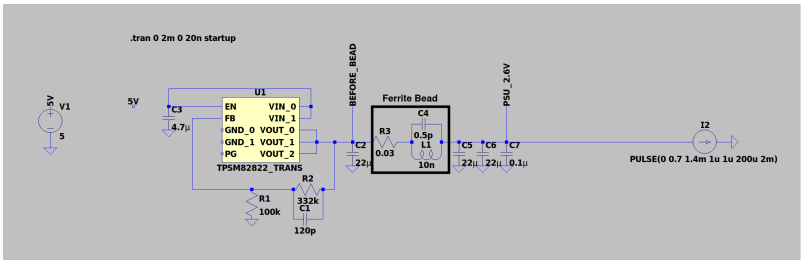


Fig. 47. LTspice simulation setup used for the verification of the 2.6 V RF supply rail.

Fig. 48 shows the transient response under the 0.7 A step; both the converter-side and load nodes hold the nominal 2.6 V level.

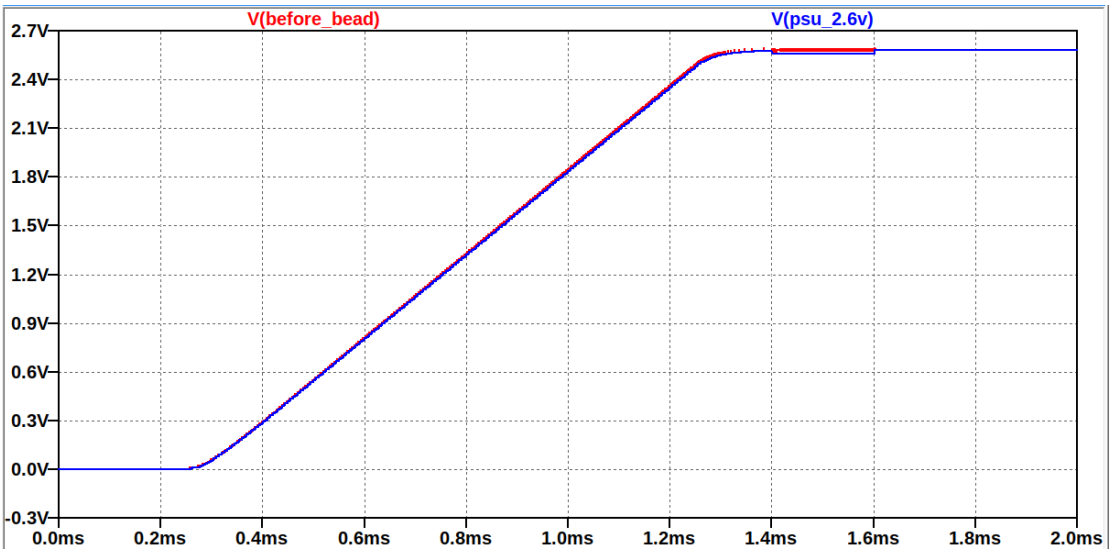


Fig. 48. Transient response of the simulated 2.6 V RF supply rail under high-load conditions.

A steady-state zoom is shown in Fig. 49. The $\sim$10 mV_{pp} converter-side ripple is filtered to sub-millivolt at the load, with a $\sim$20 mV DC offset, the largest of the four rails because the 0.7 A load develops a bigger drop across the bead resistance.

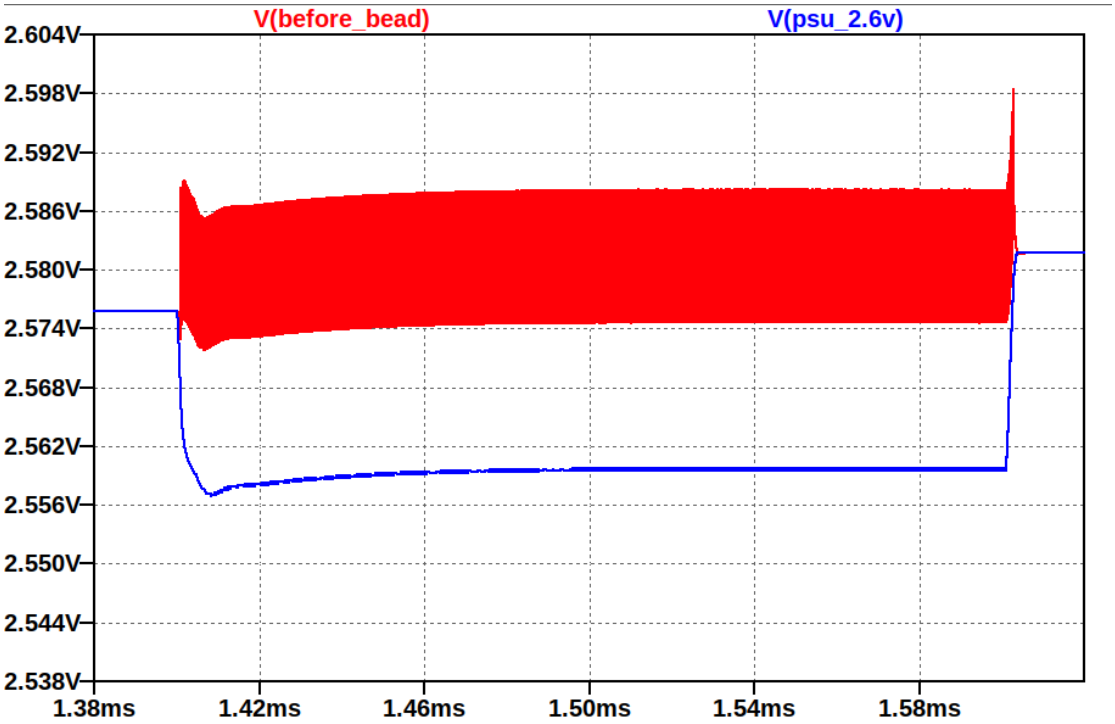


Fig. 49. Steady-state ripple comparison for the 2.6 V RF supply rail.

4.7.4 Frequency-Domain Impedance of the 2.6 V Rail

In addition to transient verification, a small-signal AC analysis was performed to evaluate the output impedance of the 2.6 V branch as a function of frequency. This complements the transient results by describing the supply branch as a frequency-dependent impedance environment seen by the load.

For this purpose, the switching regulator was replaced by an ideal DC source and a unit-amplitude AC current source was injected at the point-of-load node, as shown in Fig. 50. With this setup, the output impedance is obtained directly as $Z(f) = V(f)/I(f)$. Since the AC current amplitude was set equal to unity, the measured AC voltage corresponds numerically to the impedance magnitude.

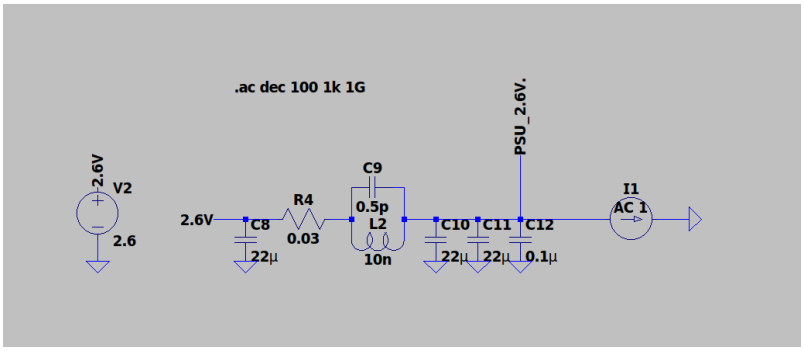


Fig. 50. AC simulation setup used to extract the output impedance of the 2.6 V rail.

The resulting impedance profile is shown in Fig. 51. The impedance remains low across a wide frequency range, with values on the order of tens of milliohms at lower frequencies and only a few milliohms in the MHz region.

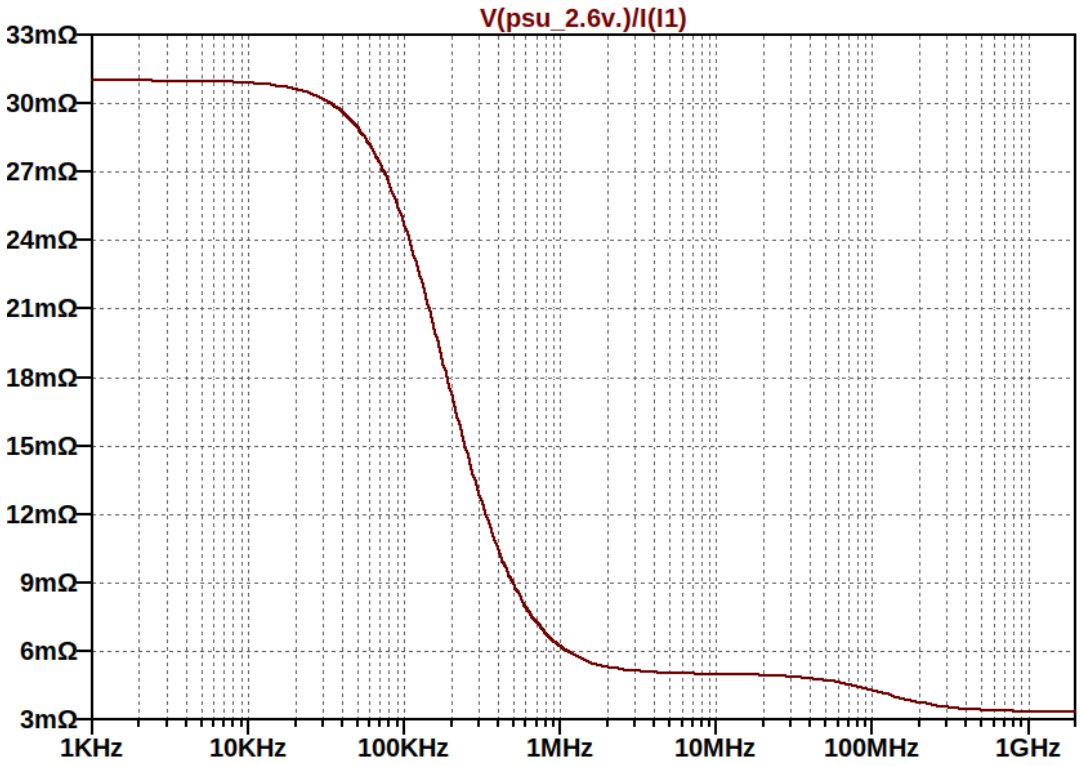


Fig. 51. Simulated output impedance of the 2.6 V rail as a function of frequency.

The ferrite bead together with the local output capacitors forms a secondary LC network between the converter output and the final load node, so the post-bead node is better filtered while keeping a low impedance over the frequency range relevant to the RF load. This agrees with the transient results: the bead-plus-decoupling structure delivers an electrically conditioned local supply node for the 2.6 V branch.

4.7.5 Verification of the -2.5 V FDA Supply Rail

The negative FDA rail required separate verification because it is generated by an inverting converter topology rather than by a conventional positive buck stage. Here, the -2.5 V branch is implemented using the TPS63710, followed by a secondary ferrite-bead filtering stage and a local decoupling network at the final point-of-load node.

The corresponding simulation setup is shown in Fig. 52. In this case, the main regulated negative output node is established directly by the inverting converter, its switching node, its inductor, and the main output capacitors. The ferrite bead is then placed after this regulated node, so that it acts as a secondary filtering stage for the local load supply, not as part of the primary converter loop.

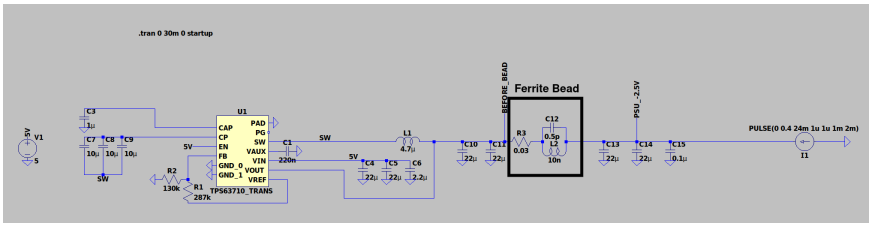


Fig. 52. Transient simulation setup used for the verification of the -2.5 V FDA supply rail.

The startup response in Fig. 53 shows that the output swings from its initial condition down to the −2.5 V target, as expected for the inverting topology.

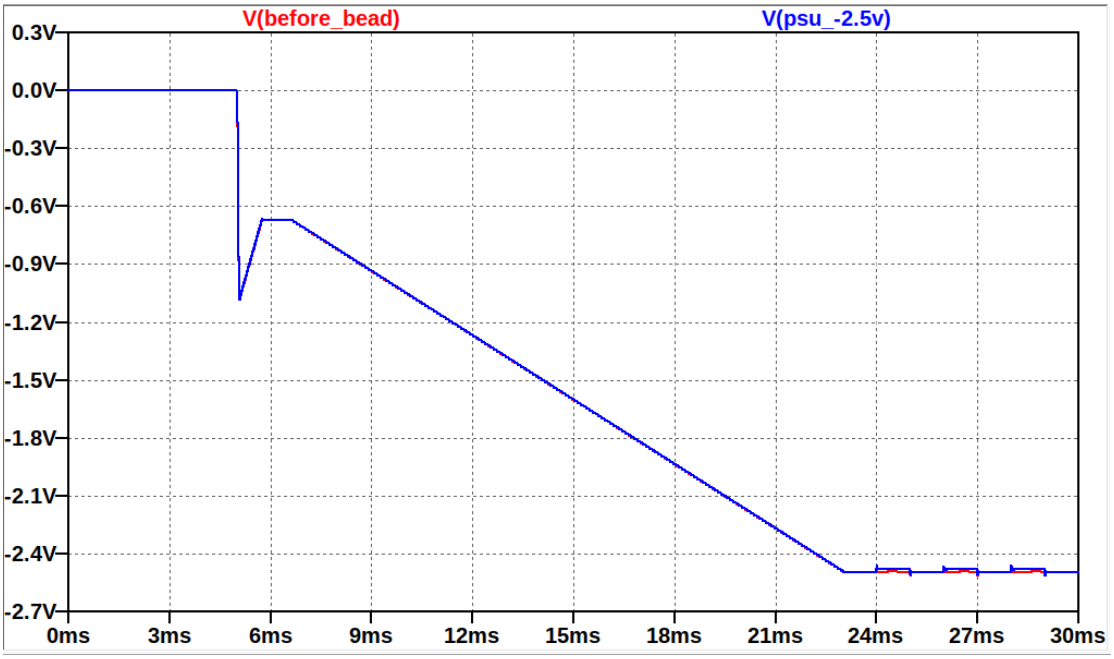


Fig. 53. Startup response of the simulated -2.5 V rail.

Compared with the positive rails, the startup time of the negative branch is slower. This matches the inverting topology behavior and it is not a design fault. The converter must establish a regulated negative output through the switching node, the inductor, and the output capacitors, so its transient evolution differs naturally from that of the positive buck-based rails.

A steady-state zoom of the -2.5 V rail is shown in Fig. 54. The converter-side node carries about $\sim$2 $\mathrm{mV_{pp}}$ of ripple, the lowest of the set, which falls to sub-millivolt after the bead, with a $\sim$12 mV offset across the bead resistance.

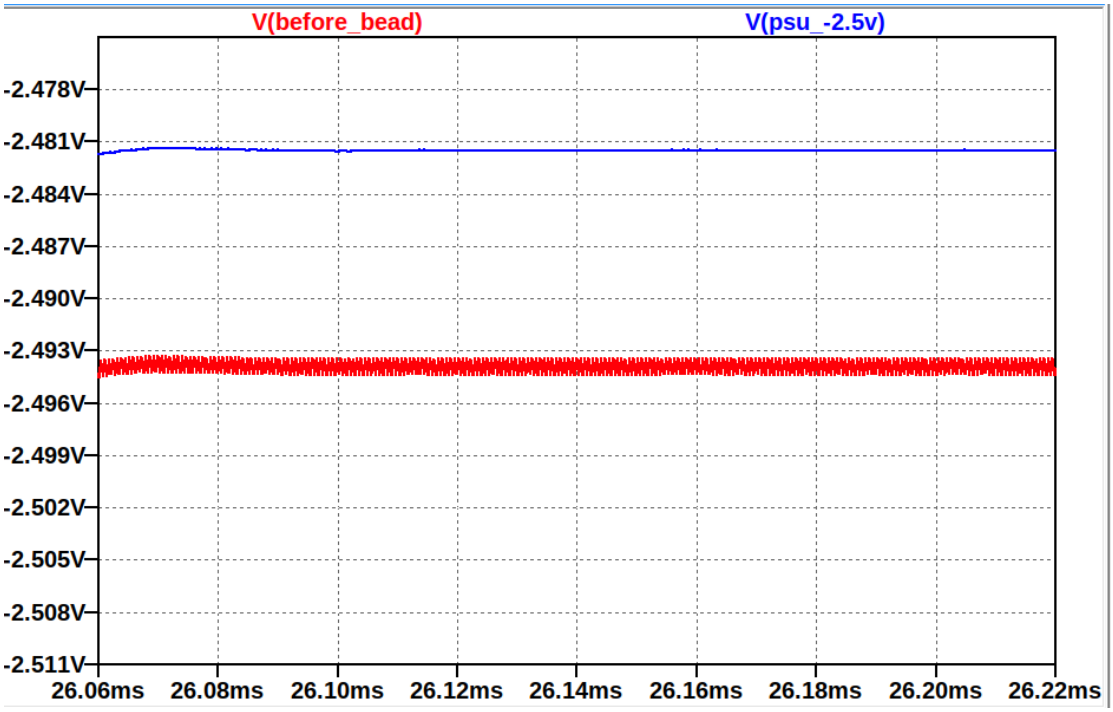


Fig. 54. Transient behavior of the $-2.5\,\mathrm{V}$ rail during load variation, comparing the converter-side node and the final post-bead node.

Despite the inverting topology, the −2.5 V branch behaves like the positive rails: effective high-frequency filtering at the cost of a small, load-dependent DC offset, and a clean post-bead node suitable for the analog load.

4.7.6 TX Baseband Interface Simulation and Impedance Verification

LTspice was used to check the TX baseband interface under the source, biasing, and loading conditions used in the board design. The verification focused on the conversion of the single-ended Red Pitaya DAC signal to a differential FDA output, the output common-mode level, the source-side input impedance, the delivered signal at the PCB interface, and the frequency response of the path.

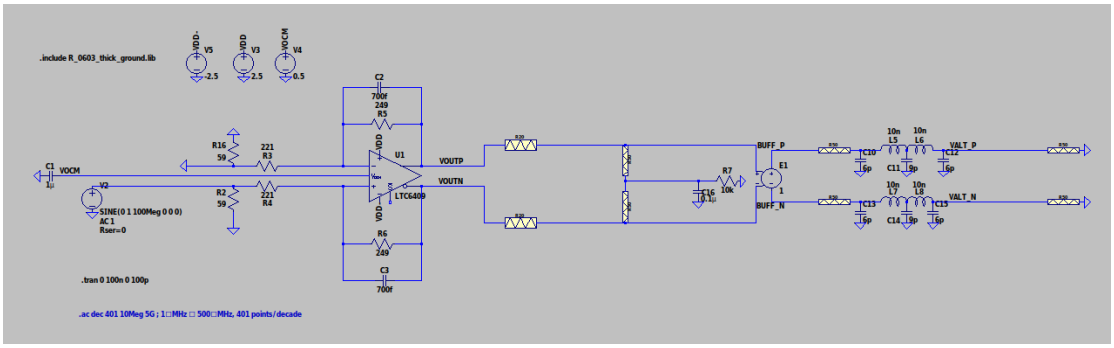


Fig. 55. LTspice setup of the TX FDA interface, showing the LTC6409 stage, the `VOCM` bias network, the input resistor network, the split termination, the series damping resistors, and the simplified transceiver-side continuation.

The LTC6409 [14] fits this interface for three practical reasons. Its 10 GHz gain-bandwidth product is well above the baseband range of interest, it can operate from the available ± 2.5 V FDA rails, and the selected resistor network allows the 2 V_{pp} single-ended Red Pitaya DAC swing to be applied at the source-side port while keeping the FDA input and output nodes inside the modeled linear range.

Fig. 55 separates the external PCB interface from a simple continuation beyond the transceiver input boundary. The external part contains the LTC6409 stage, the input network, the output split termination, and the damping resistors. Since the Red Pitaya output is single-ended, only one FDA input branch is driven. The complementary branch is not driven by a signal source; it is tied to a defined AC return path through the matching input resistor and the local shunt network. The same condition is used in the input-impedance calculation later in this section.

At the PCB boundary, the ALTUS28TR baseband input is treated as a high-impedance node. The differential voltage delivered by the external interface is sensed by the ideal source E_1 and applied to a fifth-order Chebyshev low-pass filter with 0.5 dB ripple and a cutoff near 900 MHz, followed by finite termination. The block is only a finite-bandwidth continuation of the PCB interface, not a detailed model of the ALTUS28TR internal circuitry. The nodes `BUFF_P` and `BUFF_N` are the PCB-side output nodes, while `VALT_P` and `VALT_N` belong to the simplified transceiver-side continuation.

The TX-path resistors were modeled with first-order parasitics instead of ideal lumped resistance. The model is based on published KOA data for 0603 thick-film chip resistors [32], using about 1.1 nH total series inductance and 50 fF parasitic capacitance. In LTspice, each resistor is represented by terminal inductances, a split resistive body, a small inter-terminal capacitance, and weak shunt capacitances to ground. The model is not an exact fit to a specific part, but it gives a more realistic GHz-range response than an ideal resistor.

The time-domain result verifies the voltage-domain operation of the TX stage. The source is driven with a 2 V_{pp} single-ended sinusoid, which corresponds to the nominal full-scale Red Pitaya DAC swing. This uses the available DAC dynamic range before the output network reduces and biases the signal toward the ALTUS28TR TX baseband input.

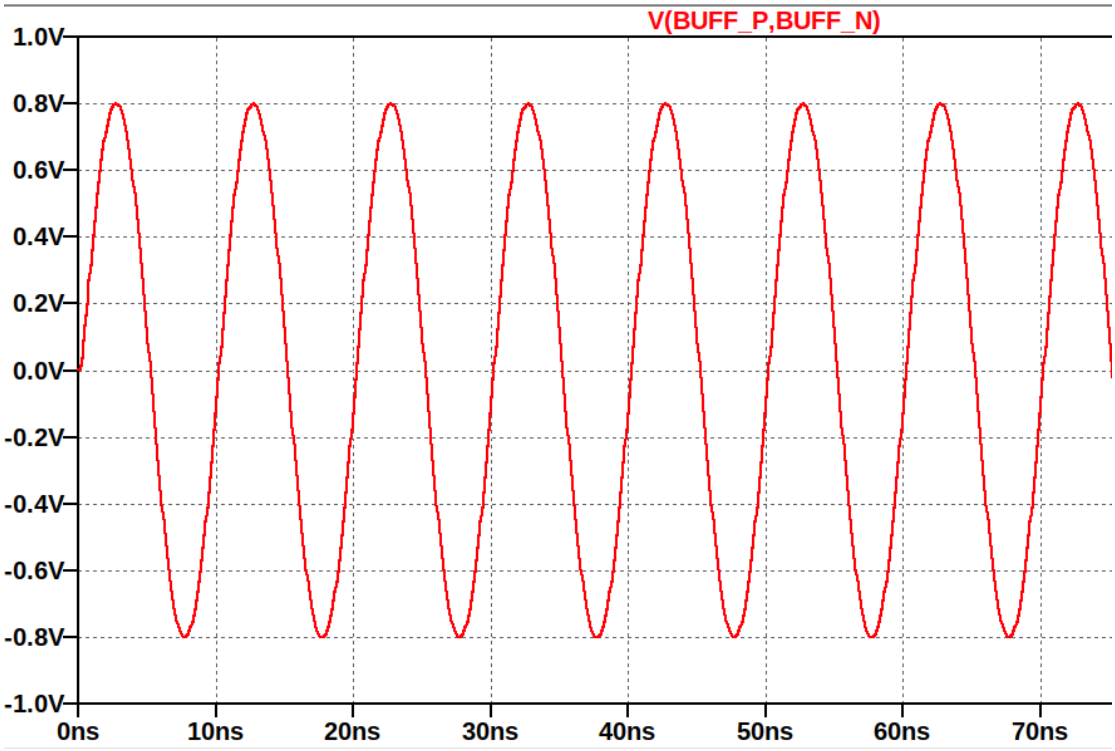


Fig. 56. Differential signal at the buffered TX output node, corresponding to the signal delivered by the external PCB interface.

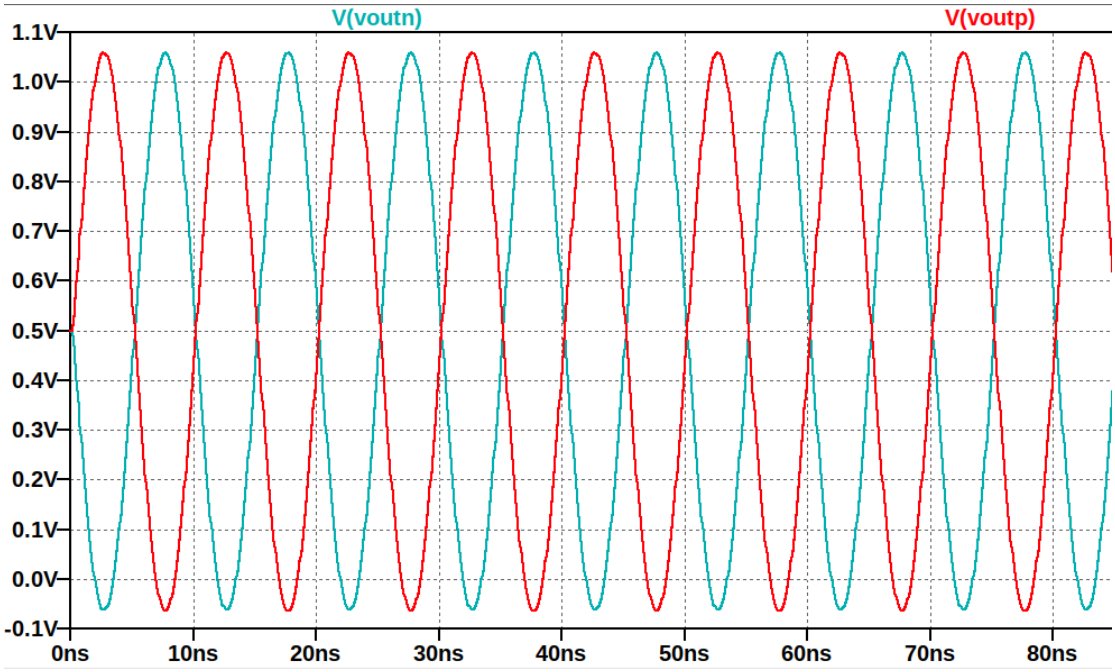


Fig. 57. Voltages at the LTC6409 outputs.

With a 1 V_{peak} single-ended input sinusoid, the FDA outputs settle around a common-mode level of 0.499 V, in line with the applied `VOCM`. The branch amplitudes at the FDA core are 0.562 V_{peak} at `VOUTP` and 0.561 V_{peak} at `VOUTN`. The FDA-core differential output is therefore 1.12 V_{peak}, or 2.24 V_{pp}. The branch mismatch is about 0.1%,

which is small for the target baseband operation.

At the PCB interface output, the two branches remain equal in amplitude and opposite in phase. Each branch is close to 0.401 V_{peak}, or 0.802 V_{pp} single-ended. The delivered differential signal $V(\text{BUFF_P}, \text{BUFF_N})$ reaches 0.802 V_{peak}, or 1.60 V_{pp}. The mismatch at the `BUFF_P`/`BUFF_N` pair is negligible.

Thus, the Red Pitaya signal is converted to a balanced differential pair, the FDA outputs settle at the selected common-mode level, and the PCB-side network reduces the swing without visible branch imbalance.

The same operating point gives the voltage gain of the stage. For the FDA core, the nominal closed-loop differential gain is set by the feedback ratio,

$$A_{v,\text{FDA}} \approx \frac{R_F}{R_G} = \frac{249}{221} \approx 1.127\ \text{V/V}, \tag{41}$$

or about 1.04 dB. The FDA-core differential gain in the simulation is

$$A_{v,\text{core,diff}} \approx \frac{1.122}{1.0} = 1.122\ \text{V/V}, \tag{42}$$

which agrees with Eq. (41). The delivered gain from the single-ended source to the PCB-side differential output is lower,

$$A_{v,\text{PCB,diff}} \approx \frac{0.802}{1.0} = 0.802\ \text{V/V}, \tag{43}$$

or about −1.92 dB. The lower delivered gain comes from the output network, since the split termination and the series damping resistors reduce the swing before the signal reaches the PCB interface output.

The frequency response of the same path is shown in Fig. 58.

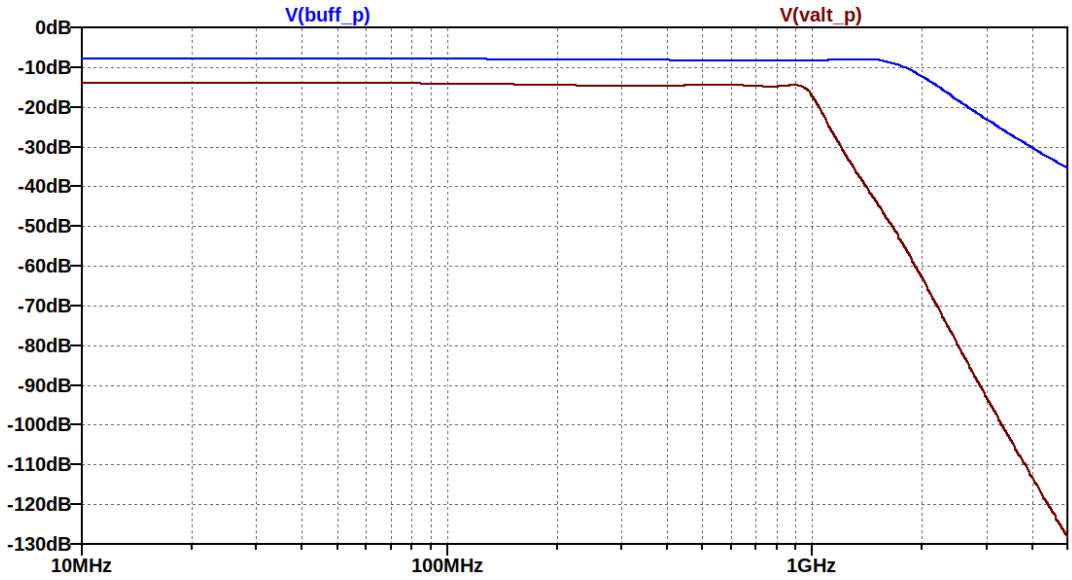


Fig. 58. AC response of the TX FDA interface, including the external FDA stage, the non-ideal resistor models, the PCB launch network, and the simplified transceiver-side continuation.

At low frequency, the AC magnitudes match the time-domain values. The FDA core branch is about 0.562 V/V at VOUTP, while the PCB-side branch is about 0.401 V/V at BUFF_P. The corresponding differential magnitudes are 1.12 V/V at the FDA core and 0.803 V/V at the PCB interface output. The FDA core sets the active gain, while the PCB-side output network sets the final delivered swing.

The extracted -3 dB points are listed in Table 7.

Tab. 7. Representative -3 dB frequencies of the simulated TX interface.

Response	-3 dB frequency
Buffered PCB-side branch, V(BUFF_P)	1.8 GHz
PCB output plus simplified transceiver-side continuation	1 GHz

The high-frequency dip and reshaping in Fig. 58 do not indicate FDA instability. They come from the combined response of the simplified transceiver-side filter, the finite amplifier bandwidth, the resistor network, and the first-order resistor parasitic models. The two -3 dB values refer to different observation points. The buffered PCB-side branch V(BUFF_P) rolls off near 1.8 GHz, while the response including the simplified transceiver-side continuation rolls off near 1 GHz because of the added low-pass filter. Both frequencies are well above the baseband bandwidth used in this work. The Red Pitaya DAC runs at 125 MS/s, giving a Nyquist limit of 62.5 MHz for directly generated waveforms, and the maximum ALTUS28TR transmit baseband filter setting is 720 MHz.

The Red Pitaya DAC output drives a 50 Ω load, so the FDA input network was sized around this condition. Let V_{in} denote the source-side input voltage. The driven branch contains

$$R_{\text{in}} = 221\,\Omega. \tag{44}$$

The complementary branch is not actively driven in this simulation. Its small-signal return path is represented by the corresponding 221 Ω input resistor and the local 59 Ω shunt,

$$R_{\text{gnd}} = 221\,\Omega + 59\,\Omega = 280\,\Omega. \tag{45}$$

The feedback resistance in each FDA loop is

$$R_F = 249\,\Omega. \tag{46}$$

Under the ideal FDA approximation, the two input nodes sit at the same small-signal voltage V_x, and the two outputs are equal in magnitude and opposite in phase. KCL at the grounded branch gives

$$\frac{V_x}{R_{\text{gnd}}} + \frac{V_x - V_{\text{out}-}}{R_F} = 0, \tag{47}$$

so

$$V_{\text{out}-} = V_x \left(1 + \frac{R_F}{R_{\text{gnd}}}\right). \tag{48}$$

For a balanced differential output,

$$V_{\text{out}+} = -V_x \left(1 + \frac{R_F}{R_{\text{gnd}}}\right). \tag{49}$$

KCL at the driven branch gives

$$\frac{V_x - V_{\text{in}}}{R_{\text{in}}} + \frac{V_x - V_{\text{out}+}}{R_F} = 0. \tag{50}$$

Substitution gives

$$\frac{V_x - V_{\text{in}}}{R_{\text{in}}} + \frac{V_x + V_x \left(1 + \dfrac{R_F}{R_{\text{gnd}}}\right)}{R_F} = 0. \tag{51}$$

After rearrangement, the equivalent impedance looking into the FDA-side network is

$$Z_{\mathrm{FDA}} = R_{\mathrm{in}} + \left(\frac{R_F}{2} \parallel R_{\mathrm{gnd}} \right). \tag{52}$$

With the selected resistor values,

$$\frac{R_F}{2} = \frac{249}{2} = 124.5\,\Omega, \tag{53}$$

$$\left(\frac{R_F}{2} \parallel R_{\mathrm{gnd}} \right) = (124.5\,\Omega \parallel 280\,\Omega) \approx 86.18\,\Omega, \tag{54}$$

and therefore

$$Z_{\mathrm{FDA}} = 221\,\Omega + 86.18\,\Omega \approx 307\,\Omega. \tag{55}$$

The Red Pitaya does not see this FDA-side value directly, because the driven-side 59 Ω shunt is connected at the source node. The source-side input impedance is

$$Z_{\mathrm{in,tot}} = 59\,\Omega \parallel 307\,\Omega \approx 49\,\Omega. \tag{56}$$

This is the relevant value for the Red Pitaya DAC, since it is the load seen at the source-side input port.

The source-side impedance was also extracted directly in LTspice. The original Red Pitaya voltage source was replaced by a unit-amplitude AC current source at the input node, and the output-side differential test source used for the output-impedance check was not present in this run. With the injected current set to unity, the measured input-node voltage gives

$$Z_{\mathrm{in}}(f) = \frac{V_{\mathrm{in}}(f)}{I_{\mathrm{test}}(f)}. \tag{57}$$

The simulated input impedance remains close to 50 Ω over the frequency range of interest.

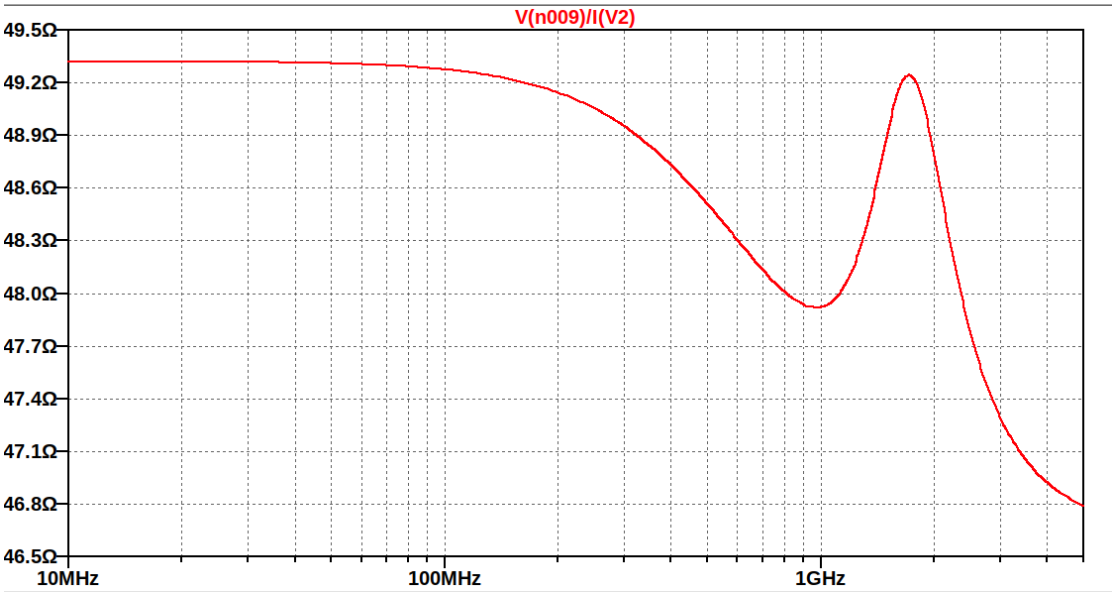


Fig. 59. Simulated input impedance of the TX FDA interface as seen by the single-ended Red Pitaya source.

At the PCB-side launch node, two 50 Ω resistors form the split termination,

$$Z_{\text{diff,split}} = 50 + 50 = 100\,\Omega. \tag{58}$$

This value describes the split network itself. The LTC6409 output pins do not see only 100 Ω, because the two 20 Ω series resistors and the downstream continuation are also part of the load presented to the amplifier.

The midpoint of the split termination sets the common-mode reference for the launched differential signal,

$$V_{\text{CM}} = \frac{V_{\text{OUT+}} + V_{\text{OUT−}}}{2}. \tag{59}$$

During normal FDA-driven operation, the FDA output common-mode is still controlled by the `VOCM` pin. The split midpoint defines the common-mode environment at the PCB-side interface and becomes useful in the alternative current-injection mode.

The 20 Ω series resistors at the FDA outputs are damping elements, not matching elements. They reduce ringing from the interaction between the high-bandwidth FDA output, PCB trace inductance, resistor parasitics, and the capacitive environment downstream. Their role is time-domain signal integrity, not strict impedance matching.

The split network also allows a current-injection test mode without changing the board topology. If the FDA stage is omitted, a differential signal can be generated by injecting equal and opposite currents into the two branches of the split termi-

nation, for example from an external current-output DAC. With each branch driven by a current I_{DAC} into a split resistor R_{split},

$$V_{\mathrm{diff}} \approx 2 I_{\mathrm{DAC}} R_{\mathrm{split}}. \tag{60}$$

For the 50 Ω+50 Ω split structure,

$$V_{\mathrm{diff}} \approx 100 I_{\mathrm{DAC}}. \tag{61}$$

In this mode, the common-mode level can be imposed directly at the midpoint of the split termination from an external bias source.

The differential output impedance was extracted with a small-signal current source between the two output branches,

$$Z_{\mathrm{out,diff}}(f) = \frac{V_{\mathrm{out,diff}}(f)}{I_{\mathrm{test}}(f)}, \tag{62}$$

where $V_{\mathrm{out,diff}} = V(\mathrm{VOUTP}, \mathrm{VOUTN})$. A unit-amplitude AC test current was used, so the measured differential voltage corresponds directly to the output impedance magnitude. The setup and result are shown in Fig. 60 and Fig. 61.

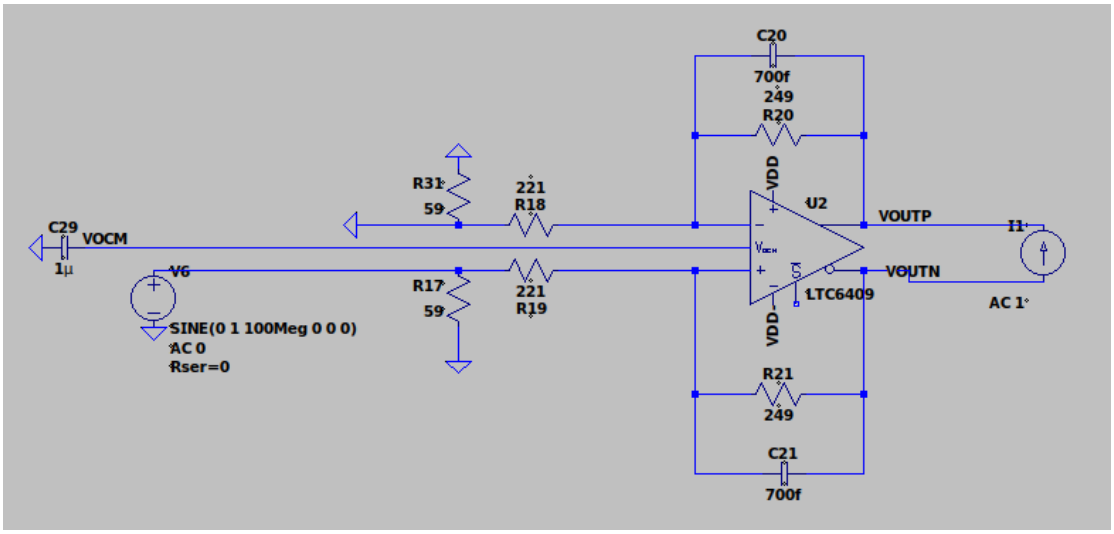


Fig. 60. Simulation setup used for extracting the differential output impedance of the TX FDA interface.

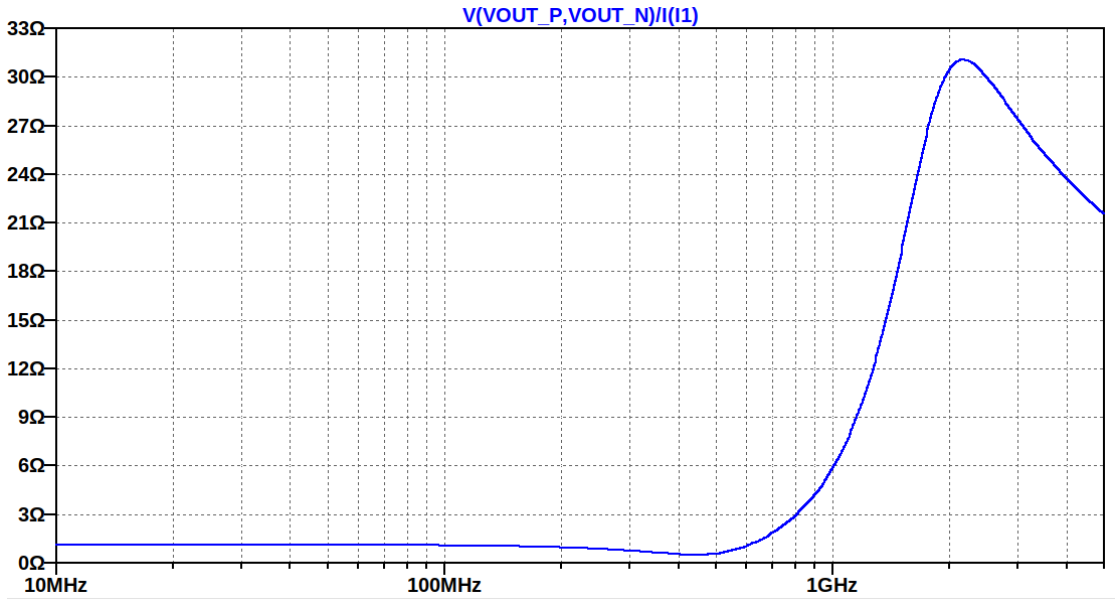


Fig. 61. Simulated differential output impedance of the TX FDA interface.

The differential output impedance stays low at low and intermediate frequencies, where the FDA still has sufficient loop gain. At higher frequencies it rises as loop gain falls and the passive output network becomes more dominant. This rise occurs outside the baseband range used here, so it does not set the practical bandwidth limit of the TX interface.

In the simulated TX configuration, the PCB-side output reaches 1.60 V_{pp} differential with branch mismatch close to 0.1%. The input network presents approximately 49 Ω to the Red Pitaya DAC, matching the target 50 Ω source environment. The buffered PCB-side response rolls off near 1.8 GHz, above the baseband bandwidth used in this work and above the maximum 720 MHz ALTUS28TR transmit baseband filter setting.

4.7.7 RX Baseband Interface Simulation

The RX baseband interface was checked in LTspice using the signal levels and loading conditions of the board design. The aim was to examine the conversion from the differential ALTUS28TR `RX_BUFF` output to a single-ended Red Pitaya ADC signal, the current drawn from the receiver buffer, the output swing at the ADC-side node, and the bandwidth of the ADA4927 stage.

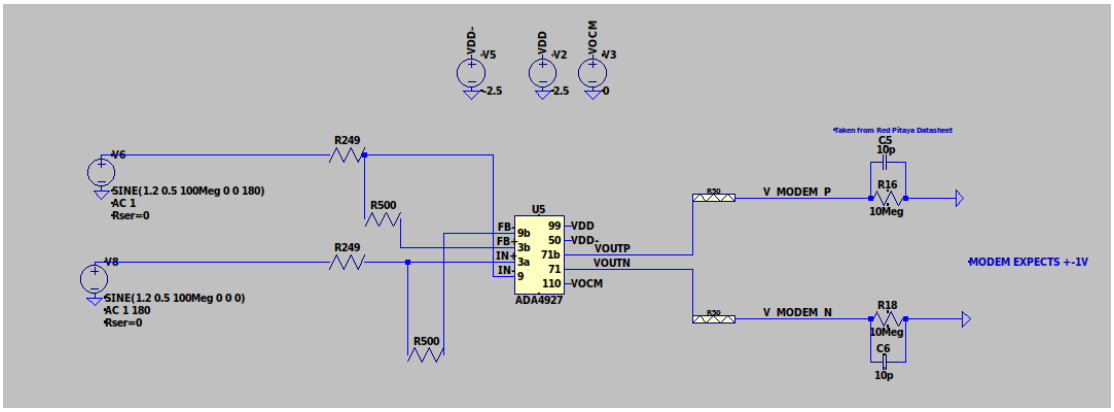


Fig. 62. LTspice setup of the RX FDA interface based on the ADA4927, showing the differential ALTUS28TR receiver-buffer excitation, the feedback network, the Red Pitaya ADC-side output network, and the simplified Red Pitaya input model.

Fig. 62 models the ALTUS28TR `RX_BUFF` output as a differential buffered source centered around its receiver-side common-mode level. The ADA4927 translates this differential signal to a single-ended voltage for the Red Pitaya ADC-side connection. The `VOCM` pin is set near 0 V, so the used output branch is centered around ground and can use the bipolar input range of the Red Pitaya.

Only one FDA output branch is used as the active Red Pitaya ADC drive in this configuration. The complementary output is still kept in the feedback and output network, rather than being left as an undefined node. It also remains available through the corresponding output path, so a future version of the setup could use a differential downstream interface if needed.

The ADA4927 [15] was chosen for this stage because it supports wide bandwidth at moderate closed-loop gain, its input common-mode range is compatible with the 1.2 V common-mode level of the ALTUS28TR `RX_BUFF` output, and its output swing is compatible with the ± 2.5 V FDA supplies used on the board. In the modeled configuration, these conditions allow a ground-centered ADC-side signal close to the 2 V_{pp} input range of the Red Pitaya.

The feedback network uses

$$R_G = 249\,\Omega, \qquad R_F = 499\,\Omega. \tag{63}$$

For the differential FDA core, the nominal closed-loop differential gain is

$$A_{v,\mathrm{diff}} = \frac{R_F}{R_G} = \frac{499}{249} \approx 2.0. \tag{64}$$

Since the Red Pitaya uses one output branch as a single-ended signal, the more relevant gain is the differential-input to single-ended-output gain,

$$A_{v,\mathrm{diff}\to\mathrm{SE}} = \frac{R_F}{2R_G} = \frac{499}{2 \times 249} \approx 1.0. \tag{65}$$

With this scaling, a $2\,V_{pp}$ differential signal from the receiver buffer is translated to about $2\,V_{pp}$ single-ended at the Red Pitaya ADC-side node. This uses the available ADC range without adding unnecessary gain in the receiver interface.

The ADA4927 is a current-feedback fully differential amplifier, so the feedback resistance and load conditions are central to stable operation. No extra compensation capacitors were added around the feedback network. The resistor values used here set the required gain and keep the modeled response well behaved over the receiver bandwidth used by the platform.

The ALTUS28TR receiver output is a buffered source, not a $50\,\Omega$ instrument source. The relevant loading check is therefore the current drawn from the `RX_BUFF` branches. With a nominal branch amplitude of $0.5\,V_{peak}$, the first-order current through the $249\,\Omega$ input resistor is

$$I_{\mathrm{RXBUF,peak}} \approx \frac{0.5}{249} \approx 2.0\,\mathrm{mA}. \tag{66}$$

This level does not heavily load the receiver buffer under the modeled signal conditions.

The time-domain waveform at the Red Pitaya ADC-side node is shown in Fig. 63.

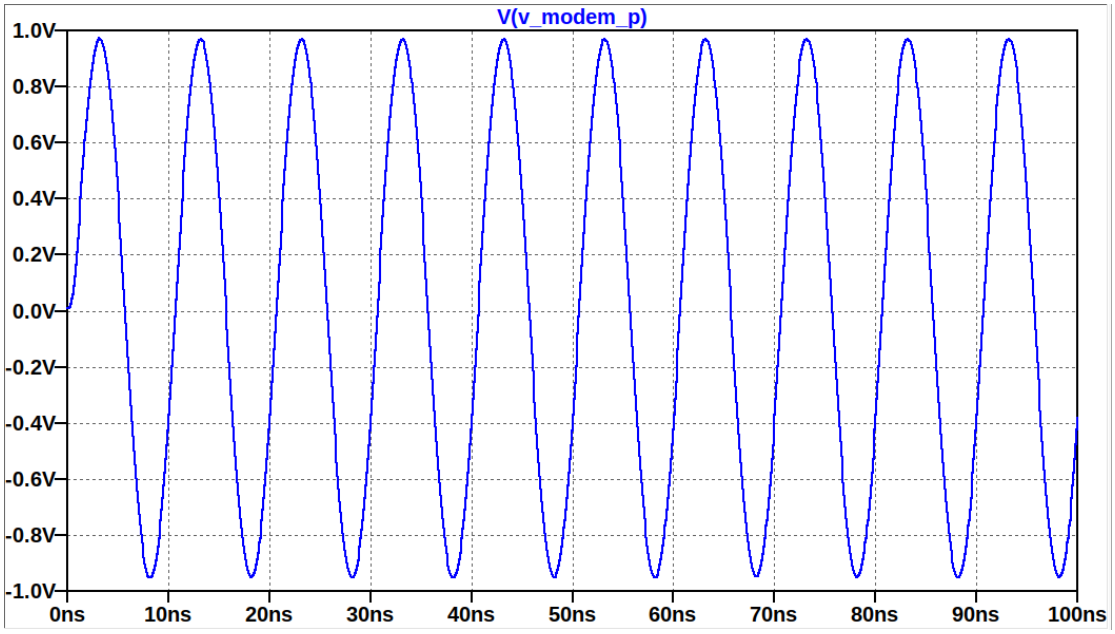


Fig. 63. Transient response of the RX FDA interface at the `V_MODEM_P` output node.

The `V_MODEM_P` waveform is centered near ground and reaches about $\pm 1\,V$, or $2\,V_{pp}$. The stage therefore performs the required differential-to-single-ended conversion and delivers a signal level compatible with the Red Pitaya ADC input range.

The frequency response at the same ADC-side node is shown in Fig. 64.

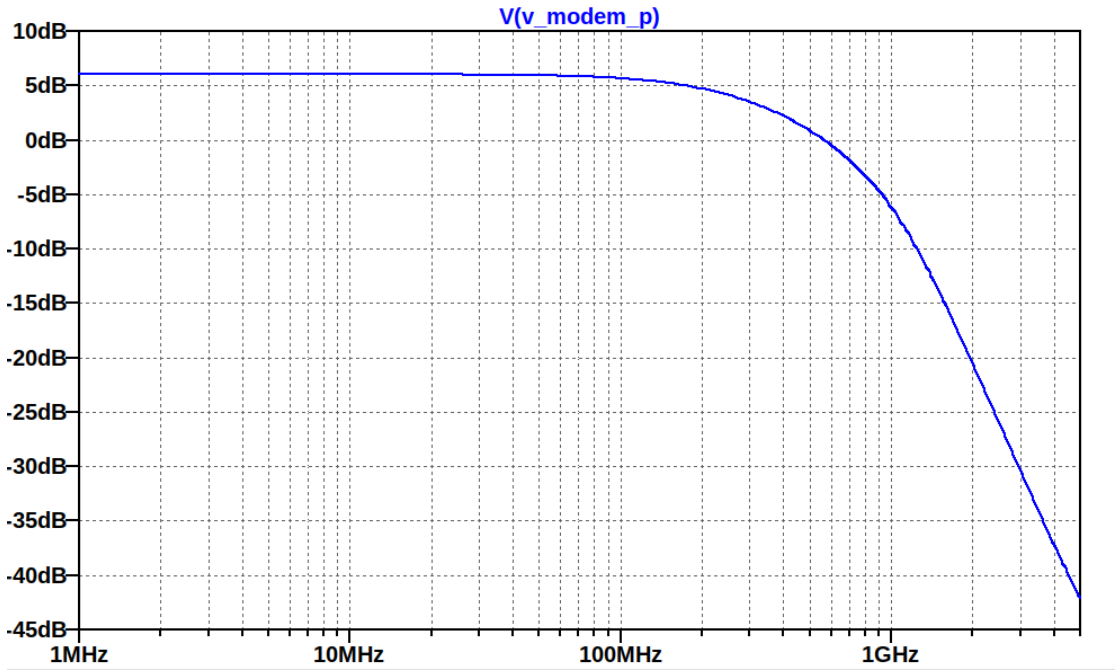


Fig. 64. AC response of the RX FDA interface at the Red Pitaya ADC-side node `V_MODEM_P`.

At low frequency, the magnitude of $V(\text{V_MODEM_P})$ is about 2.0 V/V when referenced to the amplitude of one input branch. Since the two `RX_BUFF` branches are driven with equal amplitude and opposite phase, this corresponds to about unity gain from differential input voltage to the used single-ended output branch, as expected from $A_{v,\text{diff}\rightarrow\text{SE}}$.

The extracted -3 dB frequency is about 0.34 GHz, or 340 MHz. This is above the maximum ALTUS28TR RX baseband filter setting of 240 MHz, so the ADA4927 interface does not set the bandwidth limit in the receiver path. The available margin is also above the bandwidth used by the Red Pitaya-based receiver configuration. The response shows no low-frequency peaking or obvious instability in the modeled operating condition.

No split termination was used at the RX FDA output. The receiver path is not meant to support the current-injection test mode used on the TX side, and the Red Pitaya connection uses one single-ended branch. A 50 Ω series element is kept in each output branch for isolation, launch damping, and reduced sensitivity to the capacitive environment at the ADC-side connection. The complementary branch is retained for symmetry and possible future access, but it is not part of the active single-ended Red Pitaya measurement path.

4.7.8 Generation and Buffering of the Common-Mode Reference

A dedicated common-mode reference network was implemented to provide the DC operating levels required by the baseband signal-conditioning stages. As shown

in Fig. 65, two reference-generation paths were used. The first sets a dedicated 0.5 V reference for the implemented TX FDA stage, while the second provides selectable buffered levels of 0.5 V, 1 V, and 1.5 V for auxiliary common-mode biasing of the split-termination network in the TX baseband path.

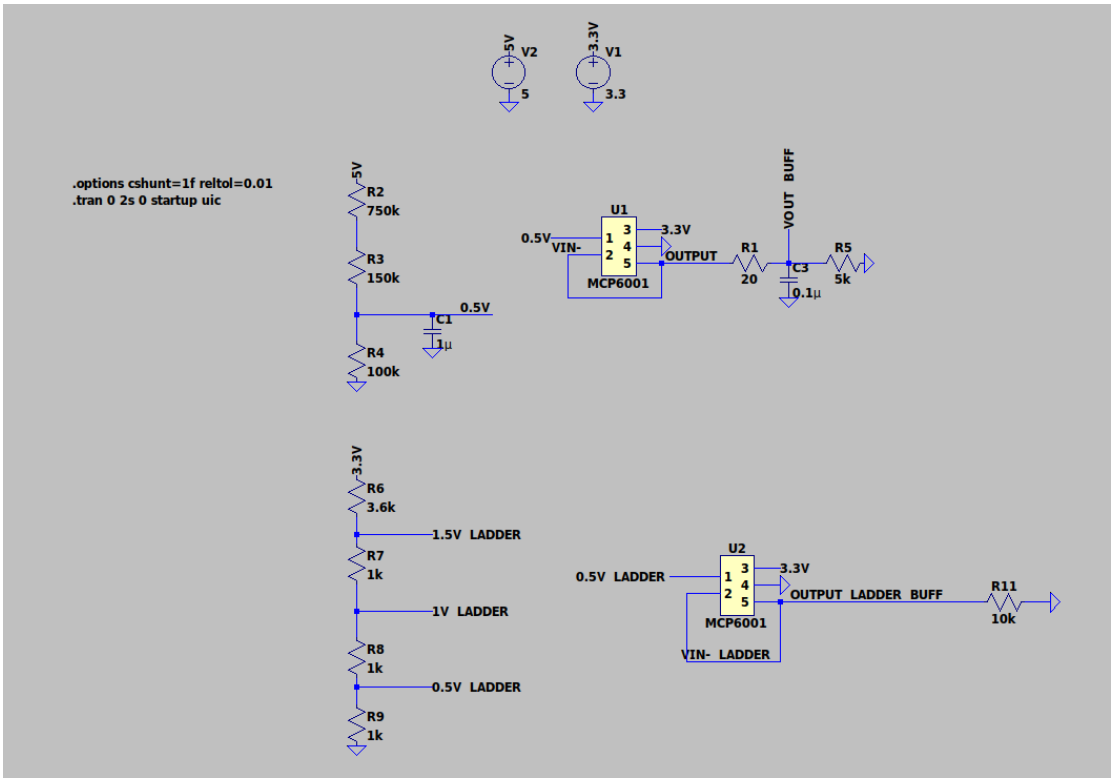


Fig. 65. Common-mode reference generation network, including the dedicated 0.5 V FDA reference branch and the selectable buffered ladder used for auxiliary common-mode biasing.

The dedicated 0.5 V reference is generated from the 5 V rail by means of a resistive divider and then buffered by a unity-gain operational amplifier before distribution to the corresponding baseband common-mode nodes. Buffering prevents the divider from being loaded by the downstream analog stages and PCB parasitics, while also providing a low-impedance reference source. A 20 Ω series resistor is placed at the buffer output to isolate the amplifier from the capacitive load associated with routing and local decoupling. Its main purpose is to stabilize the output, not to match the impedance.

In the final implementation, only one of the two initially considered 1 µF capacitors was retained on the dedicated 0.5 V reference branch. Removing the second capacitor reduced the settling time of the reference node significantly while preserving adequate local filtering for this biasing function. The simulated transient response of the buffered node is shown in Fig. 66. The result confirms monotonic settling of the buffered output toward the intended 0.5 V level, without overshoot or unstable behavior.

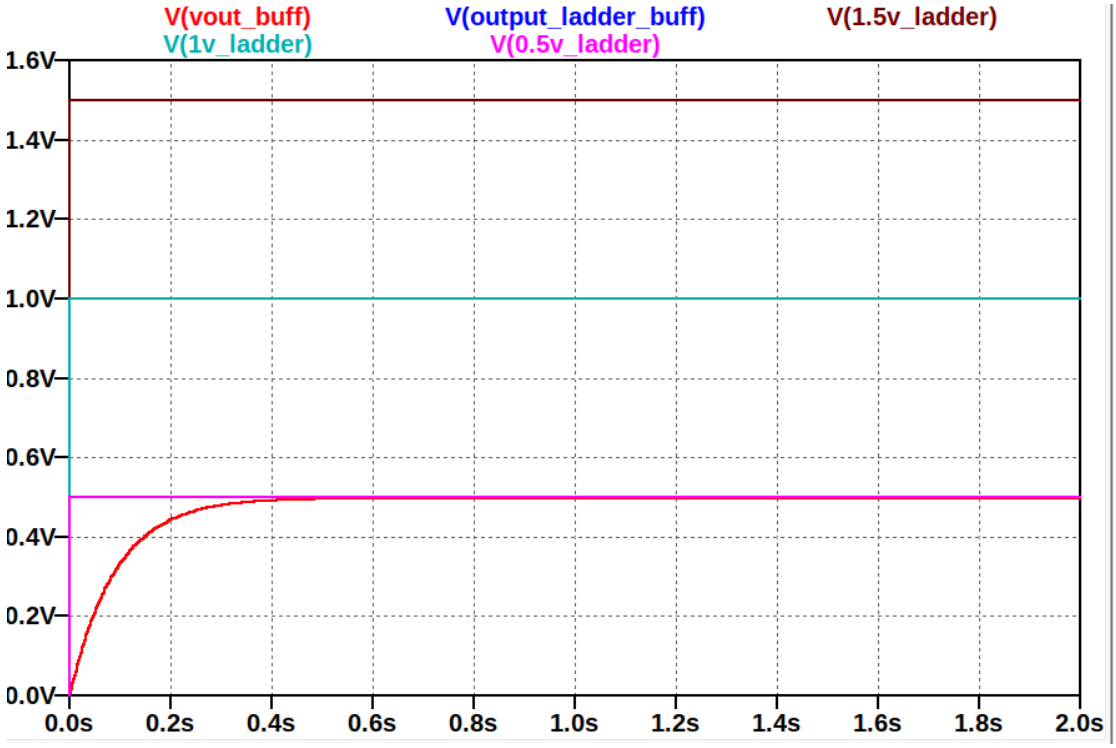


Fig. 66. Transient response of the common-mode reference network, showing correct settling of the buffered 0.5 V reference and the nominal selectable ladder levels.

The second reference path is implemented as a selectable resistive ladder derived from the 3.3 V rail. With the chosen resistor ratios, the ladder sets nominal tap voltages of 1.5 V, 1 V, and 0.5 V, which can be buffered and routed according to the required operating condition. This ladder is not required for the nominal Red Pitaya-to-FDA signal path, but it supports alternative biasing of the TX split-termination midpoint, especially in the current-injection operating mode discussed previously. In that operating mode, the common-mode level of the split-termination midpoint must be defined explicitly, while the differential signal itself is synthesized by external current injection. The selectable ladder therefore offers a practical means of imposing the required DC bias level at the corresponding common-mode control node without acting as an active signal driver.

4.8 Software Control of the Baseband Interface

A Python software layer was developed to exercise the host-side control of the Red Pitaya baseband interface. This software layer does not modify the RF or PCB design, but it provides the host-side framework through which waveform generation, TCP streaming, GPIO-based switch control, and TDD sequencing are coordinated. The five modules of this framework, the TCP-port mapping, and the three GPIO backends are documented in detail in Appendix A.

The code is organized around five responsibilities: transceiver configuration, bidirectional I/Q streaming, waveform generation, GPIO-based RF-switch control, and

frame-level TDD orchestration. Each responsibility is handled by a dedicated module. Transmission and reception use the TCP ports exposed by the Pavel Demin wideband SDR image, and three GPIO backends (memory-mapped, SCPI, and SSH) cover the different ways the Red Pitaya may be reached in practice. The framework was developed and verified in software simulation, including waveform generation, TCP streaming, and the four-state TDD state machine. Bring-up on the assembled PCB lies outside the scope of this work.

The controller advances through four states per frame: TX, guard, RX, guard. The TX path is disabled before RX is enabled, and the RF switch is updated during the guard interval. Since the sequencing runs in host software, OS scheduling introduces timing jitter on the order of milliseconds, which stretches the frame duration but does not cause TX/RX overlap, because the sequential logic enforces safe ordering by construction. Sub-microsecond deterministic timing would require FPGA-level control. A representative software-simulated frame sequence, including baseband activity, GPIO switch control, frame states, and the resulting signal-power envelope, is shown in Fig. 67.

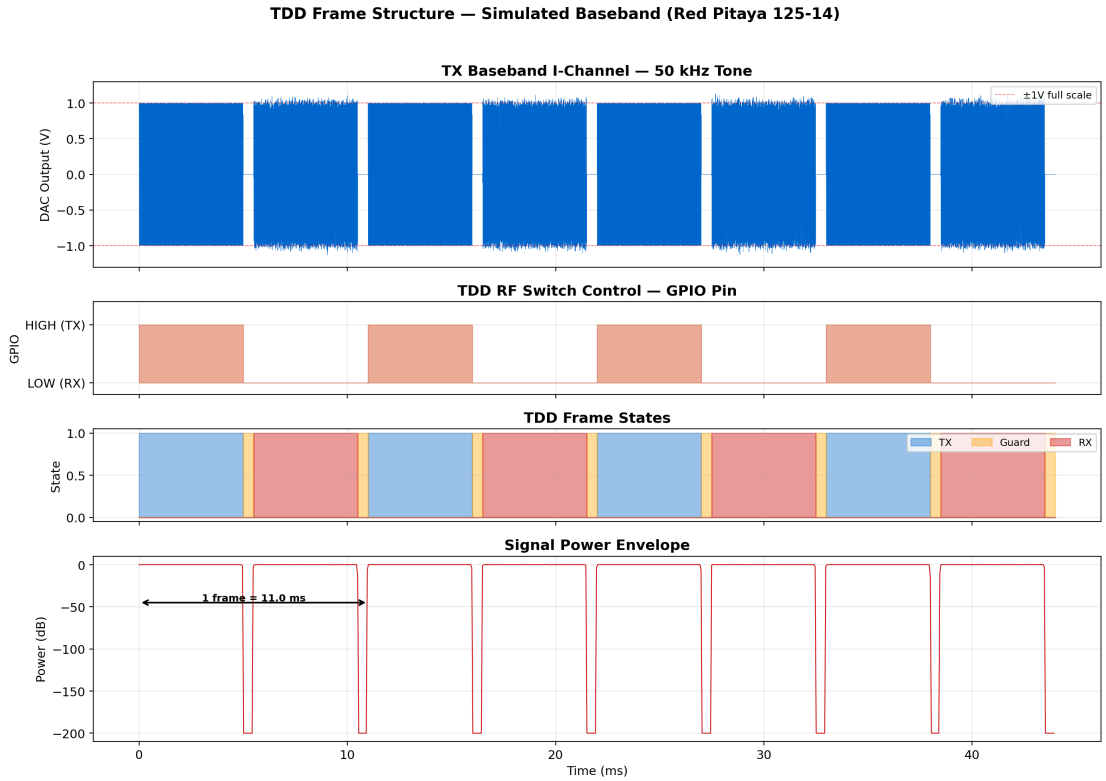


Fig. 67. Software-controlled TDD frame sequencing, showing baseband activity, GPIO switch control, frame states, and the resulting signal-power envelope.

Additional simulated software-side results, including waveform spectra, converter-mapping verification, and illustrative processing examples, are provided in Appendix C.

4.9 Electromagnetic Verification of Critical High-Frequency GCPW Structures

Selected high-frequency PCB interconnects were checked in RFPro [33] together with the circuit-level LTspice simulations. The extracted structures are local GCPW regions from the final PCB layout. The aim was not to simulate the full assembled board, but to examine areas where bends, vias, pads, nearby ground features, or layer transitions could introduce mismatch, loss, or coupling that a transmission-line estimate would not capture.

The EM models include the routed metal, pads, vias, ground pours, and dielectric stackup of the selected regions. They do not include active-device behavior, connector measurement fixtures, or package-level parasitics of the RF ICs and switches. The results should therefore be read as local PCB-interconnect checks, not as full assembled-front-end validation.

4.9.1 EM Verification Flow

The EM verification flow started from the realized KiCad layout. For each selected structure, the corresponding Gerber and drill files were imported into KLayout and then exported in GDSII format. The resulting GDSII data were subsequently imported into Virtuoso and used as the layout basis for RFPro. In this way, the EM model remained tied to the routed PCB geometry instead of an idealized transmission-line sketch.

After the geometry import, the board stackup was recreated in RFPro using the selected PCB materials, dielectric thicknesses, conductor layers, and vertical layer order. The substrate definition in Fig. 68 and the stackup views in Figs. 69 and 70 were used to check that the EM environment matched the selected hybrid stackup.

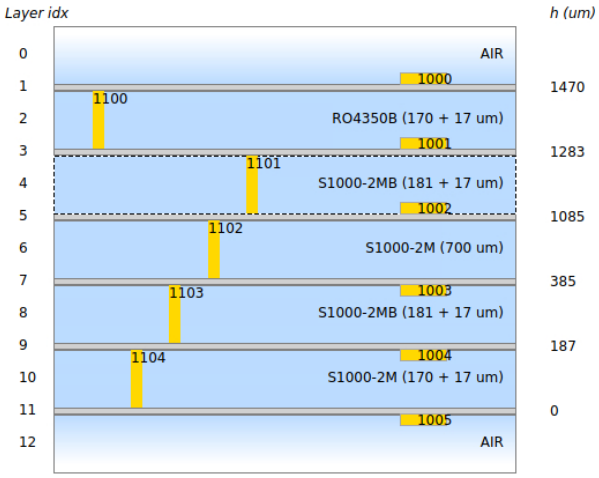


Fig. 68. Substrate definition used in RFPro for the EM verification of the selected high-frequency GCPW structures. The implemented dielectric materials, conductor layers, and their vertical arrangement were recreated in the EM environment.

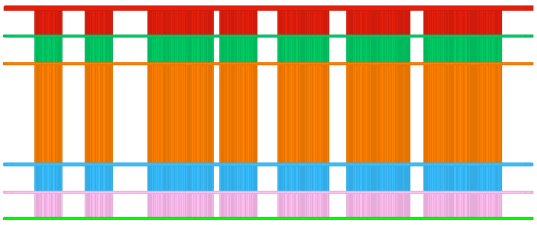


Fig. 69. Three-dimensional representation of the stackup recreated in the RFPro EM environment for the extracted PCB structures.

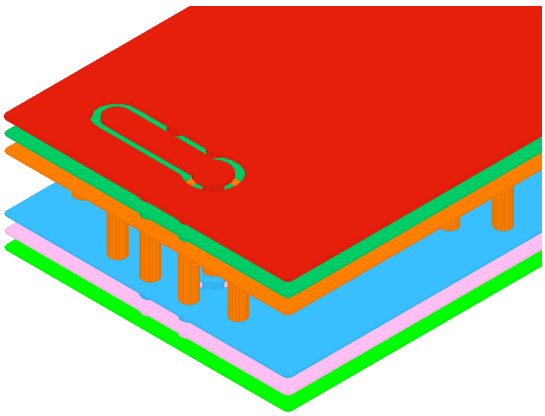


Fig. 70. Side view of the stackup recreated in the RFPro EM environment, showing the conductive and dielectric layers used in the verification setup.

For all simulated structures, excitation was defined using a ground–signal–ground (GSG) port configuration. The signal pin was placed at the physical signal-launch location of each extracted region, while the ground pins were assigned to the adjacent coplanar ground regions, which are tied to the underlying ground plane through stitching vias. Fig. 71 gives the GSG port convention, and Fig. 72 gives an example placement on extracted layout geometry.

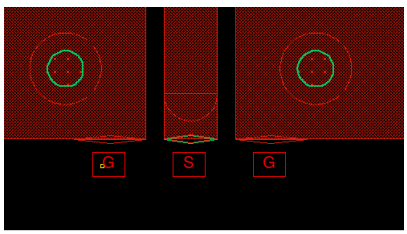


Fig. 71. Representative ground–signal–ground (GSG) port configuration used for the EM excitation of the selected GCPW structures.

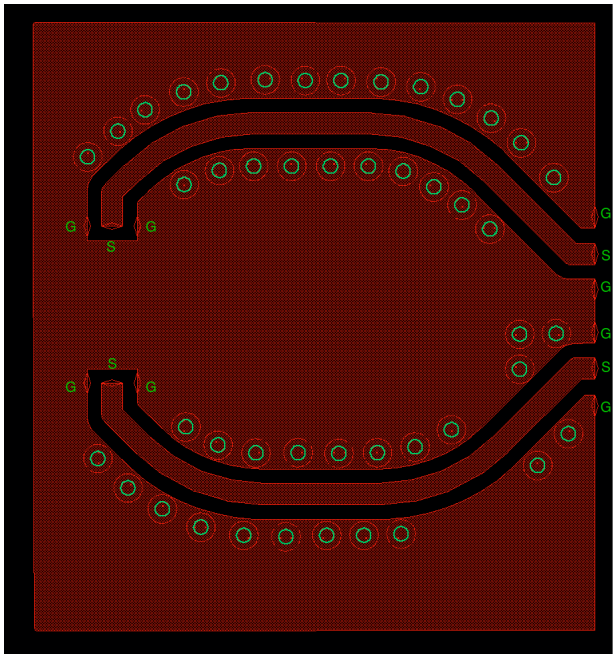


Fig. 72. Example of GSG port placement on an extracted EM structure. The signal pin is placed at the physical signal-launch location, while the ground pins are assigned to the adjacent coplanar ground regions tied to the underlying ground plane through stitching vias.

The full-wave simulations were run in RFPro using the Momentum RF solver at an ambient temperature of 25 °C. A direct-dense matrix solution was used. The via-conductor model was set to 3D, while the remaining physical-model and preprocessing options were kept at their RFPro defaults. Automatic meshing was used with a mesh density of 10 cells per wavelength.

Each extraction region was cropped to retain the local signal path and the surrounding coplanar ground structure. About 200 µm of additional metal was kept around the selected region to capture local field interaction with nearby ground. After port definition, the mesh was inspected around bends, vias, pads, and ground-fence regions before the result was accepted; Fig. 73 gives a representative example.

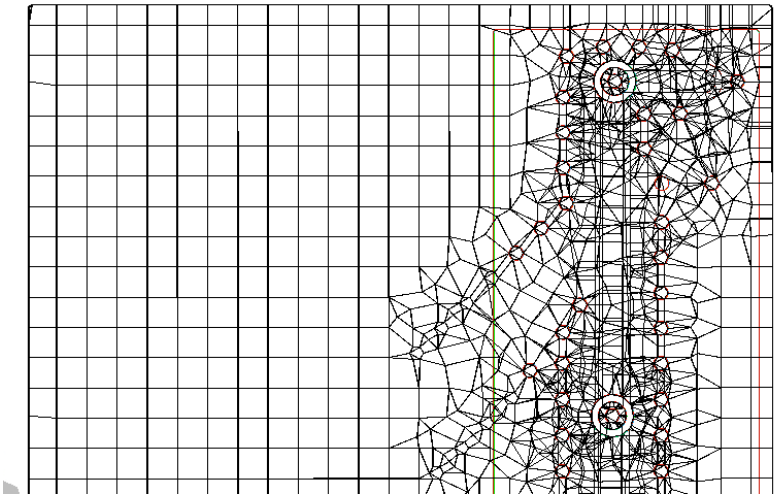


Fig. 73. Representative RFPro meshing view of one of the extracted high-frequency structures, showing local refinement around the signal path, via regions, and nearby grounding features.

After completion of the EM analysis, the extracted response was exported as an S-parameter file and evaluated in a separate N-port testbench. The generic evaluation structure used for this purpose is shown in Fig. 74. Each extracted PCB segment was then evaluated as a frequency-dependent N-port block, allowing its reflection and transmission behavior to be examined directly through its S-parameters.

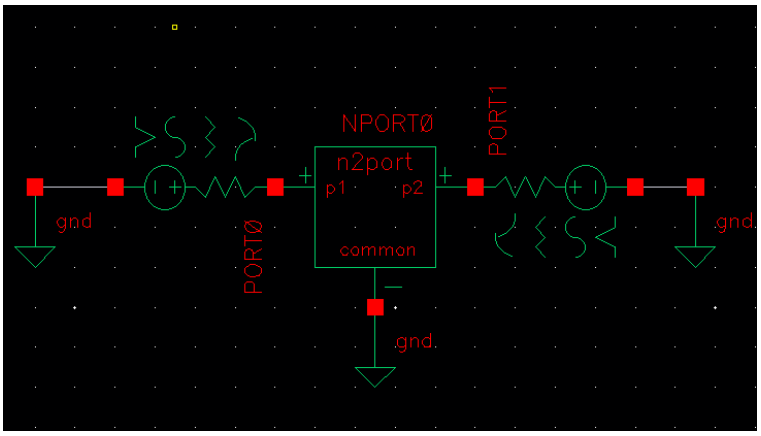


Fig. 74. Generic N-port testbench used to evaluate the exported S-parameter files of the EM-extracted structures in a circuit-level environment.

The extracted S-parameters were evaluated with a standard 50 Ω port reference impedance, over a frequency sweep extending from the low-frequency limit of the solver setup to 35 GHz using 50 linear points. The numerical values reported below are representative in-band figures, not exact point-by-point minima or maxima. The extracted EM models include only the passive PCB geometry of the selected regions: the realized metal traces, pads, vias, and surrounding ground structure. Internal package-level and active-device parasitics were not included. The EM results therefore verify the selected PCB interconnects locally; they do not constitute a full-board electromagnetic model of the assembled front-end.

Because the extracted PCB structures contain only linear reciprocal conductors and dielectrics, the corresponding S-parameter matrix is reciprocal, $S_{ij} = S_{ji}$, and therefore only one direction is shown explicitly. The output-side reflection responses, S_{22} and S_{44}, were also inspected and found to be comparable to their input-side counterparts, S_{11} and S_{33}. They are omitted here for brevity.

The discussion that follows focuses on driven-port matching, main-path transmission, and branch-to-branch coupling, because these responses capture the matching, through-path loss, and coupling behavior of the implemented interconnects.

4.9.2 LO Transition Structure

The first extracted structure was the final LO routing segment from the last splitter stage toward the two ALTUS28TR branches. This part of the LO path was of particular interest because it contains the most significant layer transition of the implemented design.

Figure 75 shows the KiCad layout excerpt used for the LO EM extraction. The vertical LO routing segment between the two transceiver branches is highlighted.

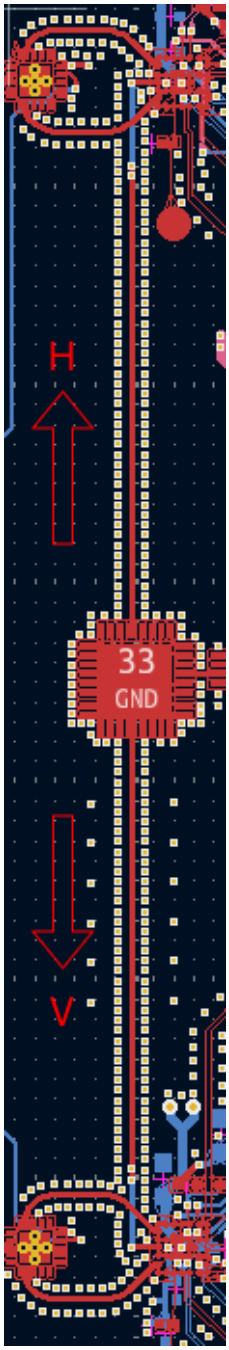


Fig. 75. KiCad layout excerpt of the LO distribution segment selected for EM verification. The extracted region corresponds to the final fan-out path from the last splitter stage toward the two ALTUS28TR branches and includes the critical vertical LO routing section.

Figure 76 shows the extracted LO geometry in the EM environment.

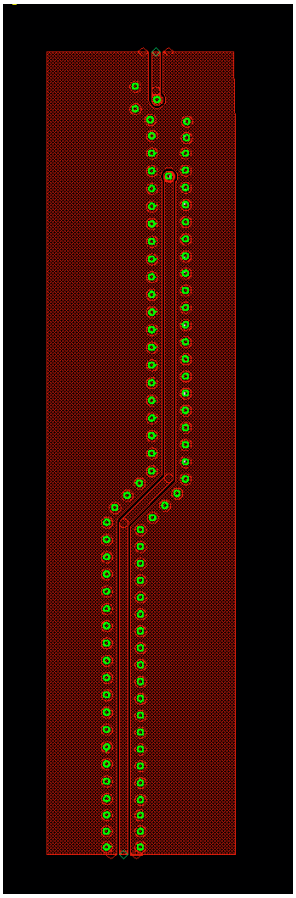


Fig. 76. Extracted LO transition structure in the EM environment.

The resulting S_{11} and S_{21} responses of the LO transition are shown in Figs. 77 and 78, respectively.

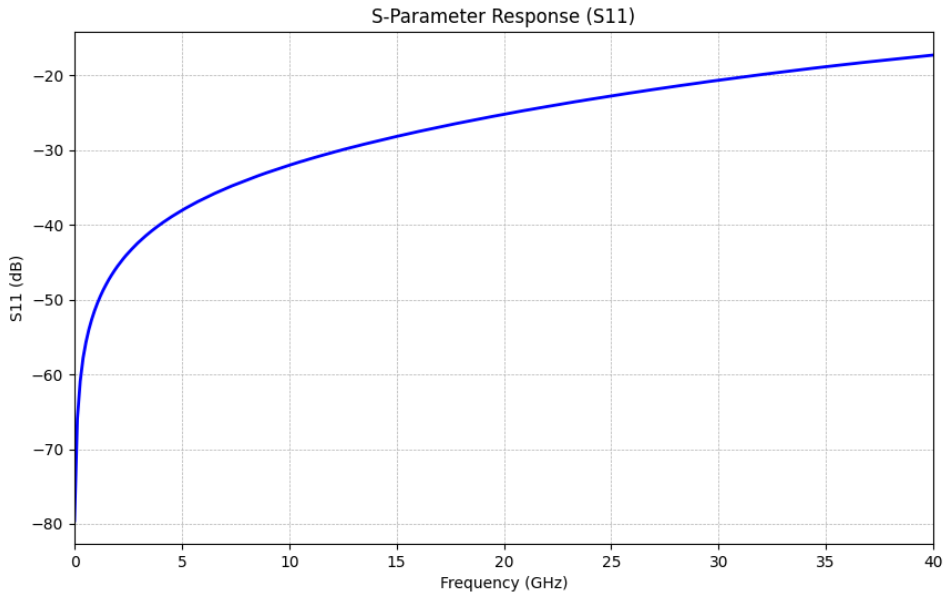


Fig. 77. Simulated S_{11} response of the extracted LO transition structure.

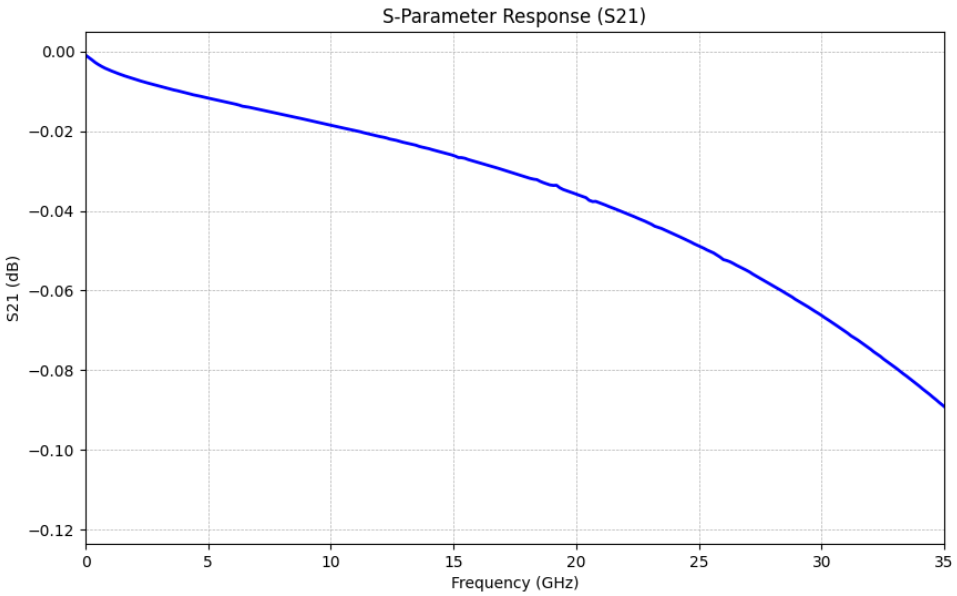


Fig. 78. Simulated S_{21} response of the extracted LO transition structure.

The total physical path length of this segment is approximately 2 mm. Within the 8 GHz to 11 GHz LO range, the simulated S_{11} remains close to −30 dB, and S_{21} stays above approximately −0.01 dB. For this short local LO segment, the extracted response is consistent with a well-matched transition and negligible PCB-only insertion loss over the LO band.

4.9.3 RF/TDD Interconnect Around the SPDT Switch

The RF/TDD switch region was extracted separately. It is longer and less uniform than the LO segment, about 8 mm per branch versus 2 mm for the LO transition, and electrically more complex: it combines curved routing, closely spaced grounded-coplanar sections, and a multi-port signal path around the SPDT switch footprint and the ALTUS28TR-side RF connections.

The corresponding KiCad layout excerpt is shown in Fig. 79. This figure identifies the actual layout region selected for EM extraction around the switching element and the transceiver-side RF/TDD routing.

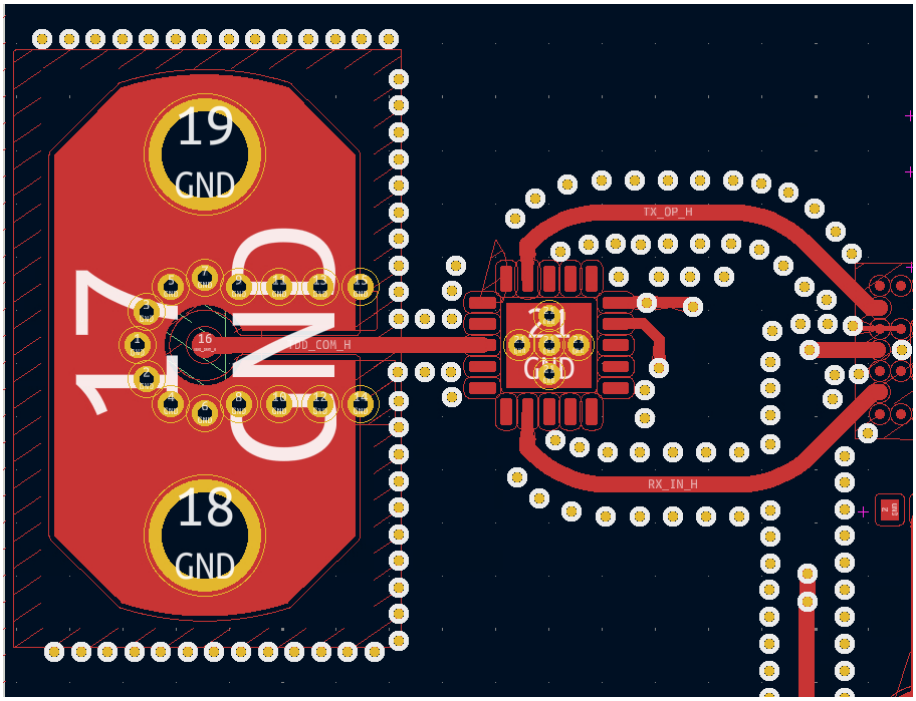


Fig. 79. KiCad layout excerpt of the RF/TDD interconnect region around the SPDT switch footprint and the ALTUS28TR-side RF connections, selected for EM verification.

The extracted geometry used in the EM environment is shown in Fig. 80.

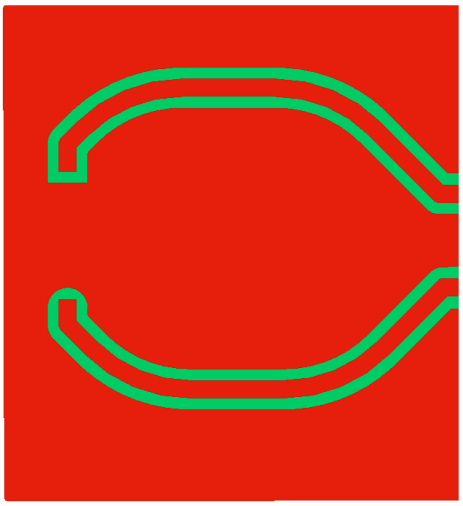


Fig. 80. Extracted RF/TDD interconnect geometry in the EM environment.

For this extracted multi-port structure, the exported S-parameter file used a four-port convention because the EM region contains two independent signal paths that share the PCB area around the SPDT switch footprint: the RX path between `RX_IN+` and `RX_OUT+`, and the TX path between `TX_IN+` and `TX_OUT+`. The active SPDT switch element itself is not part of the EM extraction; the goal here is to characterize the passive PCB routing, curved GCPW segments, pad transitions, and via

environment that surrounds the switch.

Accordingly, the ports are assigned as Port 1 to `RX_IN+`, Port 2 to `RX_OUT+`, Port 3 to `TX_IN+`, and Port 4 to `TX_OUT+`. Under this convention, the relevant S-parameters fall into three groups. The RX-path matching and insertion loss are described by S_{11} and S_{21}, the TX-path matching and insertion loss are described by S_{33} and S_{43}, and the branch-to-branch PCB coupling is captured by S_{31}, which represents input-side coupling between the two branches near the switch footprint, and S_{42}, which represents output-side coupling between the two ALTUS28TR-side connection points. The first two groups therefore characterize the two main signal paths through the extracted switching region, while the third group is used to assess coupling between the adjacent branches.

The last pair characterizes passive electromagnetic coupling between adjacent PCB traces, not the active TX/RX isolation provided by the SPDT switch in operation. Switch isolation is a property of the switch device itself, whereas the EM analysis here focuses on the PCB layout that connects to the switch.

The input-side return and transmission responses referenced to the RX branch are shown in Figs. 81 and 82. The TX-side return and transmission responses are shown in Figs. 83 and 84.

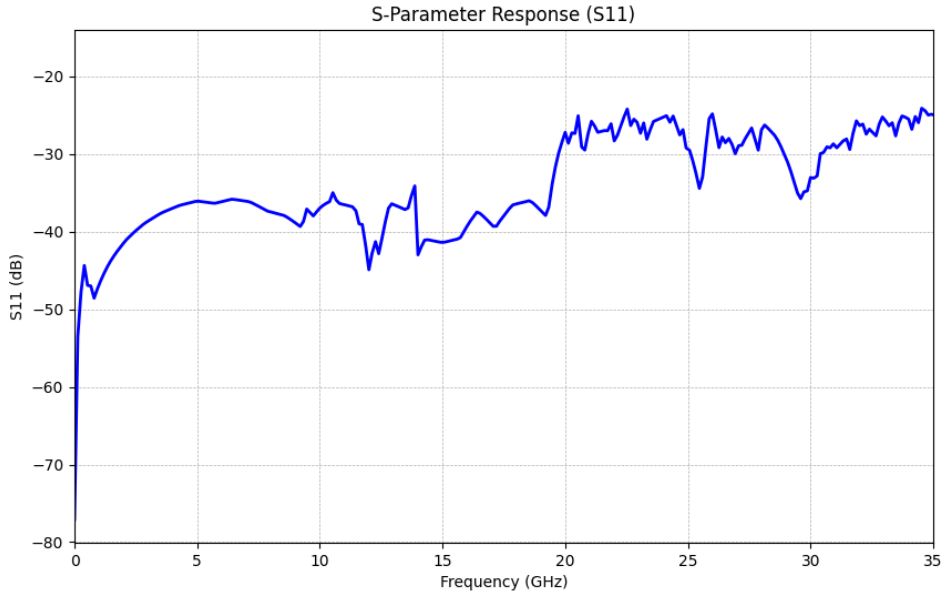


Fig. 81. Simulated S_{11} response of the extracted RF/TDD interconnect around the SPDT switch.

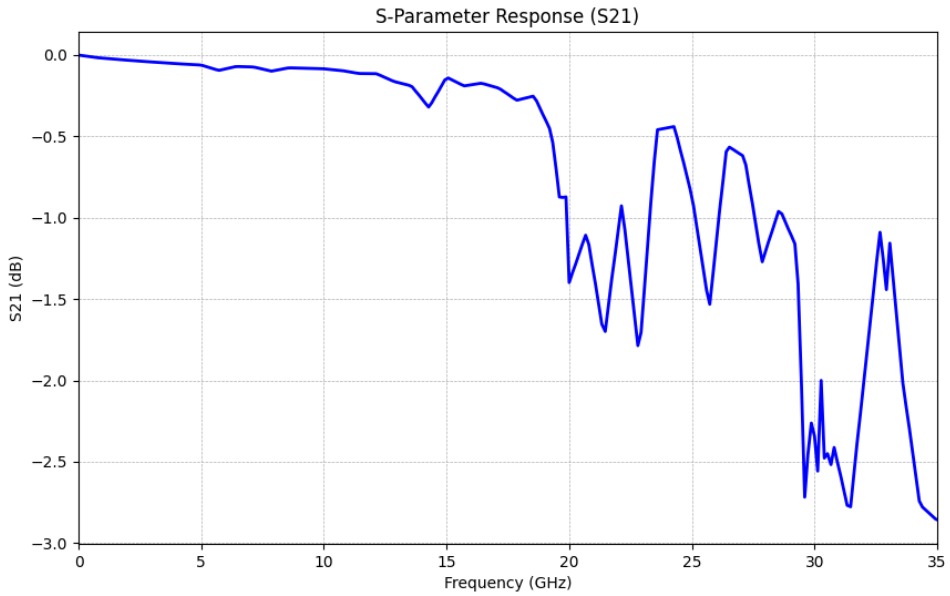


Fig. 82. Simulated S_{21} response of the extracted RF/TDD interconnect around the SPDT switch.

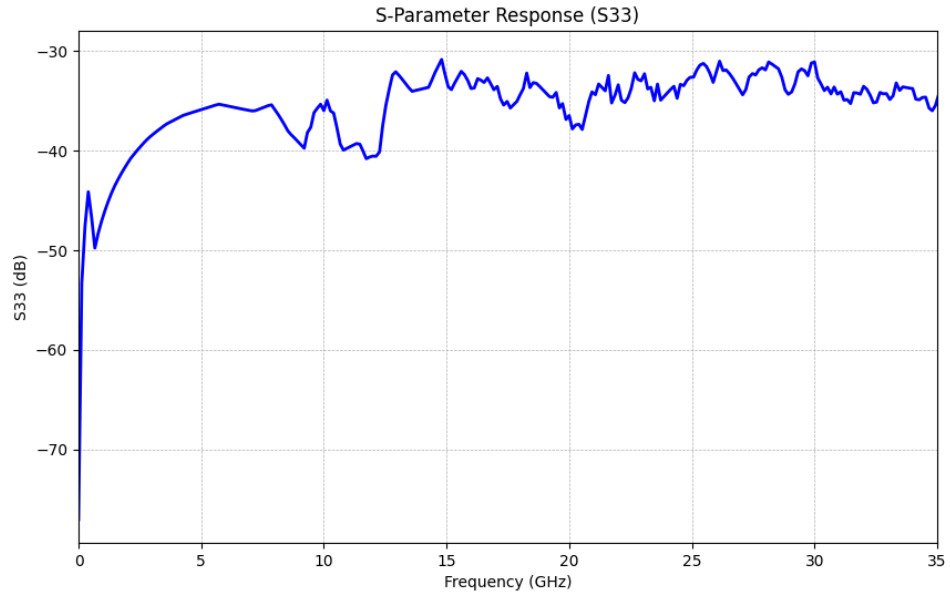


Fig. 83. Simulated S_{33} response of the extracted RF/TDD interconnect around the SPDT switch.

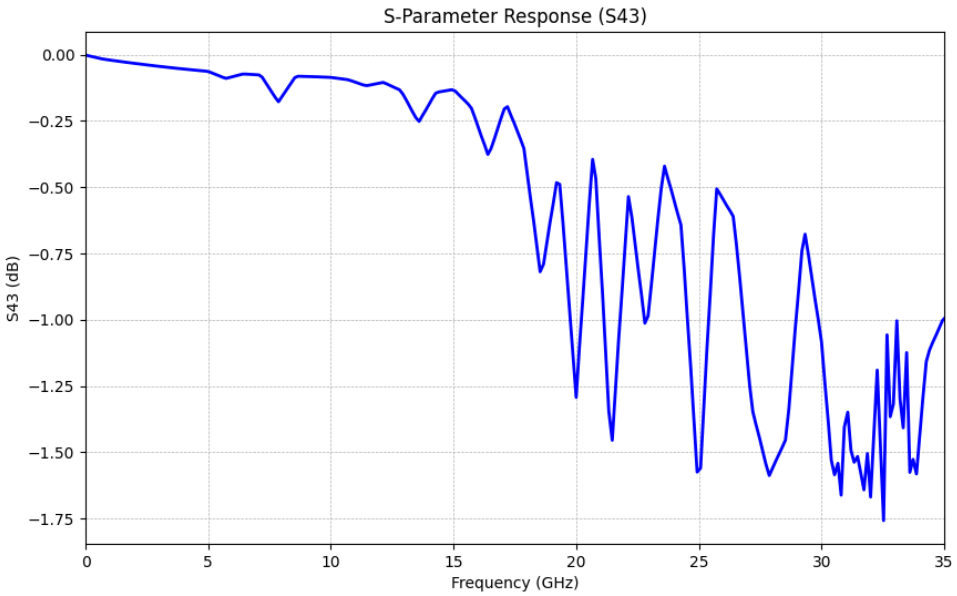


Fig. 84. Simulated S_{43} response of the extracted RF/TDD interconnect around the SPDT switch.

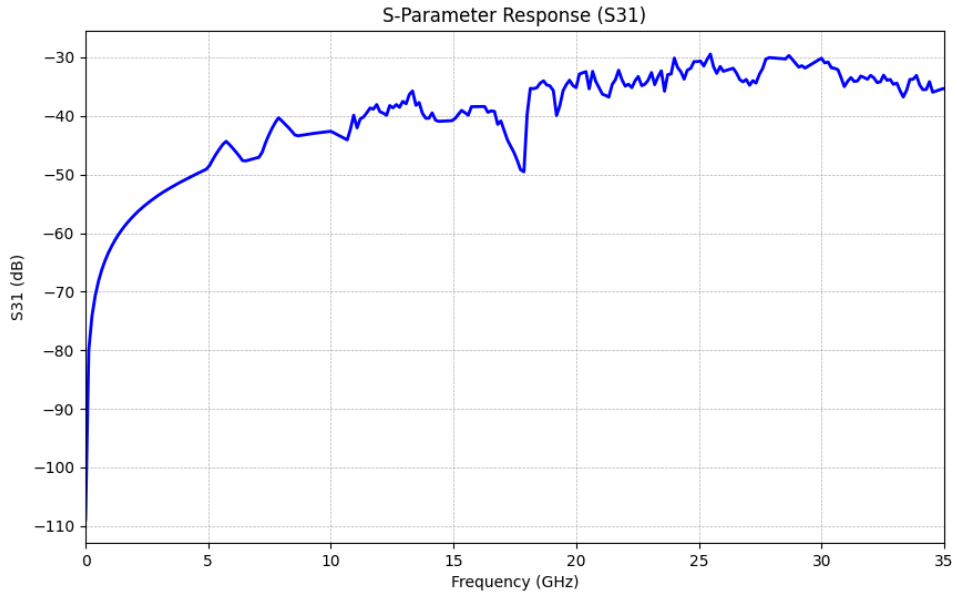


Fig. 85. Simulated S_{31} response of the extracted RF/TDD interconnect around the SPDT switch.

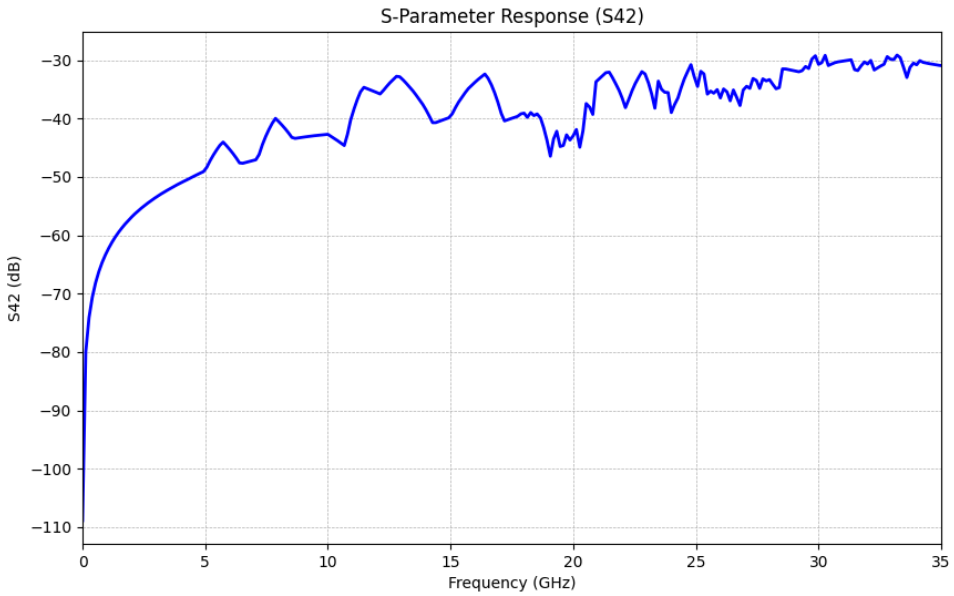


Fig. 86. Simulated S_{42} response of the extracted RF/TDD interconnect around the SPDT switch.

Compared with the short LO transition, the RF/TDD interconnect shows stronger dependence on local layout discontinuities. On the RX side, the reflection remains low but is more frequency dependent, while S_{21} shows in-band attenuation and ripple. Across the 28 GHz to 31 GHz RF band, S_{21} lies approximately between −1 dB and −3 dB, with an interpolated value near 30 GHz of about −2.5 dB. The TX-side response is slightly less degraded: S_{33} indicates good input matching, and S_{43} is about −1.5 dB near 30 GHz.

The RX and TX paths are not geometrically identical around the switch: the two branches follow different bend sequences and see different pad and via arrangements in the compact switching region. The observed loss is not pure distributed line attenuation, but the combined effect of conductor and dielectric loss plus the localized discontinuities introduced by that layout geometry.

Both coupling terms remain well below the main transmission paths. The corresponding responses are shown in Fig. 85 for the RX-side coupling S_{31} and in Fig. 86 for the TX-side coupling S_{42}. In the same 28 GHz to 31 GHz region, both responses remain in the -35 to −30 dB range, indicating limited transfer toward the adjacent branch on both sides of the extracted structure.

As a reference for interpreting the extracted insertion loss, a first-order attenuation estimate was computed for a straight GCPW section on the implemented stackup. The RF signal layer uses RO4350B ($\varepsilon_r = 3.48$, $\tan\delta = 0.0037$, $h = 0.17$ mm) with a trace width of 0.3 mm and a coplanar gap of 0.1 mm. At 30 GHz, the combined dielectric and conductor losses, including a first-order allowance for surface roughness, yield an estimated total attenuation on the order of 0.04 dB/mm to 0.07 dB/mm for a straight, discontinuity-free GCPW run.

For the 2 mm LO transition operating in the 8 GHz to 11 GHz band, the extracted loss of about 0.01 dB agrees with the short physical length and the lower operating frequency compared with the 30 GHz RF/TDD paths. The 30 GHz straight-line estimate is used here mainly as a reference for the RF/TDD interconnect. For the approximately 8 mm RF/TDD paths, the 30 GHz straight-line estimate gives about 0.32 dB to 0.56 dB. The extracted RF/TDD losses are higher than this estimate, indicating that the dominant contribution is not the straight-line GCPW loss alone. This excess is attributed to the curved routing geometry, the closely spaced multi-port layout around the switching region, and the associated pad and ground-fence discontinuities.

Tab. 8. Summary of the local EM-extracted PCB interconnect responses in their operating bands. The values are representative in-band readings from the simulated S-parameter responses.

Metric	Operating-band range	Representative value	Interpretation
LO Transition, 8 GHz *to* 11 GHz			
S_{11}	−22 dB to −20 dB	about −21 dB	low LO reflection
S_{21}	−0.1 dB or better	about −0.07 dB	low LO-path loss
RF/TDD Interconnect, 28 GHz *to* 31 GHz			
S_{11}	−35 dB to −25 dB	−34 dB near 30 GHz	low RX-side reflection
S_{21}	−1 dB to −3 dB	−2.5 dB near 30 GHz	RX-side through loss
S_{33}	−35 dB to −31 dB	−32 dB near 30 GHz	low TX-side reflection
S_{43}	−1.0 dB to −1.7 dB	−1.5 dB near 30 GHz	TX-side through loss
S_{31}	−35 dB to −30 dB	−31 dB near 30 GHz	limited input-side coupling
S_{42}	−35 dB to −30 dB	−30 dB near 30 GHz	limited output-side coupling

The EM results lead to two main observations. The LO transition shows the expected low-loss behavior across the 8 GHz to 11 GHz LO band, with low reflection and low insertion loss over its approximately 2 mm path length. The RF/TDD interconnect, by contrast, shows in-band loss above the straight-line GCPW estimate at 30 GHz. This difference is attributed to the curved routing, pad transitions, via environment, and compact multi-port geometry around the SPDT switch footprint, rather than to substrate loss alone. The EM results therefore support the LO routing choice directly and identify the RF/TDD switching region as the most critical high-frequency layout region among the structures examined.

5 Conclusions and Future work

5.1 Conclusions

This thesis developed a PCB design for a dual-polarization TDD mmWave transceiver platform intended for operation from 28 GHz to 31 GHz. The work started from the selected system architecture and ended in a routed mixed-signal board. The difficult part was the full transfer from architecture to PCB. The component choices, baseband interfaces, supply rails, RF and LO routing, digital control, and simulation checks all had to remain consistent with the same system assumptions.

The final design uses two ALTUS28TR transceivers, one shared NEXUS28Z-based LO path, and Red Pitaya boards for analog baseband access. The board has a size of 135 mm $\times$ 150 mm and uses a six-layer hybrid stackup. The RF and LO traces are routed on the top layer as grounded coplanar waveguides, with the adjacent internal layer used as the ground reference. The digital-control area was kept away from the RF and analog sections where the layout allowed it.

The stackup was a design choice, not only a fabrication detail. At 28 GHz to 31 GHz, material loss, trace geometry, and the return path affect the RF result. RO4350B was used in the outer dielectric regions, where the RF and LO traces are placed. The inner S1000-2M and S1000-2MB sections were used for the power planes, control routing, and board thickness. This kept the mmWave paths on a low-loss dielectric without forcing the whole board to use RF laminate.

The EM results showed that the high-frequency sections did not have the same margin. The LO route was easier to control. It stayed short, and only one high-frequency layer transition was kept. In RFPro, this extracted LO section gave S_{11} close to −20 dB and insertion loss below about 0.1 dB from 8 GHz to 11 GHz. The RF/TDD switch region was tighter. The SPDT switch, connector transition, local ground shape, and nearby branches all affected the same small area. In that section, S_{21} stayed between about −1 dB and −3 dB across the 28 GHz to 31 GHz RF band. The extracted coupling terms were about −35 dB to −30 dB. The switch region is therefore the part of the RF layout that should be revisited first in a later board version.

The PDN simulations gave stable results for the modeled load cases. The checked rails settled close to their nominal values. The ferrite-bead and local-decoupling stages reduced the simulated converter-side ripple from about 2 mV to 10 mV peak-to-peak to sub-millivolt levels at the point-of-load nodes. These results do not replace hardware measurements, but they show that the rail structure and post-regulation filtering are reasonable before fabrication.

The baseband interface met its design targets in the simulated setup. The LTC6409 transmit stage presents an approximately 50 Ω load to the Red Pitaya DAC, converts the signal to a differential pair, and sets the ALTUS28TR input common-mode level near 0.5 V. The ADA4927 receive stage drives the Red Pitaya ADC with a ground-centered signal of about 2 V_{pp}. In the simulation, this was done without excessive loading of the receiver-side buffer. These results apply to the interface circuits and their modeled loads, not to measured ALTUS28TR behavior.

The thesis produced a routed PCB design package with LTspice and RFPro checks on the main design risks. The board was not fabricated. For that reason, the work does not claim measured RF performance, measured timing margins, or over-the-air operation.

5.2 Limitations of the current study

The thesis is completed at PCB design and simulation level. Fabrication, assembly, bring-up, and laboratory measurements are the next stage of the work. The results therefore describe the expected behavior of the designed board before manufacturing.

The EM verification is local rather than full-board. It checks the routed LO layer transition and the RF/TDD switching area, because these regions depend strongly on the actual PCB geometry. Pads, bends, vias, nearby ground metal, and compact multi-port routing cannot be judged reliably from a simple transmission-line estimate. Connector launch behavior, soldering, package tolerances, material variation, and manufacturing spread remain outside the simulated results.

Some circuit models are also simplified. The PDN ferrite beads are represented with RLC equivalents instead of full vendor broadband models. This captures the intended filtering behavior at design level, but not vendor-specific resonances or DC-bias effects. The baseband simulations use simplified continuation models beyond the PCB boundary. These models give the FDA stages a more realistic load than an open circuit, but they do not reproduce the exact ALTUS28TR input and output behavior.

The onboard LO source does not cover the full target band by itself. The NEXUS28Z PLL output spans from 7.9 GHz to 9.8 GHz. With the internal $\times 3$ multiplication of the ALTUS28TR, this corresponds to about 23.7 GHz to 29.4 GHz at RF. Operation above about 29.4 GHz therefore requires the external LO input through the dedicated SMA connector, which is already included in the PCB design.

The Python control framework is intended for bring-up and functional testing. It can configure the Red Pitaya boards, generate the basic TDD sequence, handle GPIO-based RF-switch control, and support waveform generation and acquisition. It is not a deterministic timing controller. For tighter TDD timing, the switching sequence should move to FPGA logic or another hardware-timed controller.

5.3 Future work

The first hardware work should be bring-up, not RF testing. The supply rails, jumper-selected domains, current draw, rail indicators, and SPI communication should be checked before the RF path is exercised. These measurements are basic, but they decide whether the rest of the board can be tested safely.

The RFPro extractions give two direct measurement targets. The LO route should be measured from 8 GHz to 11 GHz to check the retained layer transition. The

RF/TDD switch region should be measured across the 28 GHz to 31 GHz band, since this was the area with the largest insertion-loss variation in simulation. Comparing these measurements with the extracted models would show whether the simulated weak point also appears on the fabricated board.

The shared LO path should be measured before dual-polarization tests. The useful quantities are amplitude balance, phase balance, external LO injection behavior, and branch-to-branch mismatch. These measurements would show how much error comes from the splitter, package transitions, and routed LO branches.

The Red Pitaya interface should be tested together with the host-side control software. Waveform generation and capture are the basic checks. TDD timing is the more important one. If the host-side sequence leaves too much timing uncertainty, the switching control should move to FPGA logic or to another hardware-timed controller.

The baseband networks may need small changes after measurement. The present resistor values and damping elements are supported by simulation, but the real transceiver-side loading may differ from the simplified models. The TX-side split-termination network also gives access for current-injection tests, which could help separate FDA behavior from the following transceiver input.

A later PCB revision should start with the RF/TDD switching region. The switch footprint transitions, pad geometry, local grounding, via placement, and branch spacing should be reviewed before larger layout changes are made. This is the area where the EM extraction showed the clearest loss and frequency dependence.

After board-level measurements, the platform can be connected to a suitable antenna or antenna array. That would move the work from PCB validation to over-the-air testing. The same hardware could then support dual-polarization transmission and reception tests, channel-sounding experiments, and TDD measurements with real guard intervals.

Appendix A Host-Side Software Framework for Baseband Control

This appendix documents the Python-based software framework developed to operate the Red Pitaya STEMlab 125-14 as a baseband generation and acquisition interface within the TDD transceiver architecture described in the main text. The framework runs on the host PC and communicates with the Red Pitaya through the TCP streaming infrastructure provided by the Pavel Demin *Wideband SDR Transceiver* FPGA image. The FPGA image was used as-is; all control is exercised from the host. The framework was not exercised on the final PCB, which remains at design and simulation level in the present work.

A.1 Design Rationale and Module Decomposition

The software is organized into functionally independent modules, each responsible for a distinct aspect of the baseband signal path. This decomposition mirrors the hardware-oriented signal chain and allows individual modules to be modified or replaced without changing the overall framework. Table 9 summarizes the module structure.

Tab. 9. Software module decomposition and functional responsibility.

Module	File	Responsibility
Configuration	`config.py`	Central parameter store (networking, timing, scaling, GPIO, decimation).
TCP Streaming	`streaming.py`	Bidirectional I/Q transport over TCP, including transceiver register configuration.
Waveform Generation	`waveforms.py`	Tone, two-tone, chirp, QPSK, and OFDM waveform synthesis.
GPIO Control	`gpio_controller.py`	RF-switch control through memory-mapped, SCPI, or SSH-based backends.
TDD Controller	`tdd_controller.py`	Frame-level orchestration of TX streaming, RX capture, and GPIO switching.

At system level, the TDD controller coordinates waveform generation, transmit streaming, receive capture, and GPIO state control within a common frame-level loop.

A.2 Converter-Side Signal Scaling

A central requirement of the software/hardware interface is the correct mapping between the normalized floating-point representation used internally and the physical voltage range at the Red Pitaya SMA connectors. In the intended low-voltage configuration, the STEMlab 125-14 DAC and ADC operate over a nominal full-scale range of ± 1 V at the SMA interface.

The underlying signed 14-bit code space spans $[-8192, +8191]$, but the implemented

software uses the practical scaling constant `FULL_SCALE = 8191` so that positive and negative excursions are symmetric and the -8192 code is never produced. The software-side mapping is therefore written as

$$x_{\text{code}} = \text{clip}(x_{\text{float}} \times 8191,\ -8191,\ +8191)\,, \tag{67}$$

where $x_{\text{float}} \in [-1.0, +1.0]$ is the normalized baseband sample and x_{code} is the integer code transmitted over the TCP link. The nominal output-voltage relation can be approximated as

$$V_{\text{out}} \approx \frac{x_{\text{code}}}{8191} \times 1.0\ \text{V}. \tag{68}$$

On the receive side, the inverse mapping is applied by dividing the signed integer ADC codes by 8191 in order to recover normalized complex floating-point samples. The nominal LSB size of the 14-bit converter is

$$\Delta V = \frac{1.0}{8192} \approx 122\ \mu\text{V}, \tag{69}$$

which reflects the converter resolution itself rather than the software clip limit.

The transmit-side conversion is implemented as follows:

Listing 1. Transmit-side scaling: normalized float to interleaved int16.

```
FULL_SCALE = 8191  # software scaling constant

def _complex_to_bytes(self, iq_complex):
        i = np.clip(np.real(iq_complex) * FULL_SCALE,
                -FULL_SCALE, FULL_SCALE).astype(np.int16)
        q = np.clip(np.imag(iq_complex) * FULL_SCALE,
                -FULL_SCALE, FULL_SCALE).astype(np.int16)
        interleaved = np.empty(2 * len(i), dtype=np.int16)
        interleaved[0::2] = i
        interleaved[1::2] = q
        return interleaved.tobytes()
```

The receive-side reconstruction is symmetric:

Listing 2. Receive-side scaling: interleaved int16 to normalized complex float.

```
def _bytes_to_complex(self, raw_bytes):
        samples = np.frombuffer(raw_bytes, dtype=np.int16)
        i_part = samples[0::2].astype(np.float64) / FULL_SCALE
        q_part = samples[1::2].astype(np.float64) / FULL_SCALE
        return i_part + 1j * q_part
```

Appendix B TDD Frame Sequencing

The TDD controller implements a software-based timing loop that cycles through four states per frame: TX active, guard (TX→RX), RX active, and guard (RX→TX). The default frame parameters are summarized in Table 10.

Tab. 10. Default TDD frame parameters.

Parameter	Value	Notes
TX slot duration	5.0 ms	4882 samples at $F_{s,\text{eff}}$
Guard interval	0.5 ms	RF-switch settling and software margin
RX slot duration	5.0 ms	4882 samples at $F_{s,\text{eff}}$
Frame period	11.0 ms	TX + Guard + RX + Guard
Effective sample rate	976 562 S/s	$125\,\text{MS/s}/128$
Usable bandwidth	∼400 kHz	After CIC roll-off

The critical aspect of the sequencing logic is the order in which the TX path, RX path, and GPIO state are updated within each transition. The implemented ordering ensures that the transmit path is always disabled before the receive path is enabled, and that the GPIO-controlled RF switch state is updated during the guard interval.

Listing 3. Core TDD timing loop (simplified).

```
while self._running:
        self._gpio.set_tx()                # initialization
        while self._running:
                # TX slot
                self._tx.enable()
                self._rx.disable()
                self._feed_tx_samples(tx_slot_sec)

                # Guard TX -> RX (GPIO updated inside guard)
                self._tx.disable()
                self._gpio.set_rx()
                time.sleep(guard_sec)

                # RX slot
                self._rx.enable()
                time.sleep(rx_slot_sec)
                rx_data = self._rx.read_samples(RX_SLOT_SAMPLES)

                # Guard RX -> TX (GPIO updated inside guard)
                self._rx.disable()
                self._gpio.set_tx()
                time.sleep(guard_sec)
```

The loop relies on `time.sleep()`, so operating-system scheduling may introduce millisecond-level timing uncertainty on a standard host platform. This stretches the effective frame duration rather than causing TX/RX overlap, because the logic disables TX before enabling RX unconditionally. Deterministic timing would require FPGA-level control and is left for future work.

B.1 GPIO Control for TDD RF Switching

The external TDD RF switch on the PCB is controlled by a single GPIO pin of the Red Pitaya configured as a digital output. The logic convention is:

- `HIGH` (3.3 V LVCMOS): TX path enabled
- `LOW` (0 V): RX path enabled

Three control backends are implemented in order to accommodate different deployment scenarios:

1. **Memory-mapped** (`/dev/mem`): direct register access via `mmap`. This is the lowest-latency option, but it requires execution on the Red Pitaya itself with suitable privileges.
2. **SSH-based**: register writes via SSH together with the `monitor` utility. This is suitable when the Red Pitaya is controlled remotely from the host PC.
3. **SCPI-based**: GPIO commands sent through the Red Pitaya SCPI server. This backend is intended for cases in which the stock Red Pitaya OS is used instead of the Pavel Demin image.

The factory function selects the appropriate backend according to the execution environment:

Listing 4. GPIO backend selection logic.

```
def create_gpio_controller(method="auto"):
        if method == "auto":
                if USE_SCPI_GPIO:
                        return ScpiGPIO()
                if "arm" in platform.machine().lower():
                        return MemMappedGPIO()
        return SshGPIO()
```

B.2 Effective Bandwidth and Decimation Trade-off

The Pavel Demin FPGA image employs CIC-based decimation on the receive path and interpolation on the transmit path. The effective sample rate is therefore determined by the programmable decimation factor,

$$F_{s,\mathrm{eff}} = \frac{125\ \mathrm{MS/s}}{D}, \tag{70}$$

where D is the decimation factor. The usable bandwidth is lower than the nominal Nyquist limit $F_{s,\mathrm{eff}}/2$ because the CIC passband exhibits roll-off toward the band edges. Table 11 summarizes the practical trade-off for representative settings.

Tab. 11. Decimation factor versus effective bandwidth.

D	$F_{s,\text{eff}}$	Usable BW	Limiting factor
128	976 kS/s	$\sim$400 kHz	CIC roll-off
32	3.9 MS/s	$\sim$1.5 MHz	CIC roll-off
8	15.6 MS/s	$\sim$6 MHz	Host-side transport

At low decimation factors, the practical limitation is not only the CIC response but also the host-side transport and processing load associated with sustained TCP streaming.

Appendix C Simulated Baseband Results

The results presented in this appendix were generated using the offline simulation mode of the developed software framework. They are based on a baseband-equivalent channel model with additive white Gaussian noise and optional delay or frequency offset, and they do not include RF front-end impairments. Their purpose is to verify the functional correctness of waveform generation, signal scaling, and TDD framing logic at software level.

All simulations use the same nominal digital settings as the intended hardware configuration, including 14-bit DAC/ADC resolution, $F_{s,\text{eff}} = 976{,}562\,\text{S/s}$ for $D = 128$, and a ± 1 V full-scale voltage range.

C.1 Generated Waveform Spectra

Figure 87 presents the power spectral density of representative baseband waveforms generated by the software framework. The spectra are consistent with the expected waveform types, including tone, two-tone, chirp, pulse-shaped QPSK, and OFDM excitation.

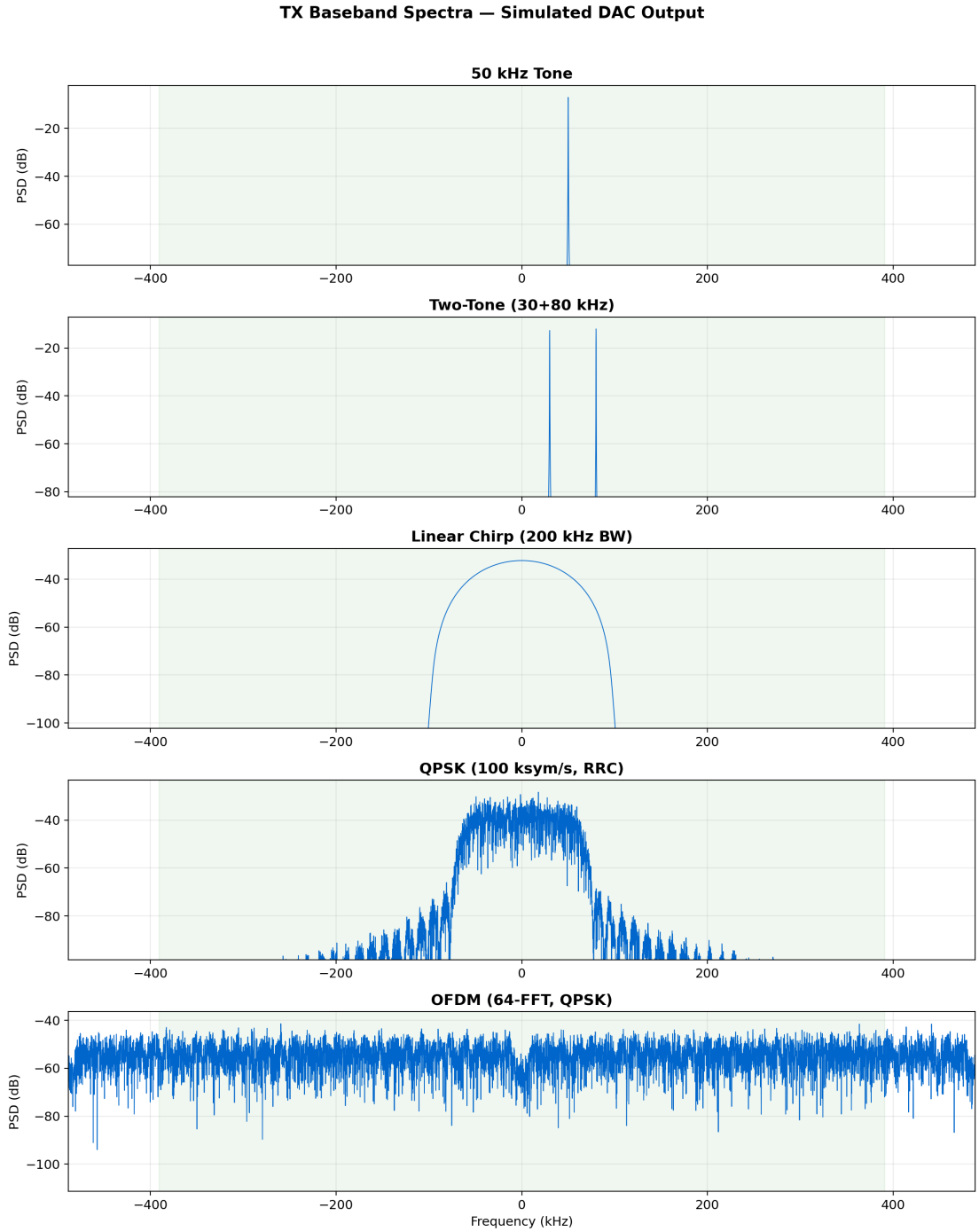


Fig. 87. Simulated power spectral density of representative waveform types generated by the software framework.

Appendix D KiCad and PCB Design Workflow

This appendix briefly describes KiCad, the open-source Electronic Design Automation (EDA) suite used to develop the PCB platform of this thesis, and summarizes the workflow followed during the design.

D.1 Overview of KiCad

KiCad is a cross-platform, open-source EDA suite released under the GNU General Public License (GPLv3). Its development has been supported, among others, by CERN and the KiCad Services Corporation, with continuous contributions from an active community. The suite integrates the tools required for a complete PCB design flow:

- **Schematic Editor (Eeschema):** hierarchical schematic capture with support for custom symbols and multi-sheet projects.
- **Symbol and Footprint Editors:** creation and management of custom component libraries, including 3D model association.
- **PCB Editor (Pcbnew):** multi-layer board layout with support for custom stackups, controlled-impedance routing, differential pairs, and length tuning.
- **PCB Calculator:** built-in transmission-line and impedance calculator for microstrip, stripline, coplanar, and grounded coplanar waveguide (GCPW) geometries, as well as differential pairs.
- **3D Viewer:** mechanical visualization of the assembled board for clearance verification and integration checks.
- **Gerber Viewer:** inspection of the manufacturing output files prior to release for fabrication.

KiCad 9.0.0 was used as the main schematic-capture and PCB-layout environment for this work. The relevant part of the workflow was not the general use of KiCad as an EDA tool, but its application to a mixed-signal mmWave board requiring custom component libraries, a controlled stackup, RF/LO net classes, GCPW routing, dense grounding, and export toward external EM verification.

D.2 PCB Design Workflow

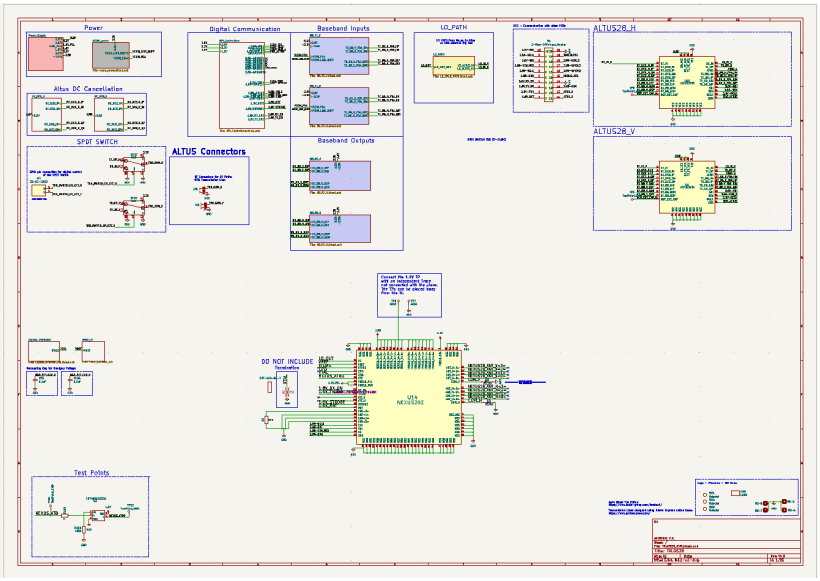
The board was developed in a sequence of stages, each producing a verified intermediate output before the next stage was started. The order followed during this thesis is summarized below.

D.2.1 Library Preparation

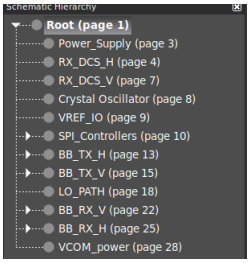
Before any schematic capture, custom symbols and footprints were created for components that were not available in the standard KiCad libraries, most notably the AKRONIC ALTUS28TR transceiver, the AKRONIC NEXUS28Z PLL, and the MASW-011145 SPDT switch, for which no manufacturer-supplied KiCad library was available at the time of layout. Symbols were drawn from the corresponding datasheets, and footprints were generated from the recommended land patterns. Each footprint was associated with a 3D model to enable subsequent mechanical verification in the 3D viewer.

D.2.2 Hierarchical Schematic Capture

The schematic was organized into hierarchical sheets, one per functional block: power distribution network, RF front-end, LO generation and distribution, baseband signal conditioning, and digital control. This separation kept each sheet readable and allowed independent review of each subsystem. Net classes were defined at this stage to distinguish power, ground, low-speed digital, high-speed digital (SPI), and RF/LO nets, so that the appropriate clearance and width rules could later be applied automatically in the PCB editor.



(a) Top-level hierarchical schematic of the project in the KiCad Schematic Editor, showing the main functional blocks and their interconnection at system level.



(b) Schematic hierarchy panel in KiCad, listing the top-level sheet and the underlying hierarchical sub-sheets of the design.

Fig. 88. Hierarchical schematic organization of the project in KiCad. The top-level sheet is shown together with the hierarchy panel, which lists the functional subsheets used in the design.

D.2.3 Electrical Rule Check

After the schematic was complete, the Electrical Rule Check (ERC) was run to detect unconnected pins, conflicting drivers, and inconsistent power flags. All ERC violations were resolved before the netlist was passed to the PCB editor.

D.2.4 Stackup Definition

Inside the PCB editor, the six-layer hybrid stackup was defined: RO4350B for the outer dielectrics and S1000-2MB (prepreg) and S1000-2M (core) for the inner dielectrics, for a total board thickness of 1.5832 mm. The dielectric constants and loss tangents were entered per layer, so that the controlled-impedance calculations performed inside KiCad would reflect the actual material properties of the board.



Fig. 89. Stackup configuration in the KiCad Board Setup dialog, showing the RO4350B and S1000-2M hybrid construction with per-layer thickness and dielectric properties.

D.2.5 Transmission-Line Geometry

The KiCad PCB Calculator was used to size the GCPW traces for the RF and LO paths and the differential pairs for the baseband signals. For the GCPW, a trace width of 0.30 mm with a gap of 0.10 mm to the side ground floods on F.Cu, referenced to In1.Cu.GND, gave $Z_0 \approx 50\,\Omega$. For the differential baseband traces, widths

of 0.14 mm with a separation of 0.10 mm on the same layer gave a differential impedance close to 100 Ω, compatible with the split-termination network used at the FDA outputs.

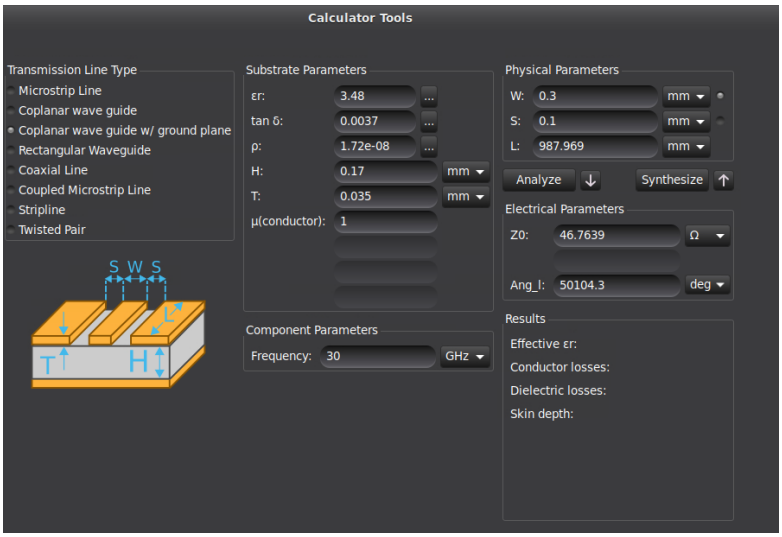


Fig. 90. KiCad PCB Calculator configured for the GCPW geometry used on F.Cu, referenced to In1.Cu.GND.

D.2.6 Component Placement

Components were placed starting from the RF front-end, since its layout is the most constraint-driven, and proceeding outward to the LO distribution, the baseband stages, the PDN, and finally the digital control subsystem. The two ALTUS28TR transceivers and the NEXUS28Z PLL were placed first to fix the main RF and LO routing channels; decoupling capacitors were then placed close to their respective supply pins, and the buck converters and LDOs were placed next so that the high-current PDN paths could be kept short and direct.

D.2.7 Routing

Routing was performed in the same order as placement. RF and LO traces were routed first as GCPW on F.Cu with the solder mask removed over them, using two parallel vias at every layer transition to reduce inductance and smooth the impedance discontinuity. Differential baseband pairs were routed next, with length matching applied where required. The digital SPI buses and GPIO control lines were routed on B.Cu.CNTRL, and the PDN was completed last using the In2.Cu.PWR and In3.Cu.PWR planes for power distribution and the In1.Cu.GND and In4.Cu.GND planes as solid references.

D.2.8 Copper Zones and Ground Pours

Copper zones were defined on the ground and power planes, as well as on the side floods of the GCPW traces on F.Cu. The zone properties dialog was used to set the clearance to neighboring nets, the minimum width, the pad connection style (thermal relief for through-hole pads and solid for surface-mount RF pads), and the fill type. The side floods of the RF and LO traces were stitched to the underlying In1.Cu.GND plane with a regular pattern of ground vias, in order to suppress parasitic modes between the floods and the reference plane.

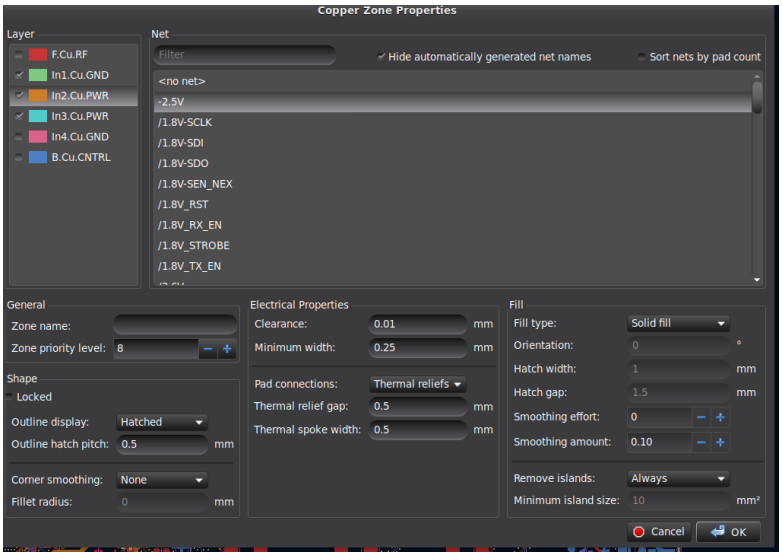


Fig. 91. Copper zone properties dialog in the KiCad PCB Editor, used to configure clearance, minimum width, pad connection style, and fill type for the ground and power planes.

D.2.9 Design Rule Check

Once routing was complete, the Design Rule Check (DRC) was run to verify clearances, drill sizes, annular rings, and net connectivity against the rules defined for each net class. The final design was checked with ERC and DRC. Remaining warnings, if any, were reviewed against the intended schematic and layout constraints, so that no known rule violation affecting fabrication release or circuit connectivity was left unresolved.

D.2.10 Mechanical Verification

The 3D viewer was used to confirm that the placement of connectors, shielding cans, and tall components did not conflict with the intended enclosure. The board outline and mounting holes were checked against the mechanical drawings of the connectors and the Red Pitaya STEMlab carrier board.

D.2.11 Manufacturing Outputs

The final step was the generation of the manufacturing outputs: Gerber X2 files for fabrication, an Excellon drill file, a STEP file for mechanical integration, and the BOM and component placement files for assembly. Although fabrication and assembly are out of the scope of this thesis, the full manufacturing data set was produced and verified in the Gerber viewer to ensure that the design is ready for release.

D.3 Interaction with External Tools

KiCad covers the full PCB design flow, but the electromagnetic verification of the high-frequency structures was performed in a separate, full-wave solver (RFPro), as discussed in the corresponding chapter. The relevant geometries, namely the GCPW sections, the via transitions, and the RF/TDD switching region, were exported from KiCad and imported into the EM solver, while the per-layer stackup parameters were re-entered to keep the two environments consistent. This division of labor uses KiCad for layout and design rule enforcement and the EM solver for the verification of the parts of the layout where the assumptions of the built-in transmission-line calculator (homogeneous dielectric, ideal ground reference, no edge effects) are not strictly valid.

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